

Oracle® Database

Database Installation Guide



19c for Oracle Solaris
E96434-12
April 2026

ORACLE®

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Preface

This guide explains how to install and configure single-instance Oracle Database.

This guide also provides information about Optimal Flexible Architecture, cloning an Oracle home, and how to remove the database software.

- [Audience](#)
This guide is intended for anyone responsible for installing Oracle Database 19c.
- [Documentation Accessibility](#)
- [Diversity and Inclusion](#)
- [Set Up Java Access Bridge to Implement Java Accessibility](#)
Install Java Access Bridge so that assistive technologies on Microsoft Windows systems can use the Java Accessibility API.
- [Command Syntax](#)
Refer to these command syntax conventions to understand command examples in this guide.
- [Related Documentation](#)
The related documentation for Oracle Database products includes the following manuals:
- [Conventions](#)

Audience

This guide is intended for anyone responsible for installing Oracle Database 19c.

Additional installation guides for Oracle Database, Oracle Real Application Clusters, Oracle Clusterware, Oracle Database Examples, and Oracle Enterprise Manager Cloud Control are available at the following URL:

<http://docs.oracle.com>

Documentation Accessibility

For information about Oracle's commitment to accessibility, visit the Oracle Accessibility Program website at <http://www.oracle.com/pls/topic/lookup?ctx=acc&id=docacc>.

Access to Oracle Support

Oracle customer access to and use of Oracle support services will be pursuant to the terms and conditions specified in their Oracle order for the applicable services.

Diversity and Inclusion

Oracle is fully committed to diversity and inclusion. Oracle respects and values having a diverse workforce that increases thought leadership and innovation. As part of our initiative to build a more inclusive culture that positively impacts our employees, customers, and partners, we are working to remove insensitive terms from our products and documentation. We are also mindful of the necessity to maintain compatibility with our customers' existing technologies and the need to ensure continuity of service as Oracle's offerings and industry standards evolve. Because of these technical constraints, our effort to remove insensitive terms is ongoing and will take time and external cooperation.

Set Up Java Access Bridge to Implement Java Accessibility

Install Java Access Bridge so that assistive technologies on Microsoft Windows systems can use the Java Accessibility API.

Java Access Bridge is a technology that enables Java applications and applets that implement the Java Accessibility API to be visible to assistive technologies on Microsoft Windows systems.

Refer to *Java Platform, Standard Edition Accessibility Guide* for information about the minimum supported versions of assistive technologies required to use Java Access Bridge. Also refer to this guide to obtain installation and testing instructions, and instructions for how to use Java Access Bridge.

Related Topics

- *Java Platform, Standard Edition Java Accessibility Guide*

Command Syntax

Refer to these command syntax conventions to understand command examples in this guide.

Convention	Description
\$	Bourne or BASH shell prompt in a command example. Do not enter the prompt as part of the command.
%	C Shell prompt in a command example. Do not enter the prompt as part of the command.
#	Superuser (root) prompt in a command example. Do not enter the prompt as part of the command.
monospace	UNIX command syntax
backslash \	A backslash is the UNIX and Linux command continuation character. It is used in command examples that are too long to fit on a single line. Enter the command as displayed (with a backslash) or enter it on a single line without a backslash: <code>dd if=/dev/rds/c0t1d0s6 of=/dev/rst0 bs=10b \ count=10000</code>
braces { }	Braces indicate required items: <code>.DEFINE {macro1}</code>

Convention	Description
brackets []	Brackets indicate optional items: cvtcrt termname [outfile]
ellipses ...	Ellipses indicate an arbitrary number of similar items: CHKVAL fieldname value1 value2 ... valueN
<i>italic</i>	Italic type indicates a variable. Substitute a value for the variable: <i>library_name</i>
vertical line	A vertical line indicates a choice within braces or brackets: FILE filesize [K M]

Related Documentation

The related documentation for Oracle Database products includes the following manuals:

Related Topics

- *Oracle Database Concepts*
- *Oracle Database New Features Guide*
- *Oracle Database Licensing Information*
- *Oracle Database Release Notes*
- *Oracle Grid Infrastructure Installation Guide*
- *Oracle Database Client Installation Guide for Oracle Solaris*
- *Oracle Database Examples Installation Guide*
- *Oracle Real Application Clusters Installation Guide for Linux and UNIX*
- *Oracle Database Administrator's Reference for Linux and UNIX-Based Operating Systems*
- *Oracle Automatic Storage Management Administrator's Guide*
- *Oracle Database Upgrade Guide*
- *Oracle Database 2 Day DBA*
- *Oracle Application Express Installation Guide*

Conventions

The following text conventions are used in this document:

Convention	Meaning
boldface	Boldface type indicates graphical user interface elements associated with an action, or terms defined in text or the glossary.
<i>italic</i>	Italic type indicates book titles, emphasis, or placeholder variables for which you supply particular values.
monospace	Monospace type indicates commands within a paragraph, URLs, code in examples, text that appears on the screen, or text that you enter.

Changes in this Release for Oracle Database

Learn about the new features and changes in Oracle Database Installation Guide for Oracle Database 19c.

- [New Features](#)
Review new features available with Oracle Database installation in Oracle Database 19c.
- [Deprecated Features](#)
Review features that are deprecated starting with Oracle Database 19c.
- [Other Changes](#)
Review other changes for Oracle Database 19c.

Related Topics

- *Oracle Database New Features Guide*

New Features

Review new features available with Oracle Database installation in Oracle Database 19c.

- [Root Scripts Automation Support for Oracle Database Installation](#)
- [Simplified Image-Based Oracle Database Client Installation](#)

Root Scripts Automation Support for Oracle Database Installation

Starting with Oracle Database 19c, the database installer, or setup wizard, provides options to set up permissions to run the root configuration scripts automatically, as required, during a database installation. You continue to have the option to run the root configuration scripts manually.

Setting up permissions for root configuration scripts to run without user intervention can simplify database installation and help avoid inadvertent permission errors.

Related Topics

- [Running Oracle Database Setup Wizard to Install Oracle Database](#)
Extract the database image files and use the `runInstaller` command to start the installation.

Simplified Image-Based Oracle Database Client Installation

Starting with Oracle Database 19c, the Oracle Database client software is available as an image file for download and installation. You must extract the image software into a directory where you want your Oracle home to be located, and then run the `runInstaller` script to start the Oracle Database client installation. Oracle Database client installation binaries continue to be available in the traditional format as non-image zip files.

As with Oracle Database and Oracle Grid Infrastructure image file installations, Oracle Database client image installations simplify Oracle Database client installations and ensure best practice deployments.

Related Topics

- [Installing Oracle Database Client Using Image File](#)

Deprecated Features

Review features that are deprecated starting with Oracle Database 19c.

The following feature is deprecated in this release, and may be desupported in another release. For more information about deprecated and desupported features, parameters and views, refer to *Oracle Database Upgrade Guide*.

- Deprecation of Oracle ASMFD

Oracle ASM Filter Driver (ASMFD) is deprecated on both Linux and Oracle Solaris beginning with Oracle Database 19c. It will be desupported in a future release.

- Deprecation of clone.pl

The `clone.pl` script is deprecated in Oracle Database 19c. The functionality of performing a software-only installation, using the gold image, is available in the installer wizard. The `clone.pl` script can be removed in a future release. Instead of using the `clone.pl` script, Oracle recommends that you install the extracted gold image as a home, using the installer wizard.

- Deprecation of the SERVICE_NAMES parameter

The use of the SERVICE_NAMES parameter is no longer actively supported. It must not be used for high availability (HA) deployments. It is not supported to use service names parameter for any HA operations. This restriction includes FAN, load balancing, FAILOVER_TYPE, FAILOVER_RESTORE, SESSION_STATE_CONSISTENCY, and any other uses.

Related Topics

- [Oracle Database Upgrade Guide](#)

Other Changes

Review other changes for Oracle Database 19c.

- Rapid Home Provisioning Name Change

Starting with Oracle Database 19c and Oracle Grid Infrastructure 19c, Rapid Home Provisioning is renamed to Fleet Patching and Provisioning (FPP).

1

Oracle Database Installation Checklist

Use checklists to review system requirements, and to plan and carry out Oracle Database installation.

Oracle recommends that you use checklists as part of your installation planning process. Using checklists can help you to confirm that your server hardware and configuration meet minimum requirements for this release, and can help you to ensure you carry out a successful installation.

- [Server Hardware Checklist for Oracle Database Installation](#)
Use this checklist to check hardware requirements for Oracle Database.
- [Operating System Checklist for Oracle Database on Oracle Solaris](#)
Use this checklist to check minimum operating system requirements for Oracle Database.
- [Server Configuration Checklist for Oracle Database Installation](#)
Use this checklist to check minimum server configuration requirements for Oracle Database installations.
- [Oracle User Environment Configuration Checklist for Oracle Database Installation](#)
Use this checklist to plan operating system users, groups, and environments for Oracle Database management.
- [Storage Checklist for Oracle Database Installation](#)
Use this checklist to review storage minimum requirements and assist with configuration planning.
- [Installer Planning Checklist for Oracle Database](#)
Use this checklist to assist you to be prepared before starting Oracle Universal Installer.
- [Deployment Checklist for Oracle Database](#)
Use this checklist to decide the deployment method for a single-instance Oracle Database.

Server Hardware Checklist for Oracle Database Installation

Use this checklist to check hardware requirements for Oracle Database.

Table 1-1 Server Hardware Checklist for Oracle Database Installation

Check	Task
Runlevel	3
Server Display Cards	At least 1024 x 768 display resolution, which Oracle Universal Installer requires.
Minimum network connectivity	Server is connected to a network
Minimum RAM	• At least 1 GB RAM for Oracle Database installations. 2 GB RAM recommended. • At least 8 GB RAM for Oracle Grid Infrastructure installations.

Operating System Checklist for Oracle Database on Oracle Solaris

Use this checklist to check minimum operating system requirements for Oracle Database.

Table 1-2 Operating System General Checklist for Oracle Database on Oracle Solaris

Item	Task
Operating system general requirements	Secure Shell is configured at installation for Oracle Solaris. The following Oracle Solaris on SPARC (64-Bit) kernels are supported: Oracle Solaris 11.4 (Oracle Solaris 11.4.2.0.1.3.0) or later SRUs and updates Oracle Solaris 11.3 SRU 31 (Oracle Solaris 11.3.31.6.0) or later SRUs and updates The following Oracle Solaris on x86-64 (64-Bit) kernels are supported: Solaris 11.4 (Oracle Solaris 11.4.2.0.1.3.0) or later SRUs and updates Oracle Solaris 11.3 SRU 31 (Oracle Solaris 11.3.31.6.0) or later SRUs and updates Review the system requirements section for a list of minimum package requirements.

Related Topics

- [Configuring Oracle Solaris Operating System for Oracle Database](#)
Complete operating system configuration requirements and checks for Oracle Solaris operating systems before you start installation.

Server Configuration Checklist for Oracle Database Installation

Use this checklist to check minimum server configuration requirements for Oracle Database installations.

Table 1-3 Server Configuration Checklist for Oracle Database

Check	Task
Disk space allocated to the /tmp directory	At least 1 GB of space in the temporary disk space (/tmp) directory
Swap space allocation relative to RAM (Oracle Database)	Between 1 GB and 2 GB: 1.5 times the size of the RAM Between 2 GB and 16 GB: Equal to the size of the RAM More than 16 GB: 16 GB Note: Configure swap for your expected system loads. This installation guide provides minimum values for installation only. Refer to your Oracle Solaris documentation for additional memory tuning guidance.

Table 1-3 (Cont.) Server Configuration Checklist for Oracle Database

Check	Task
Swap space allocation relative to RAM (Oracle Restart)	Between 8 GB and 16 GB: Equal to the size of the RAM More than 16 GB: 16 GB Note: Configure swap for your expected system loads. This installation guide provides minimum values for installation only. Refer to your Oracle Solaris documentation for additional memory tuning guidance.
Oracle Inventory (oraInventory) and OINSTALL Group Requirements	- For upgrades, Oracle Universal Installer (OUI) detects an existing oraInventory directory from the <code>/var/opt/oracle/oraInst.loc</code> file, and uses the existing oraInventory. - For new installs, if you have not configured an oraInventory directory, then the installer creates an Oracle inventory that is one directory level up from the Oracle base for the Oracle Grid Infrastructure install, and designates the installation owner's primary group as the Oracle Inventory group. The Oracle Inventory directory is the central inventory of Oracle software installed on your system. Users who have the Oracle Inventory group as their primary group are granted the OINSTALL privilege to write to the central inventory. The OINSTALL group must be the primary group of all Oracle software installation owners on the server. It should be writable by any Oracle installation owner.
Groups and users	Oracle recommends that you create groups and user accounts required for your security plans before starting installation. Installation owners have resource limits settings and other requirements. Group and user names must use only ASCII characters.
Mount point paths for the software binaries	Oracle recommends that you create an Optimal Flexible Architecture configuration as described in the appendix "Optimal Flexible Architecture" in <i>Oracle Database Installation Guide</i> for your platform.
Ensure that the Oracle home (the Oracle home path you select for Oracle Database) uses only ASCII characters	The ASCII character restriction includes installation owner user names, which are used as a default for some home paths, as well as other directory names you may select for paths.
Unset Oracle software environment variables	If you have an existing Oracle software installation, and you are using the same user to install this installation, then unset the following environment variables: <code>\$ORACLE_HOME</code>; <code>\$ORA_NLS10</code>; <code>\$TNS_ADMIN</code>. If you have set <code>\$ORA_CRS_HOME</code> as an environment variable, then unset it before starting an installation or upgrade. Do not use <code>\$ORA_CRS_HOME</code> as a user environment variable, except as directed by Oracle Support.
Set locale (if needed)	Specify the language and the territory, or locale, in which you want to use Oracle components. A locale is a linguistic and cultural environment in which a system or program is running. NLS (National Language Support) parameters determine the locale-specific behavior on both servers and clients. The locale setting of a component determines the language of the user interface of the component, and the globalization behavior, such as date and number formatting.

Related Topics

- Oracle Database Globalization Support Guide*

Oracle User Environment Configuration Checklist for Oracle Database Installation

Use this checklist to plan operating system users, groups, and environments for Oracle Database management.

Table 1-4 User Environment Configuration for Oracle Database

Check	Task
Review Oracle Inventory (oraInventory) and OINSTALL Group Requirements	The physical group you designate as the Oracle Inventory directory is the central inventory of Oracle software installed on your system. It should be the primary group for all Oracle software installation owners. Users who have the Oracle Inventory group as their primary group are granted the OINSTALL privilege to read and write to the central inventory. - If you have an existing installation, then OUI detects the existing oraInventory directory from the <code>/var/opt/oracle/oraInst.loc</code> file, and uses this location. - If you are installing Oracle software for the first time, then you can specify the Oracle inventory directory and the Oracle base directory during the Oracle software installation, and Oracle Universal Installer will set up the software directories for you. Ensure that the directory paths that you specify are in compliance with the Oracle Optimal Flexible Architecture recommendations. Ensure that the group designated as the OINSTALL group is available as the primary group for all planned Oracle software installation owners.
Create operating system groups and users for standard or role-allocated system privileges	Create operating system groups and users depending on your security requirements, as described in this install guide. Set resource limits settings and other requirements for Oracle software installation owners. Group and user names must use only ASCII characters.
Unset Oracle Software Environment Variables	If you have had an existing installation on your system, and you are using the same user account for this installation, then unset the ORACLE_HOME, ORACLE_BASE, ORACLE_SID, TNS_ADMIN environment variables and any other environment variable set for the Oracle installation user that is connected with Oracle software homes.
Configure the Oracle Software Owner Environment	Configure the environment of the oracle or grid user by performing the following tasks: - Set the default file mode creation mask (umask) to 022 in the shell startup file. - Set the DISPLAY environment variable.

Related Topics

- [Configuring Users, Groups and Environments for Oracle Grid Infrastructure and Oracle Database](#)
Before installation, create operating system groups and users, and configure user environments.

Storage Checklist for Oracle Database Installation

Use this checklist to review storage minimum requirements and assist with configuration planning.

Table 1-5 Storage Checklist for Oracle Database

Check	Task
Minimum local disk storage space for Oracle software	For Oracle Solaris on SPARC (64-Bit): At least 6.8 GB for an Oracle Grid Infrastructure for a standalone server installation At least 8.1 GB for Oracle Database Enterprise Edition At least 7.9 GB for Oracle Database Standard Edition 2
 Note Oracle recommends that you allocate approximately 100 GB to allow additional space for applying any future patches on top of the existing Oracle home. For specific patch-related disk space requirements, please refer to your patch documentation.	
Select Database File Storage Option	Ensure that you have one of the following storage options available: - File system mounted on the server. Oracle recommends that the file system you select is separate from the file system used by the operating system or the Oracle software. Options include the following: - A file system on a logical volume manager (LVM) volume or a RAID device - A network file system (NFS) mounted on a certified network-attached storage (NAS) device - Oracle Solaris ZFS file system. - Oracle Automatic Storage Management (Oracle ASM). Oracle ASM is installed as part of an Oracle Grid Infrastructure installation. If you plan to use Oracle ASM for storage, then you should install Oracle Grid Infrastructure before you install and create the database.
Determine your recovery plan	If you want to enable recovery during installation, then be prepared to select one of the following options: File system: Configure a fast recovery area on a file system during installation Oracle Automatic Storage Management Review the storage configuration sections of this document for more information about configuring recovery.

Installer Planning Checklist for Oracle Database

Use this checklist to assist you to be prepared before starting Oracle Universal Installer.

Table 1-6 Oracle Universal Installer Planning Checklist for Oracle Database Installation

Check	Task
Read the Release Notes	Review release notes for your platform, which are available for your release at the following URL: http://docs.oracle.com/en/database/database.html
Review Oracle Support Certification Matrix	New platforms and operating system software versions may be certified after this guide is published, review the certification matrix on the My Oracle Support website for the most up-to-date list of certified hardware platforms and operating system versions: https://support.oracle.com/ You must register online before using My Oracle Support. After logging in, from the menu options, select the Certifications tab. On the Certifications page, use the Certification Search options to search by Product, Release, and Platform. You can also search using the Certification Quick Link options such as Product Delivery, and Lifetime Support.
Review the Licensing Information	You are permitted to use only those components in the Oracle Database media pack for which you have purchased licenses. For more information about licenses, refer to the following URL: <i>Oracle Database Licensing Information</i>
Run OUI with CVU and use fixup scripts	Oracle Universal Installer is fully integrated with Cluster Verification Utility (CVU), automating many CVU prerequisite checks. Oracle Universal Installer runs all prerequisite checks and creates fixup scripts when you run the installer. You can run OUI up to the Summary screen without starting the installation. You can also run CVU commands manually to check system readiness. For more information, see: <i>Oracle Clusterware Administration and Deployment Guide</i>
Update and run Oracle ORAchk for runtime and upgrade checks, or runtime health checks	Oracle recommends that you update to the latest version of Oracle ORAchk by installing the latest version of Autonomous Health Framework (AHF). The Oracle ORAchk utility provides system checks that can help to prevent issues before and after installation. These checks include kernel requirements, operating system resource allocations, and other system requirements. Use the Oracle ORAchk Upgrade Readiness Assessment to obtain an automated upgrade-specific system health check for upgrades to 19c. For example: - Before you perform a fresh database installation: <code>./orachk -profile preinstall</code> - To upgrade your existing database to a higher version or release: <code>./orachk -pregupgrade -targetversion 19.0.0.0.0</code> The Oracle ORAchk Upgrade Readiness Assessment automates many of the manual pre- and post-upgrade checks described in Oracle upgrade documentation. Check My Oracle Support Note 2550798.1 for more information about Oracle ORAchk support. https://support.oracle.com/epmos/faces/DocContentDisplay?id=2550798.1&parent=DOCUMENTATION&sourceId=USERGUIDE

Table 1-6 (Cont.) Oracle Universal Installer Planning Checklist for Oracle Database Installation

Check	Task
Verify if Oracle Grid Infrastructure is installed	If you want to use Oracle ASM or Oracle Restart, then install Oracle Grid Infrastructure for a standalone server before you install and create the database. Otherwise, to use Oracle ASM, you must complete an Oracle Grid Infrastructure installation, and then manually register the database with Oracle Restart. For Oracle Real Application Clusters (Oracle RAC) installations, ensure that you have installed and configured Oracle Grid Infrastructure for a cluster.
Check running Oracle processes, and shut down if necessary	- On a standalone database not using Oracle ASM: You do not need to shut down the database while you install Oracle Grid Infrastructure. - On a standalone database using Oracle ASM: The Oracle ASM instances are restarted during installation. - On an Oracle RAC Database node: This installation requires an upgrade of Oracle Clusterware, as Oracle Clusterware is required to run Oracle RAC. As part of the upgrade, you must shut down the database one node at a time as the rolling upgrade proceeds from node to node.
Ensure <code>cron</code> jobs do not run during installation	If the installer is running when daily <code>cron</code> jobs start, then you may encounter unexplained installation problems if your <code>cron</code> job is performing cleanup, and temporary files are deleted before the installation is finished. Oracle recommends that you complete installation before daily <code>cron</code> jobs are run, or disable daily <code>cron</code> jobs that perform cleanup until after the installation is completed.
Obtain your My Oracle Support account information.	During installation, you may require a My Oracle Support user name and password to configure updates, download software updates, and other installation tasks. You can register for My Oracle Support at the following URL: https://support.oracle.com/
Decide Oracle Database management tool	By default, Oracle Database is managed by Oracle Enterprise Manager Database Express. If you have an existing Oracle Management Agent, and decide to use Oracle Enterprise Manager Cloud Control to centrally manage your database, then obtain the following information to enter during the database installation: - OMS host - OMS port - EM admin username - EM admin password - Specify password of ASMSNMP user For more information, see: <i>Oracle Database 2 Day DBA</i> <i>Oracle Enterprise Manager Cloud Control Administrator's Guide</i>

Table 1-6 (Cont.) Oracle Universal Installer Planning Checklist for Oracle Database Installation

Check	Task
Review memory allocation and Automatic Memory Management feature	You can enable automatic memory management either during, or after Oracle Database installation. If you enable automatic memory management after installation, then you must shut down and restart the database. If the total physical memory of your database instance is greater than 4 GB, then you cannot select the Oracle Automatic Memory Management option during database installation and creation. Instead, use automatic shared memory management. Automatic shared memory management automatically distributes the available memory among the various components as required, allowing the system to maximize the use of all available SGA memory. For more information, see: <i>Oracle Database Administrator's Guide</i>
Determine superuser (root) privilege delegation option for installation	During a database or grid infrastructure installation, you are asked to run configuration scripts as the root user. You can either run these scripts manually as root when prompted, or you can provide configuration information and passwords using a root privilege delegation option. To run root scripts automatically, select Automatically run configuration scripts during installation. • Use root user credentials Provide the superuser password for cluster member node servers. • Use Sudo Sudo is a UNIX and Linux utility that allows members of the sudoers list privileges to run individual commands as root. Provide the username and password of an operating system user that is a member of sudoers, and is authorized to run Sudo on each cluster member node. To enable Sudo, have a system administrator with the appropriate privileges configure a user that is a member of the sudoers list, and provide the username and password when prompted during installation.
Oracle Database Client and Oracle Database interoperability	For information about interoperability between Oracle Database Client and Oracle Database releases, see My Oracle Support Note 207303.1: https://support.oracle.com/rs?type=doc&id=207303.1

Related Topics

- [Installing Oracle Database](#)
Oracle Database and Oracle Grid Infrastructure installation software is available as image-based zip files and can be installed using several options.
- [Installing and Configuring Oracle Grid Infrastructure for a Standalone Server](#)
Oracle Grid Infrastructure for a standalone server includes Oracle Restart and Oracle Automatic Storage Management.

Deployment Checklist for Oracle Database

Use this checklist to decide the deployment method for a single-instance Oracle Database.

Table 1-7 Deployment Checklist for Oracle Database (single-instance)

Item	Task
To deploy single-instance Oracle Database software	Use one of the following deployment methods: • Install Oracle Database software using Oracle Universal Installer (OUI). • Provision Oracle Database software using Oracle Fleet Patching and Provisioning. • Clone Oracle Database.
To deploy single-instance Oracle Database software and create databases	Use one of the following deployment methods: • Install Oracle Database software using Oracle Universal Installer (OUI). • Provision Oracle Database software using Oracle Fleet Patching and Provisioning. • Clone Oracle Database.
To create single-instance Oracle Database in an already-installed Oracle home	• Use Oracle Database Configuration Assistant (Oracle DBCA). • Use Oracle Fleet Patching and Provisioning

Related Topics

- *Oracle Database Administrator's Guide*
- *Oracle Fleet Patching and Provisioning Administrator's Guide*

2

Checking and Configuring the Server for Oracle Database

Verify that servers where you install Oracle Database meet the minimum requirements for installation.

This section provides minimum server requirements to complete installation of Oracle Database. It does not provide system resource guidelines, or other tuning guidelines for particular workloads.

- [Logging In to a Remote System Using X Window System](#)
Use this procedure to run Oracle Universal Installer (OUI) by logging on to a remote system where the runtime setting prohibits logging in directly to a graphical user interface (GUI).
- [Checking Server Hardware and Memory Configuration](#)
Use this procedure to gather information about your server configuration.

Logging In to a Remote System Using X Window System

Use this procedure to run Oracle Universal Installer (OUI) by logging on to a remote system where the runtime setting prohibits logging in directly to a graphical user interface (GUI).

OUI is a graphical user interface (GUI) application. On servers where the runtime settings prevent GUI applications from running, you can redirect the GUI display to a client system connecting to the server.

Note

If you log in as another user (for example, `oracle` or `grid`), then repeat this procedure for that user as well.

1. Start an X Window System session. If you are using an X Window System terminal emulator from a PC or similar system, then you may need to configure security settings to permit remote hosts to display X applications on your local system.
2. Enter a command using the following syntax to enable remote hosts to display X applications on the local X server:

```
# xhost + RemoteHost
```

RemoteHost is the fully qualified remote host name. For example:

```
# xhost + somehost.example.com  
somehost.example.com being added to the access control list
```

3. If you are not installing the software on the local system, then use the ssh command to connect to the system where you want to install the software:

```
# ssh -Y RemoteHost
```

RemoteHost is the fully qualified remote host name. The -Y flag ("yes") enables remote clients to have full access to the original server display. For example:

```
# ssh -Y somehost.example.com
```

4. If you are not logged in as the root user, and you are performing configuration steps that require root user privileges, then switch the user to root.

① Note

For more information about remote login using X Window System, refer to your X server documentation, or contact your X server vendor or system administrator. Depending on the X server software that you are using, you may have to complete the tasks in a different order.

Checking Server Hardware and Memory Configuration

Use this procedure to gather information about your server configuration.

1. Use the following command to report the number of memory pages and swap-file disk blocks that are currently unused:

```
# sar -r n i
```

For example:

```
# sar -r 2 10
```

If the size of the physical RAM installed in the system is less than the required size, then you must install more memory before continuing.

2. Determine the swap space usage and size of the configured swap space:

```
# /usr/sbin/swap -s
```

If necessary, see your operating system documentation for information about how to configure additional swap space.

3. Determine the amount of space available in the /tmp directory:

```
# df -kh /tmp
```

If the free space available in the /tmp directory is less than what is required, then complete one of the following steps:

- Delete unnecessary files from the /tmp directory to meet the disk space requirement.

- When you set the Oracle user's environment, also set the TMP and TMPDIR environment variables to the directory you want to use instead of /tmp.
4. Determine the amount of free disk swap space on the system:

```
# df -kh
```

5. Determine the RAM size:

```
# /usr/sbin/prtconf | grep "Memory size"
```

6. Determine if the system architecture can run the software:

```
# /bin/isainfo -kv
```

This command displays the processor type. For example:

```
64-bit sparcv9 kernel modules
```

```
64-bit amd64 kernel modules
```

If you do not see the expected output, then you cannot install the software on this system.

Related Topics

- [Server Hardware Checklist for Oracle Database Installation](#)
Use this checklist to check hardware requirements for Oracle Database.

3

Automatically Configuring Oracle Solaris with Oracle Database Prerequisites Packages

Use the Oracle Database prerequisites group package to simplify Oracle Solaris operating system configuration in preparation for Oracle software installations.

Oracle recommends that you install the Oracle Database prerequisites group package `oracle-database-preinstall-19c` in preparation for Oracle Database and Oracle Grid Infrastructure installations.

- [About the Oracle Database Prerequisites Packages for Oracle Solaris](#)
Installing the Oracle Database prerequisites group package simplifies operating system configuration and ensures that you have the required packages.
- [Checking the Oracle Database Prerequisites Packages Installation](#)
Use this procedure to gather information about the Oracle Database prerequisites group package configuration.
- [Installing the Oracle Database Prerequisites Packages for Oracle Solaris](#)
Use this procedure to install the Oracle Database prerequisites group package for your Oracle software.

About the Oracle Database Prerequisites Packages for Oracle Solaris

Installing the Oracle Database prerequisites group package simplifies operating system configuration and ensures that you have the required packages.

Use the Oracle Database prerequisites group package `group/prerequisite/oracle/oracle-database-preinstall-19c` to ensure that all the necessary packages required for an Oracle Database and Oracle Grid Infrastructure installation are present on the system.

The `oracle-database-preinstall-19c` package also creates the `oracle` user with a home directory of `/export/home/oracle`, and creates the `oraInventory` (`oinstall`) and `OSDBA` (`dba`) groups for that user.

The `oracle-database-preinstall-19c` package consists of two packages—one that installs extra packages needed on the operating system and one that creates users and groups. The users and groups are created by the package `oracle-database-os-configuration`. If you have the `oracle` user and other user groups already defined, then you can choose not to change them.

You can install `oracle-database-preinstall-19c` even if you installed Oracle Solaris using any of the server package groups, such as `solaris-minimal-server`, `solaris-small-server`, `solaris-large-server`, or `solaris-desktop`. Oracle recommends that you install the `solaris-minimal-server` group package and then install `oracle-database-preinstall-19c`.

Configuring a server using Oracle Solaris and the Oracle Database prerequisites group package consists of the following steps:

1. Install the recommended Oracle Solaris version for Oracle Database.

2. Install the Oracle Database prerequisites group package `oracle-database-preinstall-19c`.
3. If you do not want the Oracle Database prerequisites group package to create the default users and groups for you, then manually create the role-allocated groups and users.
4. Complete network interface configuration for each cluster node candidate.
5. Complete system configuration for shared storage access as required for each standard or core node cluster candidate.

After these steps are complete, you can proceed to install Oracle Database, Oracle Grid Infrastructure, or Oracle RAC.

Related Topics

- [Oracle Solaris 11.3 Package Group Lists](#)

Checking the Oracle Database Prerequisites Packages Installation

Use this procedure to gather information about the Oracle Database prerequisites group package configuration.

1. To check if `oracle-database-preinstall-19c` is already installed:

```
$ pkg list oracle-database-preinstall-19c
```

2. To check for the latest version of `oracle-database-preinstall-19c`:

```
$ pkg list -n oracle-database-preinstall-19c
```

3. Before you install `oracle-database-preinstall-19c`:

- a. Use the `-n` option to check for errors:

```
$ pkg install -n oracle-database-preinstall-19c
```

Note

Use the `-n` option to check for installation errors. If `-n` does not display any errors, then omit the `-n` option when you install `oracle-database-preinstall-19c`.

- b. If there are no errors, then log in as root, and install the group package:

```
# pkg install oracle-database-preinstall-19c
```

- c. If you have the `oracle` user and other user groups already defined, and do not want to change them:

```
# pkg avoid oracle-database-os-configuration
```

4. To view what packages will be installed by oracle-database-preinstall-19c:

```
$ pkg contents -ro type,fmri -t depend oracle-database-preinstall-19c
```

A sample output of this command:

```
TYPE FMRI
conditional pkg:/service/oracle-rdbms-stats
group pkg:/group/prerequisite/oracle/oracle-database-os-configuration
group pkg:/system/font/truetype/arphic-uming
group pkg:/system/font/truetype/hanyang-ko-core
group pkg:/system/header
group pkg:/system/kernel/oracka
group pkg:/system/picl
group pkg:/x11/diagnostic/x11-info-clients
group pkg:/x11/library/libxi
group pkg:/x11/library/libxtst
group pkg:/x11/session/xauth
require pkg:/compress/unzip
require pkg:/developer/assembler
require pkg:/developer/build/make
require pkg:/system/dtrace
require pkg:/system/library/openmp
```

Related Topics

- [Adding and Updating Software in Oracle Solaris](#)

Installing the Oracle Database Prerequisites Packages for Oracle Solaris

Use this procedure to install the Oracle Database prerequisites group package for your Oracle software.

The group/prerequisite/oracle/oracle-database-preinstall-19c group package installs all the packages required for an Oracle Database and Oracle Grid Infrastructure installation. It also creates the oracle user and the oinstall and dba groups for that user.



Caution

If you have the oracle user and other user groups already defined, and do not want to change them, then ensure that you run the following command before you install the Oracle Database prerequisites group package:

```
# pkg avoid oracle-database-os-configuration
```

To install the oracle-database-preinstall-19c group packages, log in as root, and run the following command on Oracle Solaris 11.3 and later systems:

```
# pkg install oracle-database-preinstall-19c
```

Note the following guidelines about the preceding command:

- For more information about the history of the preceding command, use the `pkg history` command. See `pkg (1)`.
- You do not have to specify the entire package name, only the trailing portion of the name that is unique. See `pkg(5)`.
- Oracle recommends that you install the `solaris-minimal-server` group package and then install `oracle-database-preinstall-19c`.

Related Topics

- [Oracle Solaris Documentation](#)

4

Configuring Oracle Solaris Operating System for Oracle Database

Complete operating system configuration requirements and checks for Oracle Solaris operating systems before you start installation.

- [Guidelines for Oracle Solaris Operating System Installation](#)
Decide how you want to install Oracle Solaris.
- [Reviewing Operating System and Software Upgrade Best Practices](#)
These topics provide general planning guidelines and platform-specific information about upgrades and migration.
- [Reviewing Operating System Security Common Practices](#)
Secure operating systems are an important basis for general system security.
- [About Installation Fixup Scripts](#)
Oracle Universal Installer detects when the minimum requirements for an installation are not met, and creates shell scripts, called fixup scripts, to finish incomplete system configuration steps.
- [About Operating System Requirements](#)
Depending on the products that you intend to install, verify that you have the required operating system kernel and packages installed.
- [Operating System Requirements for Oracle Solaris on SPARC \(64-bit\)](#)
The kernels and packages listed in this section are supported for this release on SPARC 64-bit systems for Oracle Database and Oracle Grid Infrastructure.
- [Operating System Requirements for Oracle Solaris on x86-64 \(64-Bit\)](#)
The kernels and packages listed in this section are supported for this release on x86-64 (64-bit) systems for Oracle Database and Oracle Grid Infrastructure.
- [Additional Drivers and Software Packages for Oracle Solaris](#)
Information about optional drivers and software packages.
- [Checking the Software Requirements for Oracle Solaris](#)
Check the software requirements of your Oracle Solaris operating system to see if they meet minimum requirements for installation.
- [Confirming Host Name Resolution](#)
Check to ensure that the host name for your server is resolvable.
- [Using Automatic SSH Configuration During Installation](#)
To install Oracle software, configure secure shell (SSH) connectivity between all cluster member nodes.

Guidelines for Oracle Solaris Operating System Installation

Decide how you want to install Oracle Solaris.

Refer to your Oracle Solaris documentation to obtain information about installing Oracle Solaris on your servers. You may want to use Oracle Solaris 11 installation services, such as Oracle

Solaris Automated Installer (AI), to create and manage services to install the Oracle Solaris 11 operating system over the network.

Related Topics

- [Oracle Solaris Documentation](#)
- *Installing Oracle Solaris 11 Guide*
- [Resources for Running Oracle Database on Oracle Solaris](#)

Reviewing Operating System and Software Upgrade Best Practices

These topics provide general planning guidelines and platform-specific information about upgrades and migration.

- [General Upgrade Best Practices](#)
Be aware of these guidelines as a best practice before you perform an upgrade.
- [New Server Operating System Upgrade Option](#)
You can upgrade your operating system by installing a new operating system on a server, and then migrating your database either manually, or by using Export/Import method.
- [Oracle ASM Upgrade Notifications](#)
Understand Oracle ASM upgrade options and restrictions.

General Upgrade Best Practices

Be aware of these guidelines as a best practice before you perform an upgrade.

If you have an existing Oracle Database installation, then do the following:

- Record the version numbers, patches, and other configuration information
- Review upgrade procedures for your existing installation
- Review Oracle Database upgrade documentation before proceeding with installation, to decide how you want to proceed

 Caution

Always create a backup of existing databases before starting any configuration change.

Refer to *Oracle Database Upgrade Guide* for more information about required software updates, pre-upgrade tasks, post-upgrade tasks, compatibility, and interoperability between different releases.

Related Topics

- *Oracle Database Upgrade Guide*

New Server Operating System Upgrade Option

You can upgrade your operating system by installing a new operating system on a server, and then migrating your database either manually, or by using Export/Import method.

Note

Confirm that the server operating system is supported, and that kernel and package requirements for the operating system meet or exceed the minimum requirements for the Oracle Database release to which you want to migrate.

Manual, Command-Line Copy for Migrating Data and Upgrading Oracle Database

You can copy files to the new server and upgrade it manually. If you use this procedure, then you cannot use Oracle Database Upgrade Assistant. However, you can revert to your existing database if you encounter upgrade issues.

1. Copy the database files from the computer running the previous operating system to the one running the new operating system.
2. Re-create the control files on the computer running the new operating system.
3. Manually upgrade the database using command-line scripts and utilities.

See Also

Oracle Database Upgrade Guide to review the procedure for upgrading the database manually, and to evaluate the risks and benefits of this option

Export/Import Method for Migrating Data and Upgrading Oracle Database

You can install the operating system on the new server, install the new Oracle Database release on the new server, and then use Oracle Data Pump Export and Import utilities to migrate a copy of data from your current database to a new database in the new release. Data Pump Export and Import are recommended for higher performance and to ensure support for new data types.

See Also

Oracle Database Upgrade Guide to review the Export/Import method for migrating data and upgrading Oracle Database

Oracle ASM Upgrade Notifications

Understand Oracle ASM upgrade options and restrictions.

- You can upgrade Oracle Automatic Storage Management (Oracle ASM) 11g release 2 (11.2) and later without shutting down an Oracle RAC database by performing a rolling upgrade either of individual nodes, or of a set of nodes in the cluster. However, if you have a standalone database on a cluster that uses Oracle ASM, then you must shut down the standalone database before upgrading.
- The location of the Oracle ASM home changed in Oracle Grid Infrastructure 11g release 2 (11.2) so that Oracle ASM is installed with Oracle Clusterware in the Oracle Grid Infrastructure home (Grid home).

- Two nodes of different releases cannot run in the cluster. When upgrading from Oracle Grid Infrastructure 11g release 2 (11.2) or Oracle Grid Infrastructure 12c release 1 (12.1) to a later release, if there is an outage during the rolling upgrade, then when you restart the upgrade, ensure that you start the earlier release of Oracle Grid Infrastructure and bring the Oracle ASM cluster back in the rolling migration mode.

Reviewing Operating System Security Common Practices

Secure operating systems are an important basis for general system security.

Ensure that your operating system deployment is in compliance with common security practices as described in your operating system vendor security guide.

About Installation Fixup Scripts

Oracle Universal Installer detects when the minimum requirements for an installation are not met, and creates shell scripts, called fixup scripts, to finish incomplete system configuration steps.

If Oracle Universal Installer detects an incomplete task, then it generates fixup scripts (runfixup.sh). You can run the fixup script and click **Fix and Check Again**. The fixup script modifies both persistent parameter settings and parameters in memory, so you do not have to restart the system.

The Fixup script does the following tasks:

- Sets kernel parameters, if necessary, to values required for successful installation, including:
 - Shared memory parameters.
 - Open file descriptor and UDP send/receive parameters.
- Creates and sets permissions on the Oracle Inventory (central inventory) directory.
- Creates or reconfigures primary and secondary group memberships for the installation owner, if necessary, for the Oracle Inventory directory and the operating system privileges groups.
- Sets shell limits, if necessary, to required values.

Note

Using fixup scripts does not ensure that all the prerequisites for installing Oracle Database are met. You must still verify that all the preinstallation requirements are met to ensure a successful installation.

Oracle Universal Installer is fully integrated with Cluster Verification Utility (CVU) automating many prerequisite checks for your Oracle Grid Infrastructure or Oracle Real Application Clusters (Oracle RAC) installation. You can also manually perform various CVU verifications by running the `cluvfy` command.

Related Topics

- [Completing Preinstallation Tasks Manually](#)
You can complete the preinstallation configuration tasks manually.

- *Oracle Clusterware Administration and Deployment Guide*

About Operating System Requirements

Depending on the products that you intend to install, verify that you have the required operating system kernel and packages installed.

Requirements listed in this document are current as of the date listed on the title page.

Oracle Universal Installer performs checks on your system to verify that it meets the listed operating system package requirements. To ensure that these checks complete successfully, verify the requirements before you start OUI.

Note

Oracle does not support running different operating system versions on cluster members, unless an operating system is being upgraded. You cannot run different operating system version binaries on members of the same cluster, even if each operating system is supported.

Operating System Requirements for Oracle Solaris on SPARC (64-bit)

The kernels and packages listed in this section are supported for this release on SPARC 64-bit systems for Oracle Database and Oracle Grid Infrastructure.

The platform-specific hardware and software requirements included in this guide were current when this guide was published. However, because new platforms and operating system software versions might be certified after this guide is published, review the certification matrix on the My Oracle Support website for the most up-to-date list of certified hardware platforms and operating system versions:

<https://support.oracle.com/>

Identify the requirements for your Oracle Solaris on SPARC (64-bit) system, and ensure that you have a supported kernel and required packages installed before starting installation.

- [Supported Oracle Solaris 11 Releases for SPARC \(64-bit\)](#)
Check the supported Oracle Solaris 11 distributions and other operating system requirements.

Related Topics

- [Installation Requirements for Programming Environments for Oracle Solaris](#)
Ensure that your system meets the requirements for the programming environment you want to configure:

Supported Oracle Solaris 11 Releases for SPARC (64-bit)

Check the supported Oracle Solaris 11 distributions and other operating system requirements.

Table 4-1 Oracle Solaris 11 Releases for SPARC (64-bit) Minimum Operating System Requirements

Item	Requirements
SSH Requirement	Secure Shell is configured at installation for Oracle Solaris.
Oracle Solaris 11 operating system	Oracle Solaris 11.4 (Oracle Solaris 11.4.2.0.1.3.0) or later SRUs and updates Oracle Solaris 11.3 SRU 31 (Oracle Solaris 11.3.31.6.0) or later SRUs and updates
Packages for Oracle Solaris 11	The following packages must be installed: pkg://solaris/system/library/openmp pkg://solaris/compress/unzip pkg://solaris/developer/assembler pkg://solaris/developer/build/make pkg://solaris/system/dtrace pkg://solaris/system/header pkg://solaris/system/library pkg://solaris/system/linker pkg://solaris/system/xopen/xcu4 (If not already installed as part of standard Oracle Solaris 11 installation) pkg://solaris/x11/diagnostic/x11-info-clients pkg://solaris/system/kernel/oracka Note: Starting with Oracle Solaris 11.2, if you have performed a standard Oracle Solaris 11 installation, and installed the Oracle Database prerequisites group package <code>oracle-database-preinstall-19c</code>, then you do not have to install these packages, as <code>oracle-database-preinstall-19c</code> installs them for you.
Oracle Solaris Cluster for Oracle Solaris 11	This information applies only if you are using Oracle Solaris Cluster. Please refer to the Oracle Solaris Cluster Compatibility Guide for more information: Oracle Solaris Cluster 4 Compatibility Guide

Related Topics

- [Automatically Configuring Oracle Solaris with Oracle Database Prerequisites Packages](#)
Use the Oracle Database prerequisites group package to simplify Oracle Solaris operating system configuration in preparation for Oracle software installations.

Operating System Requirements for Oracle Solaris on x86-64 (64-Bit)

The kernels and packages listed in this section are supported for this release on x86-64 (64-bit) systems for Oracle Database and Oracle Grid Infrastructure.

The platform-specific hardware and software requirements included in this guide were current when this guide was published. However, because new platforms and operating system software versions might be certified after this guide is published, review the certification matrix on the My Oracle Support website for the most up-to-date list of certified hardware platforms and operating system versions:

<https://support.oracle.com/>

Identify the requirements for your Oracle Solaris on x86-64 (64-bit) system, and ensure that you have a supported kernel and required packages installed before starting installation.

- [Supported Oracle Solaris 11 Releases for x86-64 \(64-Bit\)](#)
Check the supported Oracle Solaris 11 distributions and other operating system requirements.

Related Topics

- [Installation Requirements for Programming Environments for Oracle Solaris](#)
Ensure that your system meets the requirements for the programming environment you want to configure:

Supported Oracle Solaris 11 Releases for x86-64 (64-Bit)

Check the supported Oracle Solaris 11 distributions and other operating system requirements.

Table 4-2 Oracle Solaris 11 Releases for x86-64 (64-Bit) Minimum Operating System Requirements

Item	Requirements
SSH Requirement	Secure Shell is configured at installation for Oracle Solaris.
Oracle Solaris 11 operating system	Oracle Solaris 11.4 (Oracle Solaris 11.4.2.0.1.3.0) or later SRUs and updates Oracle Solaris 11.3 SRU 31 (Oracle Solaris 11.3.31.6.0) or later SRUs and updates
Packages for Oracle Solaris 11	The following packages must be installed: pkg://solaris/system/library/openmp pkg://solaris/compress/unzip pkg://solaris/developer/assembler pkg://solaris/developer/build/make pkg://solaris/system/dtrace pkg://solaris/system/header pkg://solaris/system/library pkg://solaris/system/linker pkg://solaris/system/xopen/xcu4 (If not already installed as part of standard Oracle Solaris 11 installation) pkg://solaris/x11/diagnostic/x11-info-clients pkg://solaris/system/kernel/oracka Note: Starting with Oracle Solaris 11.2, if you have performed a standard Oracle Solaris 11 installation, and installed the Oracle Database prerequisites group package <code>oracle-database-preinstall-19c</code>, then you do not have to install these packages, as <code>oracle-database-preinstall-19c</code> installs them for you.
Oracle Solaris Cluster for Oracle Solaris 11	This information applies only if you are using Oracle Solaris Cluster. Please refer to the Oracle Solaris Cluster Compatibility Guide for more information: Oracle Solaris Cluster 4 Compatibility Guide

Related Topics

- [Automatically Configuring Oracle Solaris with Oracle Database Prerequisites Packages](#)
Use the Oracle Database prerequisites group package to simplify Oracle Solaris operating system configuration in preparation for Oracle software installations.

Additional Drivers and Software Packages for Oracle Solaris

Information about optional drivers and software packages.

You are not required to install additional drivers and packages, but you may choose to install or configure these drivers and packages.

- [Installing Oracle Messaging Gateway](#)
Oracle Messaging Gateway is installed with Enterprise Edition of Oracle Database. However, you may require a CSD or Fix Packs.
- [Installation Requirements for ODBC and LDAP](#)
Review these topics to install Open Database Connectivity (ODBC) and Lightweight Directory Access Protocol (LDAP).
- [Installation Requirements for Programming Environments for Oracle Solaris](#)
Ensure that your system meets the requirements for the programming environment you want to configure:
- [Installation Requirements for Web Browsers](#)
Web browsers are required only if you intend to use Oracle Enterprise Manager Database Express and Oracle Enterprise Manager Cloud Control. Web browsers must support JavaScript, and the HTML 4.0 and CSS 1.0 standards.

Installing Oracle Messaging Gateway

Oracle Messaging Gateway is installed with Enterprise Edition of Oracle Database. However, you may require a CSD or Fix Packs.

If you require a CSD or Fix Packs for IBM WebSphere MQ, then see the following website for more information:

<https://www.ibm.com/support/>

Note

- Oracle Messaging Gateway does not support the integration of Advanced Queuing with TIBCO Rendezvous on IBM: Linux on System z.
- Oracle Messaging Gateway is not supported on Linux for Arm (aarch64).

Related Topics

- *Oracle Database Advanced Queuing User's Guide*

Installation Requirements for ODBC and LDAP

Review these topics to install Open Database Connectivity (ODBC) and Lightweight Directory Access Protocol (LDAP).

- [About ODBC Drivers and Oracle Database](#)
Open Database Connectivity (ODBC) is a set of database access APIs that connect to the database, prepare, and then run SQL statements on the database.

- [Installing ODBC Drivers for Oracle Solaris](#)
If you intend to use ODBC, then install the most recent ODBC Driver Manager for Oracle Solaris.
- [About LDAP and Oracle Plug-ins](#)
Lightweight Directory Access Protocol (LDAP) is an application protocol for accessing and maintaining distributed directory information services over IP networks.
- [Installing the LDAP Package](#)
LDAP is included in a default operating system installation.

About ODBC Drivers and Oracle Database

Open Database Connectivity (ODBC) is a set of database access APIs that connect to the database, prepare, and then run SQL statements on the database.

An application that uses an ODBC driver can access non-uniform data sources, such as spreadsheets and comma-delimited files.

Installing ODBC Drivers for Oracle Solaris

If you intend to use ODBC, then install the most recent ODBC Driver Manager for Oracle Solaris.

Download and install the ODBC Driver Manager from the following website:

<http://www.unixodbc.org>

Review the minimum supported ODBC driver releases, and install ODBC drivers of the following or later releases for all Oracle Solaris distributions:

unixODBC-2.3.4 or later

About LDAP and Oracle Plug-ins

Lightweight Directory Access Protocol (LDAP) is an application protocol for accessing and maintaining distributed directory information services over IP networks.

You require the LDAP package if you want to use features requiring LDAP, including the Oracle Database scripts `odisrvreg` and `oidca` for Oracle Internet Directory, or `schemasync` for third-party LDAP directories.

Installing the LDAP Package

LDAP is included in a default operating system installation.

If you did not perform a default operating system installation, and you intend to use Oracle scripts requiring LDAP, then use a package management system for your distribution to install a supported LDAP package for your distribution, and install any other required packages for that LDAP package.

Installation Requirements for Programming Environments for Oracle Solaris

Ensure that your system meets the requirements for the programming environment you want to configure:

Table 4-3 Requirements for Programming Environments for Oracle Solaris

Programming Environments	Support Requirements
Java Database Connectivity (JDBC) / JDBC Oracle Call Interface (JDBC OCI)	JDK 8 (Java SE Development Kit) with the JNDI extension with Oracle Java Database Connectivity.
Oracle Solaris Studio	Oracle Solaris Studio 12.6 (formerly Sun Studio) Sun C 5.15 2017/05/30 pkg://solarisstudio/developer/ developerstudio-126/ Download Oracle Solaris Studio from the following URL: https://www.oracle.com/technetwork/server-storage/ developerstudio/overview/index.html
Pro*COBOL	Micro Focus Visual COBOL Development Hub 2.3 - Update 2 Micro Focus Visual COBOL v6.0
Pro*FORTRAN	Oracle Solaris Studio 12 (Fortran 95)

Note

Starting with Oracle Database 12c Release 2 (12.2), JDK 8 (32-bit) is not supported on Oracle Solaris. Features that use Java (32-bit) are not available on Oracle Solaris.

Note

Additional patches may be needed depending on applications you deploy.

Installation Requirements for Web Browsers

Web browsers are required only if you intend to use Oracle Enterprise Manager Database Express and Oracle Enterprise Manager Cloud Control. Web browsers must support JavaScript, and the HTML 4.0 and CSS 1.0 standards.

<https://support.oracle.com>

Related Topics

- *Oracle Enterprise Manager Cloud Control Basic Installation Guide*

Checking the Software Requirements for Oracle Solaris

Check the software requirements of your Oracle Solaris operating system to see if they meet minimum requirements for installation.

- [Verifying Operating System Version on Oracle Solaris](#)
To check your software to see if they meet minimum version requirements for installation, perform the following steps:
- [Verifying Operating System Packages on Oracle Solaris](#)
To check if your operating system has the required Oracle Solaris 11 packages for installation, run the following commands:

Verifying Operating System Version on Oracle Solaris

To check your software to see if they meet minimum version requirements for installation, perform the following steps:

1. To determine which version of Oracle Solaris is installed:

```
$ uname -r
```

```
5.11
```

In this example, the version shown is Oracle Solaris 11 (5.11). If necessary, refer to your operating system documentation for information about upgrading the operating system.

2. To determine the release level:

```
$ cat /etc/release
```

```
Oracle Solaris 11.4 SPARC
```

In this example, the release level shown is Oracle Solaris 11.4 SPARC.

3. To determine detailed information about the operating system version such as update level, SRU, and build:

- On Oracle Solaris 11

```
$ pkg list entire
```

```
NAME (PUBLISHER) VERSION INFO
entire (solaris) 0.5.11-0.175.3.1.0.5.0 i--
```

Verifying Operating System Packages on Oracle Solaris

To check if your operating system has the required Oracle Solaris 11 packages for installation, run the following commands:

- To determine if the required packages are installed on Oracle Solaris 11:

```
# /usr/bin/pkg verify [-Hqv] [pkg_pattern ...]
```

 - The -H option omits the headers from the verification output.
 - The -q option prints nothing but return failure if any fatal errors are found.
 - The -v option includes informational messages regarding packages.

If a package that is required for your system architecture is not installed, then download and install it from My Oracle Support:

<https://support.oracle.com>

Note

There may be more recent versions of packages listed installed on the system. If a listed patch is not installed, then determine if a more recent version is installed before installing the version listed. Refer to your operating system documentation for information about installing packages.

Related Topics

- The Adding and Updating Oracle Solaris Software Packages guide
- Oracle Solaris 11 Product Documentation
- [My Oracle Support note 1021281.1](#)

Confirming Host Name Resolution

Check to ensure that the host name for your server is resolvable.

Typically, the computer on which you want to install Oracle Database is connected to a network. Ensure that the computer host name is resolvable, either through a Domain Name System (DNS), a network information service (NIS), or a centrally-maintained TCP/IP host file, such as `/etc/hosts`. Use the `ping` command to ensure that your computer host name is resolvable. For example:

```
ping myhostname
pinging myhostname.example.com [192.0.2.2] with 32 bytes of data:
Reply from 192.0.2.2: bytes=32 time=138ms TTL=56
```

Related Topics

- [Configuring Networks for Oracle Database](#)
If you install Oracle Databases on servers with multiple Oracle homes, multiple aliases, or without a static IP address, then review these network configuration topics.

Using Automatic SSH Configuration During Installation

To install Oracle software, configure secure shell (SSH) connectivity between all cluster member nodes.

Oracle Universal Installer (OUI) uses the `ssh` and `sftp` commands during installation to run remote commands on and copy files to the other cluster nodes. You must configure SSH so that these commands do not prompt for a password.

Note

Oracle configuration assistants use SSH for configuration operations from local to remote nodes. Oracle Enterprise Manager also uses SSH. RSH is no longer supported.

You can configure SSH from the OUI interface during installation for the user account running the installation. The automatic configuration creates passwordless SSH connectivity between all cluster member nodes. Oracle recommends that you use the automatic procedure if possible.

To enable the script to run, you must remove `stty` commands from the profiles of any existing Oracle software installation owners you want to use, and remove other security measures that are triggered during a login, and that generate messages to the terminal. These messages, mail checks, and other displays prevent Oracle software installation owners from using the SSH configuration script that is built into OUI. If they are not disabled, then SSH must be configured manually before an installation can be run.

In rare cases, Oracle Clusterware installation can fail during the "AttachHome" operation when the remote node closes the SSH connection. To avoid this problem, set the timeout wait to unlimited by setting the following parameter in the SSH daemon configuration file `/etc/ssh/sshd_config` on all cluster nodes:

```
LoginGraceTime 0
```

Related Topics

- [Preventing Installation Errors Caused by Terminal Output Commands](#)
During an Oracle Grid Infrastructure installation, OUI uses SSH to run commands and copy files to the other nodes. During the installation, hidden files on the system (for example, `.bashrc` or `.cshrc`) can cause `makefile` and other installation errors if they contain terminal output commands.

5

Configuring Users, Groups and Environments for Oracle Grid Infrastructure and Oracle Database

Before installation, create operating system groups and users, and configure user environments.

- [Required Operating System Groups and Users](#)
Oracle software installations require an installation owner, an Oracle Inventory group, which is the primary group of all Oracle installation owners, and at least one group designated as a system privileges group.
- [Oracle Installations with Standard and Job Role Separation Groups and Users](#)
A job role separation configuration of Oracle Database and Oracle ASM is a configuration with groups and users to provide separate groups for operating system authentication.
- [Creating Operating System Privileges Groups](#)
The following sections describe how to create operating system groups for Oracle Grid Infrastructure and Oracle Database:
- [Creating Operating System Oracle Installation User Accounts](#)
Before starting installation, create Oracle software owner user accounts, and configure their environments.
- [Creating Oracle Database Vault User Accounts](#)
If you intend to use Oracle Database Vault by default, then you must create an Oracle Database Vault user account, and configure that user.
- [Unsetting Oracle Installation Owner Environment Variables](#)
Unset Oracle installation owner environment variables before you start the installation.

Required Operating System Groups and Users

Oracle software installations require an installation owner, an Oracle Inventory group, which is the primary group of all Oracle installation owners, and at least one group designated as a system privileges group.

Review group and user options with your system administrator. If you have system administration privileges, then review the topics in this section and configure operating system groups and users as needed.

- [Determining If an Oracle Inventory and Oracle Inventory Group Exist](#)
Determine if you have an existing Oracle central inventory, and ensure that you use the same Oracle Inventory for all Oracle software installations. Also, ensure that all Oracle software users you intend to use for installation have permissions to write to this directory.
- [Creating the Oracle Inventory Group If an Oracle Inventory Does Not Exist](#)
Create an Oracle Inventory group manually as part of a planned installation, particularly where more than one Oracle software product is installed on servers.

- [About Oracle Installation Owner Accounts](#)
Select or create an Oracle installation owner for your installation, depending on the group and user management plan you want to use for your installations.
- [Identifying an Oracle Software Owner User Account](#)
You must create at least one software owner user account the first time you install Oracle software on the system. Either use an existing Oracle software user account, or create an Oracle software owner user account for your installation.

Determining If an Oracle Inventory and Oracle Inventory Group Exist

Determine if you have an existing Oracle central inventory, and ensure that you use the same Oracle Inventory for all Oracle software installations. Also, ensure that all Oracle software users you intend to use for installation have permissions to write to this directory.

When you install Oracle software on the system for the first time, OUI creates the `oraInst.loc` file. This file identifies the name of the Oracle Inventory group (by default, `oinstall`), and the path of the Oracle central inventory directory. If you have an existing Oracle central inventory, then ensure that you use the same Oracle Inventory for all Oracle software installations, and ensure that all Oracle software users you intend to use for installation have permissions to write to this directory.

```
oraInst.loccentral_inventory_locationgroup
```

```
inventory_loc=central_inventory_location  
inst_group=group
```

Use the `more` command to determine if you have an Oracle central inventory on your system. For example:

```
# more /var/opt/oracle/oraInst.loc
```

```
inventory_loc=/u01/app/oraInventory  
inst_group=oinstall
```

Use the command `grep groupname /etc/group` to confirm that the group specified as the Oracle Inventory group still exists on the system. For example:

```
$ grep oinstall /etc/group  
oinstall:x:54321:grid,oracle
```

Note

Do not put the `oraInventory` directory under the Oracle base directory for a new installation, because that can result in user permission errors for other installations.

Creating the Oracle Inventory Group If an Oracle Inventory Does Not Exist

Create an Oracle Inventory group manually as part of a planned installation, particularly where more than one Oracle software product is installed on servers.

By default, if an `oraInventory` group does not exist, then the installer uses the primary group of the installation owner for the Oracle software being installed as the `oraInventory` group. Ensure

that this group is available as a primary group for all planned Oracle software installation owners.

```
oraInst.loc
```

```
# /usr/sbin/groupadd -g 54321 oinstall
```

About Oracle Installation Owner Accounts

Select or create an Oracle installation owner for your installation, depending on the group and user management plan you want to use for your installations.

You must create a software owner for your installation in the following circumstances:

- If an Oracle software owner user does not exist; for example, if this is the first installation of Oracle software on the system.
- If an Oracle software owner user exists, but you want to use a different operating system user, with different group membership, to separate Oracle Grid Infrastructure administrative privileges from Oracle Database administrative privileges.

In Oracle documentation, a user created to own only Oracle Grid Infrastructure software installations is called the Grid user (grid). This user owns both the Oracle Clusterware and Oracle Automatic Storage Management binaries. A user created to own either all Oracle installations, or one or more Oracle database installations, is called the Oracle user (oracle). You can have only one Oracle Grid Infrastructure installation owner, but you can have different Oracle users to own different installations.

Oracle software owners must have the Oracle Inventory group as their primary group, so that each Oracle software installation owner can write to the central inventory (oraInventory), and so that OCR and Oracle Clusterware resource permissions are set correctly. The database software owner must also have the OSDBA group and (if you create them) the OSOPER, OSBACKUPDBA, OSDGDBA, OSRACDBA, and OSKMDBA groups as secondary groups.

Identifying an Oracle Software Owner User Account

You must create at least one software owner user account the first time you install Oracle software on the system. Either use an existing Oracle software user account, or create an Oracle software owner user account for your installation.

To use an existing user account, obtain from your system administrator the name of an existing Oracle installation owner. Confirm that the existing owner is a member of the Oracle Inventory group.

```
oinstalloinstall
```

```
$ grep "oinstall" /etc/group
oinstall:x:54321:grid,oracle
```

You can then use the ID command to verify that the Oracle installation owners you intend to use have the Oracle Inventory group as their primary group. For example:

```
$ id -a oracle
uid=54321(oracle) gid=54321(oinstall) groups=54321(oinstall),54322(dba),
```

```
54323(oper),54324(backupdba),54325(dgdba),54326(kmdba),54327(asmdba),54330(rac  
dba)
```

```
$ id -a grid  
uid=54331(grid) gid=54321(oinstall) groups=54321(oinstall),54322(dba),  
54327(asmdba),54328(asmoper),54329(asmadmin),54330(racdba)
```

For Oracle Restart installations, to successfully install Oracle Database, ensure that the grid user is a member of the racdba group.

After you create operating system groups, create or modify Oracle user accounts in accordance with your operating system authentication planning.

Related Topics

- [Creating an Oracle Software Owner User](#)
If the Oracle software owner user (oracle or grid) does not exist, or if you require a new Oracle software owner user, then create it as described in this section.
- [Modifying Oracle Owner User Groups](#)
If you have created an Oracle software installation owner account, but it is not a member of the groups you want to designate as the OSDBA, OSOPER, OSDBA for ASM, ASMADMIN, or other system privileges group, then modify the group settings for that user before installation.

Oracle Installations with Standard and Job Role Separation Groups and Users

A job role separation configuration of Oracle Database and Oracle ASM is a configuration with groups and users to provide separate groups for operating system authentication.

Review the following sections to understand more about a Job Role Separation deployment:

- [About Oracle Installations with Job Role Separation](#)
Job role separation requires that you create different operating system groups for each set of system privileges that you grant through operating system authorization.
- [Standard Oracle Database Groups for Database Administrators](#)
Oracle Database has two standard administration groups: OSDBA, which is required, and OSOPER, which is optional.
- [Extended Oracle Database Groups for Job Role Separation](#)
Oracle Database provides an extended set of database groups to grant task-specific system privileges for database administration.
- [Creating an ASMSNMP User](#)
The ASMSNMP user is an Oracle ASM user with privileges to monitor Oracle ASM instances. You are prompted to provide a password for this user during installation.
- [Oracle Automatic Storage Management Groups for Job Role Separation](#)
Oracle Grid Infrastructure operating system groups provide their members task-specific system privileges to access and to administer Oracle Automatic Storage Management.

About Oracle Installations with Job Role Separation

Job role separation requires that you create different operating system groups for each set of system privileges that you grant through operating system authorization.

With Oracle Grid Infrastructure job role separation, Oracle ASM has separate operating system groups that provide operating system authorization for Oracle ASM system privileges for storage tier administration. This operating system authorization is separated from Oracle Database operating system authorization. In addition, the Oracle Grid Infrastructure installation owner provides operating system user authorization for modifications to Oracle Grid Infrastructure binaries.

With Oracle Database job role separation, each Oracle Database installation has separate operating system groups to provide authorization for system privileges on that Oracle Database. Multiple databases can, therefore, be installed on the cluster without sharing operating system authorization for system privileges. In addition, each Oracle software installation is owned by a separate installation owner, to provide operating system user authorization for modifications to Oracle Database binaries. Note that any Oracle software owner can start and stop all databases and shared Oracle Grid Infrastructure resources such as Oracle ASM or Virtual IP (VIP). Job role separation configuration enables database security, and does not restrict user roles in starting and stopping various Oracle Clusterware resources.

You can choose to create one administrative user and one group for operating system authentication for all system privileges on the storage and database tiers. For example, you can designate the `oracle` user to be the installation owner for all Oracle software, and designate `oinstall` to be the group whose members are granted all system privileges for Oracle Clusterware; all system privileges for Oracle ASM; all system privileges for all Oracle Databases on the servers; and all `OINSTALL` system privileges for installation owners. This group must also be the Oracle Inventory group.

If you do not want to use role allocation groups, then Oracle strongly recommends that you use at least two groups:

- A system privileges group whose members are granted administrative system privileges, including `OSDBA`, `OSASM`, and other system privileges groups.
- An installation owner group (the `oraInventory` group) whose members are granted Oracle installation owner system privileges (the `OINSTALL` system privilege).

Note

To configure users for installation that are on a network directory service such as Network Information Services (NIS), refer to your directory service documentation.

Related Topics

- *Oracle Database Administrator's Guide*
- *Oracle Automatic Storage Management Administrator's Guide*

Standard Oracle Database Groups for Database Administrators

Oracle Database has two standard administration groups: `OSDBA`, which is required, and `OSOPER`, which is optional.

- The `OSDBA` group (typically, `dba`)

You must create this group the first time you install Oracle Database software on the system. This group identifies operating system user accounts that have database administrative privileges (the SYSDBA privilege).

If you do not create separate OSDBA, OSOPER, and OSASM groups for the Oracle ASM instance, then operating system user accounts that have the SYSOPER and SYSASM privileges must be members of this group. The name used for this group in Oracle code examples is dba. If you do not designate a separate group as the OSASM group, then the OSDBA group you define is also by default the OSASM group.

- The OSOPER group for Oracle Database (typically, oper)

OSOPER grants the OPERATOR privilege to start up and shut down the database (the SYSOPER privilege). By default, members of the OSDBA group have all privileges granted by the SYSOPER privilege.

Extended Oracle Database Groups for Job Role Separation

Oracle Database provides an extended set of database groups to grant task-specific system privileges for database administration.

The extended set of Oracle Database system privileges groups are task-specific and less privileged than the OSDBA/SYSDBA system privileges. They are designed to provide privileges to carry out everyday database operations. Users granted these system privileges are also authorized through operating system group membership.

You do not have to create these specific group names, but during interactive and silent installation, you must assign operating system groups whose members are granted access to these system privileges. You can assign the same group to provide authorization for these privileges, but Oracle recommends that you provide a unique group to designate each privilege.

The subset of OSDBA job role separation privileges and groups consist of the following:

- OSBACKUPDBA group for Oracle Database (typically, backupdba)

Create this group if you want a separate group of operating system users to have a limited set of database backup and recovery related administrative privileges (the SYSBACKUP privilege).

- OSDGDBA group for Oracle Data Guard (typically, dgdba)

Create this group if you want a separate group of operating system users to have a limited set of privileges to administer and monitor Oracle Data Guard (the SYSDG privilege). To use this privilege, add the Oracle Database installation owners as members of this group.

- The OSKMDBA group for encryption key management (typically, kmdba)

Create this group if you want a separate group of operating system users to have a limited set of privileges for encryption key management such as Oracle Wallet Manager management (the SYSKM privilege). To use this privilege, add the Oracle Database installation owners as members of this group.

- The OSRACDBA group for Oracle Real Application Clusters Administration (typically, racdba)

Create this group if you want a separate group of operating system users to have a limited set of Oracle Real Application Clusters (RAC) administrative privileges (the SYSRAC privilege). To use this privilege:

- Add the Oracle Database installation owners as members of this group.

- For Oracle Restart configurations, if you have a separate Oracle Grid Infrastructure installation owner user (grid), then you must also add the grid user as a member of the OSRACDBA group of the database to enable Oracle Grid Infrastructure components to connect to the database.

Related Topics

- *Oracle Database Administrator's Guide*
- *Oracle Database Security Guide*

Creating an ASMSNMP User

The ASMSNMP user is an Oracle ASM user with privileges to monitor Oracle ASM instances. You are prompted to provide a password for this user during installation.

In addition to the OSASM group, whose members are granted the SYSASM system privilege to administer Oracle ASM, Oracle recommends that you create a less privileged user, ASMSNMP, and grant that user SYSDBA privileges to monitor the Oracle ASM instance. Oracle Enterprise Manager uses the ASMSNMP user to monitor Oracle ASM status.

During installation, you are prompted to provide a password for the ASMSNMP user. You can create an operating system authenticated user, or you can create an Oracle Database user called asmsnmp. In either case, grant the user SYSDBA privileges.

Oracle Automatic Storage Management Groups for Job Role Separation

Oracle Grid Infrastructure operating system groups provide their members task-specific system privileges to access and to administer Oracle Automatic Storage Management.

- The OSASM group for Oracle ASM Administration (typically, asmadmin)

Create this group as a separate group to separate administration privileges groups for Oracle ASM and Oracle Database administrators. Members of this group are granted the SYSASM system privileges to administer Oracle ASM. In Oracle documentation, the operating system group whose members are granted privileges is called the OSASM group, and in code examples, where there is a group specifically created to grant this privilege, it is referred to as asmadmin.

Oracle ASM can support multiple databases. If you have multiple databases on your system, and use multiple OSDBA groups so that you can provide separate SYSDBA privileges for each database, then you should create a group whose members are granted the OSASM/SYSASM administrative privileges, and create a grid infrastructure user (grid) that does not own a database installation, so that you separate Oracle Grid Infrastructure SYSASM administrative privileges from a database administrative privileges group.

Members of the OSASM group can use SQL to connect to an Oracle ASM instance as SYSASM using operating system authentication. The SYSASM privileges permit mounting and dismounting disk groups, and other storage administration tasks. SYSASM privileges provide no access privileges on an RDBMS instance.

If you do not designate a separate group as the OSASM group, but you do define an OSDBA group for database administration, then by default the OSDBA group you define is also defined as the OSASM group.

- The OSOPER group for Oracle ASM (typically, asmoper)

This is an optional group. Create this group if you want a separate group of operating system users to have a limited set of Oracle instance administrative privileges (the SYSOPER for ASM privilege), including starting up and stopping the Oracle ASM instance.

By default, members of the OSASM group also have all privileges granted by the SYSOPER for ASM privilege.

Creating Operating System Privileges Groups

The following sections describe how to create operating system groups for Oracle Grid Infrastructure and Oracle Database:

- [Creating the OSDBA for ASM Group](#)
You must designate a group as the OSDBA for ASM (asmdba) group during installation. Members of this group are also allowed to access the Oracle Automatic Storage Management (Oracle ASM) disk devices.
- [Creating the OSOPER for ASM Group](#)
You can choose to designate a group as the OSOPER for ASM group (asmoper) during installation. Members of this group are granted startup and shutdown privileges to Oracle Automatic Storage Management.
- [Creating the OSDBA Group for Database Installations](#)
Each Oracle Database requires an operating system group to be designated as the OSDBA group. Members of this group are granted the SYSDBA system privileges to administer the database.
- [Creating an OSOPER Group for Database Installations](#)
Create an OSOPER group only if you want to identify a group of operating system users with a limited set of database administrative privileges (SYSOPER operator privileges).
- [Creating the OSBACKUPDBA Group for Database Installations](#)
You must designate a group as the OSBACKUPDBA group during installation. Members of this group are granted the SYSBACKUP privileges to perform backup and recovery operations using RMAN or SQL*Plus.
- [Creating the OSDGDBA Group for Database Installations](#)
You must designate a group as the OSDGDBA group during installation. Members of this group are granted the SYSDG privileges to perform Data Guard operations.
- [Creating the OSKMDBA Group for Database Installations](#)
You must designate a group as the OSKMDBA group during installation. Members of this group are granted the SYSKM privileges to perform Transparent Data Encryption keystore operations.
- [Creating the OSRACDBA Group for Database Installations](#)
You must designate a group as the OSRACDBA group during database installation. Members of this group are granted the SYSRAC privileges to perform day-to-day administration of Oracle databases on an Oracle RAC cluster.

Creating the OSDBA for ASM Group

You must designate a group as the OSDBA for ASM (asmdba) group during installation. Members of this group are also allowed to access the Oracle Automatic Storage Management (Oracle ASM) disk devices.

The Oracle ASM disk devices should be owned by this group.

Create an OSDBA for ASM group using the group name asmdba unless a group with that name already exists:

```
# /usr/sbin/groupadd -g 54327 asmdba
```

Creating the OSOPER for ASM Group

You can choose to designate a group as the OSOPER for ASM group (asmoper) during installation. Members of this group are granted startup and shutdown privileges to Oracle Automatic Storage Management.

If you want to create an OSOPER for ASM group, use the group name asmoper unless a group with that name already exists:

```
# /usr/sbin/groupadd -g 54328 asmoper
```

Creating the OSDBA Group for Database Installations

Each Oracle Database requires an operating system group to be designated as the OSDBA group. Members of this group are granted the SYSDBA system privileges to administer the database.

You must create an OSDBA group in the following circumstances:

- An OSDBA group does not exist, for example, if this is the first installation of Oracle Database software on the system
- An OSDBA group exists, but you want to give a different group of operating system users database administrative privileges for a new Oracle Database installation

Create the OSDBA group using the group name dba, unless a group with that name already exists:

```
# /usr/sbin/groupadd -g 54322 dba
```

Creating an OSOPER Group for Database Installations

Create an OSOPER group only if you want to identify a group of operating system users with a limited set of database administrative privileges (SYSOPER operator privileges).

For most installations, it is sufficient to create only the OSDBA group. However, to use an OSOPER group, create it in the following circumstances:

- If an OSOPER group does not exist; for example, if this is the first installation of Oracle Database software on the system
- If an OSOPER group exists, but you want to give a different group of operating system users database operator privileges in a new Oracle installation

If the OSOPER group does not exist, or if you require a new OSOPER group, then create it. Use the group name oper unless a group with that name already exists. For example:

```
# groupadd -g 54323 oper
```

Creating the OSBACKUPDBA Group for Database Installations

You must designate a group as the OSBACKUPDBA group during installation. Members of this group are granted the SYSBACKUP privileges to perform backup and recovery operations using RMAN or SQL*Plus.

Create the OSBACKUPDBA group using the group name backupdba, unless a group with that name already exists:

```
# /usr/sbin/groupadd -g 54324 backupdba
```

Creating the OSDGDBA Group for Database Installations

You must designate a group as the OSDGDBA group during installation. Members of this group are granted the SYSDG privileges to perform Data Guard operations.

Create the OSDGDBA group using the group name dgdba, unless a group with that name already exists:

```
# /usr/sbin/groupadd -g 54325 dgdba
```

Creating the OSKMDBA Group for Database Installations

You must designate a group as the OSKMDBA group during installation. Members of this group are granted the SYSKM privileges to perform Transparent Data Encryption keystore operations.

If you want a separate group for Transparent Data Encryption, then create the OSKMDBA group using the group name kmdba unless a group with that name already exists:

```
# /usr/sbin/groupadd -g 54326 kmdba
```

Creating the OSRACDBA Group for Database Installations

You must designate a group as the OSRACDBA group during database installation. Members of this group are granted the SYSRAC privileges to perform day-to-day administration of Oracle databases on an Oracle RAC cluster.

Create the OSRACDBA group using the groups name racdba unless a group with that name already exists:

```
# /usr/sbin/groupadd -g 54330 racdba
```

Creating Operating System Oracle Installation User Accounts

Before starting installation, create Oracle software owner user accounts, and configure their environments.

Oracle software owner user accounts require resource settings and other environment configuration. To protect against accidents, Oracle recommends that you create one software installation owner account for each Oracle software program you install.

- [Creating an Oracle Software Owner User](#)
If the Oracle software owner user (oracle or grid) does not exist, or if you require a new Oracle software owner user, then create it as described in this section.
- [Environment Requirements for Oracle Software Owners](#)
You must make the following changes to configure Oracle software owner environments:
- [Procedure for Configuring Oracle Software Owner Environments](#)
Configure each Oracle installation owner user account environment:
- [Modifying Oracle Owner User Groups](#)
If you have created an Oracle software installation owner account, but it is not a member of the groups you want to designate as the OSDBA, OSOPER, OSDBA for ASM, ASMAADMIN, or other system privileges group, then modify the group settings for that user before installation.
- [Preventing Installation Errors Caused by Terminal Output Commands](#)
During an Oracle Grid Infrastructure installation, OUI uses SSH to run commands and copy files to the other nodes. During the installation, hidden files on the system (for example, .bashrc or .cshrc) can cause makefile and other installation errors if they contain terminal output commands.

Creating an Oracle Software Owner User

If the Oracle software owner user (oracle or grid) does not exist, or if you require a new Oracle software owner user, then create it as described in this section.

The following example shows how to create the user oracle with the user ID 54321; with the primary group oinstall; and with secondary groups dba, asmdba, backupdba, dgdba, kmdba, and racdba:

```
# /usr/sbin/useradd -u 54321 -g oinstall -G  
dba,asmdba,backupdba,dgdba,kmdba,racdba oracle
```

The following example shows how to create the user grid with the user ID 54331; with the primary group oinstall; and with secondary groups asmadmin, asmdba, and racdba:

```
# /usr/sbin/useradd -u 54331 -g oinstall -G asmadmin,asmdba,racdba grid
```

You must note the user ID number for installation users, because you need it during preinstallation.

Environment Requirements for Oracle Software Owners

You must make the following changes to configure Oracle software owner environments:

- Set the installation software owner user (grid, oracle) default file mode creation mask (umask) to 022 in the shell startup file. Setting the mask to 022 ensures that the user performing the software installation creates files with 644 permissions.
- Set ulimit settings for file descriptors and processes for the installation software owner (grid, oracle).
- Set the DISPLAY environment variable in preparation for running an Oracle Universal Installer (OUI) installation.

 Caution

If you have existing Oracle installations that you installed with the user ID that is your Oracle Grid Infrastructure software owner, then unset all Oracle environment variable settings for that user.

Related Topics

- [Unsetting Oracle Installation Owner Environment Variables](#)
Unset Oracle installation owner environment variables before you start the installation.

Procedure for Configuring Oracle Software Owner Environments

Configure each Oracle installation owner user account environment:

1. Start an X terminal session (xterm) on the server where you are running the installation.
2. Enter the following command to ensure that X Window applications can display on this system, where *hostname* is the fully qualified name of the local host from which you are accessing the server:

```
$ xhost + hostname
```

3. If you are not logged in as the software owner user, then switch to the software owner user you are configuring. For example, with the user *grid*:

```
$ su - grid
```

On systems where you cannot run *su* commands, use *sudo* instead:

```
$ sudo -u grid -s
```

4. To determine the default shell for the user, enter the following command:

```
$ echo $SHELL
```

5. Open the user's shell startup file in any text editor:

- Bash shell (bash):

```
$ vi .bash_profile
```

- Bourne shell (sh) or Korn shell (ksh):

```
$ vi .profile
```

- C shell (csh or tcsh):

```
% vi .login
```

6. Enter or edit the following line, specifying a value of 022 for the default file mode creation mask:

```
umask 022
```

7. If the ORACLE_SID, ORACLE_HOME, or ORACLE_BASE environment variables are set in the file, then remove these lines from the file.
8. Save the file, and exit from the text editor.
9. To run the shell startup script, enter one of the following commands:

- Bash shell:

```
$ . ~/.bash_profile
```

- Bourne, Bash, or Korn shell:

```
$ . ~/.profile
```

- C shell:

```
% source ~/.login
```

10. Use the following command to check the PATH environment variable:

```
$ echo $PATH
```

Remove any Oracle environment variables.

11. Unset any Oracle environment variables.

If you have an existing Oracle software installation, and you are using the same user to install this installation, then unset the \$ORACLE_HOME, \$ORA_NLS10, and \$TNS_ADMIN environment variables.

If you have set \$ORA_CRS_HOME as an environment variable, then unset it before starting an installation or upgrade. Do not use \$ORA_CRS_HOME as a user environment variable, except as directed by Oracle Support.

12. If you are not installing the software on the local system, then enter a command similar to the following to direct X applications to display on the local system:

- Bourne, Bash, or Korn shell:

```
$ export DISPLAY=local_host:0.0
```

- C shell:

```
% setenv DISPLAY local_host:0.0
```

In this example, local_host is the host name or IP address of the system (your workstation, or another client) on which you want to display the installer.

13. If the /tmp directory has less than 1 GB of free space, then identify a file system with at least 1 GB of free space and set the TMP and TMPDIR environment variables to specify a temporary directory on this file system:

Note

You cannot use a shared file system as the location of the temporary file directory (typically /tmp) for Oracle RAC installations. If you place /tmp on a shared file system, then the installation fails.

- a. Use the `df -h` command to identify a suitable file system with sufficient free space.
- b. If necessary, enter commands similar to the following to create a temporary directory on the file system that you identified, and set the appropriate permissions on the directory:

```
$ sudo -s
# mkdir /mount_point/tmp
# chmod 775 /mount_point/tmp
# exit
```

- c. Enter commands similar to the following to set the TMP and TMPDIR environment variables:

Bourne, Bash, or Korn shell:

```
$ TMP=/mount_point/tmp
$ TMPDIR=/mount_point/tmp
$ export TMP TMPDIR
```

C shell:

```
% setenv TMP /mount_point/tmp
% setenv TMPDIR /mount_point/tmp
```

14. To verify that the environment has been set correctly, enter the following commands:

```
$ umask
$ env | more
```

Verify that the `umask` command displays a value of 22, 022, or 0022 and that the environment variables you set in this section have the correct values.

Modifying Oracle Owner User Groups

If you have created an Oracle software installation owner account, but it is not a member of the groups you want to designate as the OSDBA, OSOPER, OSDBA for ASM, ASMADMIN, or other system privileges group, then modify the group settings for that user before installation.

ⓘ Warning

Each Oracle software owner must be a member of the same central inventory group. Do not modify the primary group of an existing Oracle software owner account, or designate different groups as the OINSTALL group. If Oracle software owner accounts have different groups as their primary group, then you can corrupt the central inventory.

During installation, the user who is installing the software should have the OINSTALL group as its primary group, and it must be a member of the operating system groups appropriate for your installation. For example:

```
# /usr/sbin/usermod -g oinstall -G
dba,asmdba,backupdba,dgdba,kmdba,racdba[,oper] oracle
```

Preventing Installation Errors Caused by Terminal Output Commands

During an Oracle Grid Infrastructure installation, OUI uses SSH to run commands and copy files to the other nodes. During the installation, hidden files on the system (for example, `.bashrc` or `.cshrc`) can cause `makefile` and other installation errors if they contain terminal output commands.

To avoid this problem, you must modify hidden files in each Oracle installation owner user home directory to suppress all output on STDOUT or STDERR (for example, `stty`, `xtitle`, and other such commands) as in the following examples:

Bourne, Bash, or Korn shell:

```
if [ -t 0 ]; then
    stty intr ^C
fi
```

C shell:

```
test -t 0
if ($status == 0) then
    stty intr ^C
endif
```

Note

If the remote shell can load hidden files that contain `stty` commands, then OUI indicates an error and stops the installation.

Creating Oracle Database Vault User Accounts

If you intend to use Oracle Database Vault by default, then you must create an Oracle Database Vault user account, and configure that user.

You must create the Database Vault Owner account before you can use Oracle Database Vault. You can also create a Database Vault Account Manager administrative account.

Oracle Database Vault installs a baseline database auditing policy. This policy covers the access control configuration information stored in Oracle Database Vault database tables, information stored in Oracle Catalog (rollback segments, tablespaces, and so on), the use of system privileges, and Oracle Label Security configuration. When you install Oracle Database Vault, the security specific database initialization parameters are initialized with default values.

Related Topics

- *Oracle Database Vault Administrator's Guide*

Unsetting Oracle Installation Owner Environment Variables

Unset Oracle installation owner environment variables before you start the installation.

The environment variables you have set for the Oracle installation owner account you use to run the installation can cause issues if they are set to values that conflict with the values needed for installation.

If you have set `ORA_CRS_HOME` as an environment variable, following instructions from Oracle Support, then unset it before starting an installation or upgrade. You should never use `ORA_CRS_HOME` as an environment variable except under explicit direction from Oracle Support.

If you have had an existing installation on your system, and you are using the same user account to install this installation, then unset the following environment variables: `ORA_CRS_HOME`, `ORACLE_HOME`, `ORA_NLS10`, `TNS_ADMIN`, and any other environment variable set for the Oracle installation user that is connected with Oracle software homes.

Also, ensure that the `$ORACLE_HOME/bin` path is removed from your `PATH` environment variable.

6

Configuring Networks for Oracle Database

If you install Oracle Databases on servers with multiple Oracle homes, multiple aliases, or without a static IP address, then review these network configuration topics.

If you are installing Oracle Database on a server with a static host name and IP address and at least one network interface, then no special network configuration is required.

Note

If you have a firewall running, ensure that all the required ports are open. See, [Oracle Database Component Port Numbers and Protocols](#) for a list of port numbers and protocols configured for Oracle Database components.

- [About Oracle Database Network Configuration Options](#)
You can enable database clients to connect to servers associated with multiple IP addresses, and you can install Oracle Database on servers with no network connections, and set up database services after installation.
- [About Assigning Global Database Names During Installation](#)
The database name input field is used to set the DB_NAME, DB_UNIQUE_NAME, and DB_DOMAIN Oracle initialization parameter values.
- [Network Configuration for Computers Completed After Installation](#)
You must confirm that a non-networked computer can connect to itself to ensure that you can configure client network resolution after installation. A non-networked computer is a computer that does not have a fixed network address, such as a computer using DHCP.
- [Network Configuration for Multihomed Computers](#)
You must set the ORACLE_HOSTNAME environment variable to install Oracle Database on a multihomed computer. A multihomed computer is associated with multiple IP addresses.
- [Setting the ORACLE_HOSTNAME Environment Variable](#)
Run the commands shown in this example as the Oracle user account to set the ORACLE_HOSTNAME environment variable.
- [Network Configuration for Computers with Multiple Aliases](#)
You must set the ORACLE_HOSTNAME environment variable to install Oracle Database on a multialias computer. A multialias computer is one to which multiple aliases resolve.

About Oracle Database Network Configuration Options

You can enable database clients to connect to servers associated with multiple IP addresses, and you can install Oracle Database on servers with no network connections, and set up database services after installation.

Typically, the computer on which you want to install Oracle Database is a server running a single database instance, with a single host name that is resolvable on a network. Oracle Universal Installer uses the host name and Oracle Database instance information to set up

network services automatically. The database provides database services to clients using a connect descriptor that resolves to the host name where the database instance is running.

However, you can configure Oracle Database on servers with the following nonstandard configurations:

- **Multihomed Computers:** Servers with multiple Oracle Database installations
- **Multiple Alias Computers:** Servers with multiple aliases, so that more than one host name resolves to the computer
- **Non-Networked computers:** Servers that do not have network connectivity at the time of installation

About Assigning Global Database Names During Installation

The database name input field is used to set the DB_NAME, DB_UNIQUE_NAME, and DB_DOMAIN Oracle initialization parameter values.

The Oracle Database software identifies a database by its global database name. A global database name consists of the database name and database domain. Usually, the database domain is the same as the network domain, but it need not be. The global database name uniquely distinguishes a database from any other database in the same network. You specify the global database name when you create a database during the installation or using the Oracle Database Configuration Assistant.

`sales.us.example.com`

Here:

- `sales.us` is the name of the database. The database name, DB_UNIQUE_NAME, portion is a string of no more than 30 characters that can contain alphanumeric characters, underscore (_), dollar sign (\$), and pound sign (#) but must begin with an alphabetic character. No other special characters are permitted in a database name.
- `sales.us` is also the DB_NAME. The DB_NAME initialization parameter specifies a database identifier of up to eight characters.
- `example.com` is the database domain in which the database is located. In this example, the database domain equals the network domain. Together, the database name and the database domain make the global database name unique. The domain portion is a string of no more than 128 characters that can contain alphanumeric characters, underscore (_), and pound sign (#). The DB_DOMAIN initialization parameter specifies the database domain name.

However, the DB_NAME parameter need not necessarily be the first eight characters of DB_UNIQUE_NAME.

The DB_NAME parameter and the DB_DOMAIN parameter combine to create the global database name value.

The system identifier (SID) identifies a specific database instance. The SID uniquely distinguishes the instance from any other instance on the same computer. Each database instance requires a unique SID and database name. In most cases, the SID equals the database name portion of the global database name.

Related Topics

- *Oracle Database Reference*

- *Oracle Database Administrator's Guide*

Network Configuration for Computers Completed After Installation

You must confirm that a non-networked computer can connect to itself to ensure that you can configure client network resolution after installation. A non-networked computer is a computer that does not have a fixed network address, such as a computer using DHCP.

You can install Oracle Database on a non-networked computer. If the computer, such as a laptop, is configured for DHCP and you plan to connect the computer to the network after the Oracle Database installation, then use the `ping` command on the computer on which you want to install the database to check if the computer can connect to itself. Perform this step by first using only the host name and then using the fully qualified name, which should be in the `/etc/hosts` file.

If you connect the computer to a network after installation, then the Oracle Database instance on the computer can work with other instances on the network. The computer can use a static IP or DHCP, depending on the network to which you are connected.

When you run the `ping` command on the computer itself, the `ping` command should return the IP address of the computer. If the `ping` command fails, then contact your network administrator.

Network Configuration for Multihomed Computers

You must set the `ORACLE_HOSTNAME` environment variable to install Oracle Database on a multihomed computer. A multihomed computer is associated with multiple IP addresses.

Typically, a server configured to run multiple Oracle Database Oracle homes is configured with multiple network interface cards. A host name resolves to an IP address configured for one network card for each Oracle Database. You can also set up aliases for host names. By default, during installation, Oracle Universal Installer uses the value set for the environment variable `ORACLE_HOSTNAME` set for the Oracle installation user account running the installation to find the host name. If the user environment variable `ORACLE_HOSTNAME` is not set for the Oracle user, and you are installing on a computer that has multiple network cards, then Oracle Universal Installer determines the host name from the `/etc/hosts` file and the information you provide during the installation session.

Oracle Database clients connecting to the database must be able to access the computer by using either the alias for the host name, or by using the host name associated with that instance. To verify that the client can resolve to the database using both alias and host name, use the `ping` command to check connectivity to the host name both for the database on the server (host name only), and for the fully qualified domain name (host name and domain name).

Note

Clients must be able to obtain a response using the `ping` command both for the host name and for the fully qualified domain name. If either test fails, then contact your network administrator to resolve the issue.

Setting the ORACLE_HOSTNAME Environment Variable

Run the commands shown in this example as the Oracle user account to set the ORACLE_HOSTNAME environment variable.

The following example shows the commands to run on the Oracle user account to set the ORACLE_HOSTNAME environment variable. In this example, the fully qualified host name is `somehost.example.com`.

Bourne, Bash or Korn Shell

```
$ export ORACLE_HOSTNAME=somehost.example.com
```

C Shell

```
% setenv ORACLE_HOSTNAME somehost.example.com
```

Network Configuration for Computers with Multiple Aliases

You must set the ORACLE_HOSTNAME environment variable to install Oracle Database on a multialias computer. A multialias computer is one to which multiple aliases resolve.

A computer with multiple aliases is a computer that is registered with the naming service under a single IP address, but which resolves multiple aliases to that address. The naming service resolves any of those aliases to the same computer. Before installing Oracle Database on such a computer, set the Oracle installation owner environment variable ORACLE_HOSTNAME to the computer whose host name you want to use.

Supported Storage Options for Oracle Database and Oracle Grid Infrastructure

Review supported storage options as part of your installation planning process.

- [Supported Storage Options for Oracle Database](#)
The following table shows the storage options supported for Oracle Database binaries and files:
- [About Oracle Grid Infrastructure for a Standalone Server](#)
If you plan to use Oracle Automatic Storage Management (Oracle ASM), then you must install Oracle Restart before installing your database.
- [About Upgrading Existing Oracle Automatic Storage Management Instances](#)
Oracle Automatic Storage Management (Oracle ASM) upgrades are carried out during an Oracle Grid Infrastructure upgrade.
- [About Managing Disk Groups for Older Database Versions](#)
When you install earlier Oracle Database releases on Oracle Grid Infrastructure, use Oracle ASM Configuration Assistant (Oracle ASMCA) to create and modify disk groups.
- [Oracle ACFS and Oracle ADVM](#)
Oracle Automatic Storage Management Cluster File System (Oracle ACFS) extends Oracle ASM technology to support of all of your application data in both single instance and cluster configurations.
- [File System Options for Oracle Database](#)
If you install Oracle Database files on a file system, then Oracle Database Configuration Assistant creates the database files in a directory on a file system mounted on the computer.
- [Guidelines for Placing Oracle Database Files On a File System or Logical Volume](#)
If you choose to place the Oracle Database files on a file system, then use the following guidelines when deciding where to place them:
- [About NFS Storage for Data Files](#)
Review this section for NFS storage configuration guidelines.
- [About Direct NFS Client Mounts to NFS Storage Devices](#)
Direct NFS (dNFS) Client integrates the NFS client functionality directly in the Oracle software to optimize the I/O path between Oracle and the NFS server. This integration can provide significant performance improvements.

Supported Storage Options for Oracle Database

The following table shows the storage options supported for Oracle Database binaries and files:

Table 7-1 Supported Storage Options for Oracle Database

Storage Option	Oracle Database Binaries	Oracle Database Data Files	Oracle Database Recovery Files
Oracle Automatic Storage Management (Oracle ASM) Note: Loopback devices are not supported for use with Oracle ASM	No	Yes	Yes
Oracle Automatic Storage Management Cluster File System (Oracle ACFS)	Yes	Yes (Oracle Database 12c Release 1 (12.1) and later)	Yes (Oracle Database 12c Release 1 (12.1) and later)
Local file system	Yes	Yes, but not recommended	Yes, but not recommended
Network file system (NFS) on a certified network-attached storage (NAS) filer	Yes	Yes	Yes

Guidelines for Storage Options

Use the following guidelines when choosing storage options:

- Oracle strongly recommends that you use a dedicated set of disks for Oracle ASM.
- You can choose any combination of the supported storage options for each file type provided that you satisfy all requirements listed for the chosen storage options.
- You can use Oracle ASM to store Oracle Clusterware files.
- Direct use of raw or block devices is not supported. You can only use raw or block devices under Oracle ASM.

Related Topics

- *Oracle Database Upgrade Guide*

About Oracle Grid Infrastructure for a Standalone Server

If you plan to use Oracle Automatic Storage Management (Oracle ASM), then you must install Oracle Restart before installing your database.

Oracle Grid Infrastructure for a standalone server is a version of Oracle Grid Infrastructure that supports single instance databases. This support includes volume management, file system, and automatic restart capabilities. Oracle Grid Infrastructure for a standalone server includes Oracle Restart and Oracle Automatic Storage Management. Oracle combined the two infrastructure products into a single set of binaries that is installed into an Oracle Restart home.

Oracle Restart is a feature provided as part of Oracle Grid Infrastructure. Oracle Restart monitors and can restart Oracle Database instances, Oracle Net Listeners, and Oracle ASM instances. Oracle Restart is currently restricted to manage single instance Oracle Databases and Oracle ASM instances only.

Oracle Automatic Storage Management is a volume manager and a file system for Oracle Database files that supports single-instance Oracle Database and Oracle Real Application

Clusters (Oracle RAC) configurations. Oracle Automatic Storage Management also supports a general purpose file system for your application needs, including Oracle Database binaries. Oracle Automatic Storage Management is Oracle's recommended storage management solution that provides an alternative to conventional volume managers, and file systems.

Oracle Restart improves the availability of your Oracle database by providing the following services:

- When there is a hardware or a software failure, Oracle Restart automatically starts all Oracle components, including the Oracle database instance, Oracle Net Listener, database services, and Oracle ASM.
- Oracle Restart starts components in the proper order when the database host is restarted.
- Oracle Restart runs periodic checks to monitor the status of Oracle components. If a check operation fails for a component, then the component is shut down and restarted.

Note the following restrictions for using Oracle Restart:

- You can neither install Oracle Restart on an Oracle Grid Infrastructure cluster member node, nor add an Oracle Restart server to an Oracle Grid Infrastructure cluster member node. Oracle Restart supports single-instance databases on one server, while Oracle Grid Infrastructure for a Cluster supports single-instance or Oracle RAC databases on a cluster.
- If you want to use Oracle ASM or Oracle Restart, then you should install Oracle Grid Infrastructure for a standalone server before you install and create the database. Otherwise, you must install Oracle Restart, and then manually register the database with Oracle Restart.
- You can use the Oracle Restart implementation of Oracle Grid Infrastructure only in single-instance (nonclustered) environments. Use Oracle Grid Infrastructure with Oracle Clusterware for clustered environments.

About Upgrading Existing Oracle Automatic Storage Management Instances

Oracle Automatic Storage Management (Oracle ASM) upgrades are carried out during an Oracle Grid Infrastructure upgrade.

If you are upgrading from Oracle ASM 11g Release 2 (11.2.0.4) or later, then Oracle ASM is always upgraded with Oracle Grid Infrastructure as part of the upgrade, and Oracle Automatic Storage Management Configuration Assistant (Oracle ASMCA) is started by the root scripts during upgrade. Subsequently, you can use Oracle ASMCA (located in `Grid_home/bin`) to configure failure groups, Oracle ASM volumes, and Oracle Automatic Storage Management Cluster File System (Oracle ACFS).

Oracle ASMCA cannot perform a separate upgrade of Oracle ASM from a prior release to the current release.

Upgrades of Oracle ASM from releases prior to 11g Release 2 (11.2) are not supported.

Related Topics

- *Oracle Automatic Storage Management Administrator's Guide*
- *Oracle Database Upgrade Guide*

About Managing Disk Groups for Older Database Versions

When you install earlier Oracle Database releases on Oracle Grid Infrastructure, use Oracle ASM Configuration Assistant (Oracle ASMCA) to create and modify disk groups.

Releases prior to Oracle Database 11g Release 2 used Oracle Database Configuration Assistant (Oracle DBCA) to perform administrative tasks on Oracle ASM. Starting with Oracle Database 11g Release 2 (11.2), Oracle ASM is installed as part of an Oracle Grid Infrastructure installation. You can no longer use Oracle DBCA to perform administrative tasks on Oracle ASM.

See Also

Oracle Automatic Storage Management Administrator's Guide for details about configuring disk group compatibility for databases using Oracle Database 11g software with this release of Oracle Grid Infrastructure.

Oracle ACFS and Oracle ADVM

Oracle Automatic Storage Management Cluster File System (Oracle ACFS) extends Oracle ASM technology to support of all of your application data in both single instance and cluster configurations.

Oracle Automatic Storage Management Dynamic Volume Manager (Oracle ADVM) provides volume management services and a standard disk device driver interface to clients. Oracle ACFS communicates with Oracle ASM through the Oracle ADVM interface.

- [Oracle ACFS and Oracle ADVM Support on Oracle Solaris](#)
Oracle ACFS and Oracle ADVM are supported on Oracle Solaris.
- [Restrictions and Guidelines for Oracle ACFS](#)
Review these topics as part of your storage plan for using Oracle ACFS for single instance and cluster configurations.

Related Topics

- *Oracle Automatic Storage Management Administrator's Guide*

Oracle ACFS and Oracle ADVM Support on Oracle Solaris

Oracle ACFS and Oracle ADVM are supported on Oracle Solaris.

Table 7-2 Platforms That Support Oracle ACFS and Oracle ADVM

Platform / Operating System	Support Information
Oracle Solaris 11	Supported.
Oracle Solaris containers	Not supported.

① See Also

- My Oracle Support Note 1369107.1 for more information about platforms and releases that support Oracle ACFS and Oracle ADVM:
<https://support.oracle.com/rs?type=doc&id=1369107.1>
- Patch Set Updates for Oracle Products (My Oracle Support Note 854428.1) for current release and support information:
<https://support.oracle.com/rs?type=doc&id=854428.1>

Restrictions and Guidelines for Oracle ACFS

Review these topics as part of your storage plan for using Oracle ACFS for single instance and cluster configurations.

- Oracle Automatic Storage Management Cluster File System (Oracle ACFS) provides a general purpose file system.
- You can only use Oracle ACFS when Oracle ASM is configured.
- Note the following general guidelines and restrictions for placing Oracle Database and Oracle Grid Infrastructure files on Oracle ACFS:
 - You can place Oracle Database binaries, data files, and administrative files (for example, trace files) on Oracle ACFS.
 - Oracle ACFS does not support encryption or replication with Oracle Database data files, tablespace files, control files, redo logs, archive logs, RMAN backups, Data Pump dumpsets, and flashback files.
 - You can place Oracle Database homes on Oracle ACFS only if the database release is Oracle Database 11g Release 2, or more recent releases. You cannot install earlier releases of Oracle Database on Oracle ACFS.
 - For installations on Oracle Clusterware, you cannot place Oracle Clusterware files on Oracle ACFS.
- Oracle Restart does not support root-based Oracle Clusterware resources. For this reason, the following restrictions apply if you run Oracle ACFS on an Oracle Restart Configuration:
 - Starting with Oracle Database 18c, configuration assistants do not allow the creation of Oracle Database homes on Oracle ACFS in an Oracle Restart configuration.
 - Oracle Restart does not support Oracle ACFS resources on all platforms.
 - Starting with Oracle Database 12c, Oracle Restart configurations do not support the Oracle ACFS registry.
 - On Linux, Oracle ACFS provides an automated mechanism to load and unload drivers and mount and unmount Oracle ACFS file systems on system restart and shutdown. However, Oracle ACFS does not provide automated recovery of mounted file systems when the system is running. Other than Linux, Oracle ACFS does not provide this automated mechanism on other operating systems.
 - Creating Oracle data files on an Oracle ACFS file system is not supported in Oracle Restart configurations. Creating Oracle data files on an Oracle ACFS file system is supported on Oracle Grid Infrastructure for a cluster configurations.
- Oracle ACFS and Oracle ADVM are not supported on IBM AIX Workload Partitions (WPARs).

Related Topics

- *Oracle Automatic Storage Management Administrator's Guide*

File System Options for Oracle Database

If you install Oracle Database files on a file system, then Oracle Database Configuration Assistant creates the database files in a directory on a file system mounted on the computer.

Oracle recommends that the file system be separate from the file systems used by the operating system or the Oracle Database software.

The file system can be any of the following:

Standard Oracle Database Creation Options

- A file system on a disk that is physically attached to the system.
If you are creating a database on basic disks that are not logical volumes or RAID devices, then Oracle recommends that you follow the Optimal Flexible Architecture (OFA) recommendations and distribute the database files over many disks.
- A file system on a logical volume manager (LVM) volume or a RAID device.
If you are using multiple disks in an LVM or RAID configuration, then Oracle recommends that you use the Stripe and Mirror Everything (S.A.M.E) methodology to increase performance and reliability. Using this methodology, you do not have to specify multiple file system mount points for the database storage.
- A network file system (NFS) mounted from a certified network-attached storage (NAS) device. You also have the option to use Direct NFS Client, which simplifies the administration of NFS configurations and also improves performance.

Advanced Oracle Database Creation Options

- The three file system options available to standard Oracle Database installations.
- With Oracle Managed Files, you specify file system directories in which the database automatically creates, names, and manages files at the database object level.
If you use the Oracle Managed Files feature, then you must specify only the database object name instead of file names when creating or deleting database files.

Related Topics

- [Oracle RAC Technologies Matrix for Linux Platforms](#)
- *Oracle Database Administrator's Guide*

Guidelines for Placing Oracle Database Files On a File System or Logical Volume

If you choose to place the Oracle Database files on a file system, then use the following guidelines when deciding where to place them:

- The default path suggested by Oracle Universal Installer for the database file directory is a subdirectory of the Oracle base directory.
- You can choose either a single file system or more than one file system to store the database files:

- If you want to use a single file system, then choose a file system on a physical device that is dedicated to the database.

For best performance and reliability, choose a RAID device or a logical volume on more than one physical device, and implement a stripe-and-mirror-everything (SAME) storage policy.
- If you want to use more than one file system, then choose file systems on separate physical devices that are dedicated to the database.

This method enables you to distribute physical input-output operations and create separate control files on different devices for increased reliability. It also enables you to fully implement Oracle Optimal Flexible Architecture (OFA) guidelines. Choose the Advanced database creation option to implement this method.
- If you intend to create a preconfigured database during the installation, then the file system (or file systems) that you choose must have at least 2 GB of free disk space.

For production databases, you must estimate the disk space requirement depending on the use of the database.
- For optimum performance, the file systems that you choose must be on physical devices that are used only by the database.
- The Oracle user running the Oracle Database installation must have write permissions to create the files in the path that you specify.

About NFS Storage for Data Files

Review this section for NFS storage configuration guidelines.

Network-Attached Storage and NFS Protocol

Network-attached storage (NAS) systems use the network file system (NFS) protocol to access files over a network, which enables client servers to access files over networks as easily as to storage devices attached directly to the servers. You can store data files on supported NFS systems. NFS is a shared file system protocol, so NFS can support both single instance and Oracle Real Application Clusters databases.

Note

The performance of Oracle software and databases stored on NAS devices depends on the performance of the network connection between the servers and the network-attached storage devices. For better performance, Oracle recommends that you connect servers to NAS devices using private dedicated network connections. NFS network connections should use Gigabit Ethernet or better.

Refer to your vendor documentation to complete NFS configuration and mounting.

Requirements for Using NFS Storage

Before you start installation, NFS file systems must be mounted and available to servers.

About Direct NFS Client Mounts to NFS Storage Devices

Direct NFS (dNFS) Client integrates the NFS client functionality directly in the Oracle software to optimize the I/O path between Oracle and the NFS server. This integration can provide significant performance improvements.

Direct NFS Client supports NFSv3, NFSv4, NFSv4.1, and pNFS protocols to access the NFS server. Direct NFS Client also simplifies, and in many cases automates, the performance optimization of the NFS client configuration for database workloads.

Starting with Oracle Database 12c Release 2, when you enable Direct NFS, you can also enable the Direct NFS dispatcher. The Direct NFS dispatcher consolidates the number of TCP connections that are created from a database instance to the NFS server. In large database deployments, using Direct NFS dispatcher improves scalability and network performance. Parallel NFS deployments also require a large number of connections. Hence, the Direct NFS dispatcher is recommended with Parallel NFS deployments too.

Direct NFS Client can obtain NFS mount points either from the operating system mount entries, or from the `orantstab` file.

Direct NFS Client Requirements

- NFS servers must have write size values (`wtmax`) of 32768 or greater to work with Direct NFS Client.
- NFS mount points must be mounted both by the operating system kernel NFS client and Direct NFS Client, even though you configure Direct NFS Client to provide file service.

If Oracle Database cannot connect to an NFS server using Direct NFS Client, then Oracle Database connects to the NFS server using the operating system kernel NFS client. When Oracle Database fails to connect to NAS storage through Direct NFS Client, it logs an informational message about the Direct NFS Client connect error in the Oracle alert and trace files.

- Follow standard guidelines for maintaining integrity of Oracle Database files mounted by both operating system NFS and by Direct NFS Client.

Direct NFS Mount Point Search Order

Direct NFS Client searches for mount entries in the following order:

1. `$ORACLE_HOME/dbs/orantstab`
2. `/var/opt/oracle/orantstab`
3. `/etc/mnttab`

Direct NFS Client uses the first matching entry as the mount point.

Note

You can have only one active NFS Client implementation for each instance. Enabling Direct NFS Client on an instance prevents you from using another NFS Client implementation, such as kernel NFS Client.

Default NFS Version Selection

The Direct NFS Client uses the following rules to determine the default NFS protocol version:

1. If a matching mount entry is found in `oranstab` and `nfs_version` is specified, then dNFS uses the specified version.
2. If a matching mount entry is found in `oranstab` and `nfs_version` is not specified, then dNFS uses NFSv3.
3. If a matching mount entry is not found in `oranstab` or if `oranstab` is not present, then dNFS queries the operating system mount table (`mtab`) and uses the NFS version configured for the mount.
4. If the specified version in `oranstab` or `mtab` is NFSv4.2, then dNFS uses NFSv4.1 because NFSv4.2 is not supported.

Related Topics

- [Configuring NFS Buffer Size Parameters for Oracle Database](#)
Set the values for the NFS buffer size parameters `rsize` and `wsiz` to at least 32768.
- [Creating an `oranstab` File for Direct NFS Client](#)
Direct NFS uses a configuration file, `oranstab`, to determine the available mount points.

See Also

- *Oracle Database Reference* for information about setting the `enable_dnfs_dispatcher` parameter in the initialization parameter file to enable Direct NFS dispatcher
- *Oracle Database Performance Tuning Guide* for performance benefits of enabling Parallel NFS and Direct NFS dispatcher
- *Oracle Automatic Storage Management Administrator's Guide* for guidelines about managing Oracle Database data files created with Direct NFS Client or kernel NFS

8

Configuring File System Storage for Oracle Database

Complete these procedures to use file system storage for Oracle Database.

For optimal database organization and performance, Oracle recommends that you install data files and the Oracle Database software in different disks.

If you plan to place storage on Network File System (NFS) protocol devices, then Oracle recommends that you use Oracle Direct NFS (dNFS) to take advantage of performance optimizations built into the Oracle Direct NFS client.

- [Configuring NFS Buffer Size Parameters for Oracle Database](#)
Set the values for the NFS buffer size parameters `rsize` and `wsiz` to at least 32768.
- [Checking TCP Network Protocol Buffer for Direct NFS Client](#)
Check your TCP network buffer size to ensure that it is adequate for the speed of your servers.
- [Creating an `orantab` File for Direct NFS Client](#)
Direct NFS uses a configuration file, `orantab`, to determine the available mount points.
- [Enabling and Disabling Direct NFS Client Control of NFS](#)
By default, Direct NFS Client is installed in a disabled state with single-instance Oracle Database installations. Before enabling Direct NFS, you must configure an `orantab` file.
- [Enabling Hybrid Columnar Compression on Direct NFS Client](#)
Perform these steps to enable Hybrid Columnar Compression (HCC) on Direct NFS Client:

Related Topics

- [My Oracle Support note 1496040.1](#)

Configuring NFS Buffer Size Parameters for Oracle Database

Set the values for the NFS buffer size parameters `rsize` and `wsiz` to at least 32768.

For example, to use `rsize` and `wsiz` buffer settings with the value 32768 for an Oracle Database data files mount point, set mount point parameters to values similar to the following:

```
nfs_server:/vol/DATA/oradata /home/oracle/netapp nfs\  
rw,bg,hard,nointr,rsize=32768,wsiz=32768,tcp,actimeo=0,vers=3,timeo=600
```

Direct NFS Client issues writes at `wtmax` granularity to the NFS server.

Related Topics

- [My Oracle Support note 359515.1](#)

Checking TCP Network Protocol Buffer for Direct NFS Client

Check your TCP network buffer size to ensure that it is adequate for the speed of your servers.

By default, the network buffer size is set to 1 MB for TCP, and 2 MB for UDP. The TCP buffer size can set a limit on file transfers, which can negatively affect performance for Direct NFS Client users.

To check the current TCP buffer size on Oracle Solaris 11:

```
# ipadm show-prop -p max_buf tcp
```

Oracle recommends that you set the value based on the link speed of your servers. For example:

On Oracle Solaris 11:

```
# ipadm set-prop -p max_buf=1048576 tcp
```

Additionally, check your TCP send window size and TCP receive window size to ensure that they are adequate for the speed of your servers.

To check the current TCP send window size and TCP receive window size on Oracle Solaris 11:

```
# ipadm show-prop -p send_buf tcp  
# ipadm show-prop -p recv_buf tcp
```

Oracle recommends that you set the value based on the link speed of your servers. For example:

On Oracle Solaris 11:

```
# ipadm set-prop -p send_buf=1056768 tcp  
# ipadm set-prop -p recv_buf=1056768 tcp
```

Creating an oranfstab File for Direct NFS Client

Direct NFS uses a configuration file, `oranfstab`, to determine the available mount points.

Create an `oranfstab` file with the following attributes for each NFS server that you want to access using Direct NFS Client:

- `server`

The NFS server name.

For NFS setup with Kerberos authentication, the `server` attribute name must be the fully-qualified name of the NFS server. This `server` attribute name is used to create service principal for Ticket Granting Service (TGS) request from the Kerberos server. If you are configuring external storage snapshot cloning, then the NFS server name should be a valid host name. For all other scenarios, the NFS server name can be any unique name.

- `local`
Up to four paths on the database host, specified by IP address or by name, as displayed using the `ifconfig` command run on the database host.
- `path`
Up to four network paths to the NFS server, specified either by IP address, or by name, as displayed using the `ifconfig` command on the NFS server.
- `export`
The exported path from the NFS server.
- `mount`
The corresponding local mount point for the exported volume.
- `mnt_timeout`
Specifies (in seconds) the time Direct NFS Client should wait for a successful mount before timing out. This parameter is optional. The default timeout is 10 minutes (600).
- `nfs_version`
Specifies the NFS protocol version used by Direct NFS Client. Possible values are NFSv3, NFSv4, NFSv4.1, and pNFS. The default version is NFSv3. If you select NFSv4.x, then you must configure the value in `orafstab` for `nfs_version`.

Specify `nfs_version` as pNFS, if you want to use Direct NFS with Parallel NFS. Direct NFS supports only the default sys security authentication with Parallel NFS. Direct NFS does not support Parallel NFS when combined with any of the Kerberos authentication parameters.
- `security_default`
Specifies the default security mode applicable for all the exported NFS server paths for a server entry. This parameter is optional. `sys` is the default value. See the description of the `security` parameter for the supported security levels for the `security_default` parameter.
- `security`
Specifies the security level, to enable security using Kerberos authentication protocol with Direct NFS Client. This optional parameter can be specified per export-mount pair. The supported security levels for the `security_default` and `security` parameters are:
 - `sys`: UNIX level security AUTH_UNIX authentication based on user identifier (UID) and group identifier (GID) values. This is the default value for security parameters.
 - `krb5`: Direct NFS runs with plain Kerberos authentication. Server is authenticated as the real server which it claims to be.
 - `krb5i`: Direct NFS runs with Kerberos authentication and NFS integrity. Server is authenticated and each of the message transfers is checked for integrity.
 - `krb5p`: Direct NFS runs with Kerberos authentication and NFS privacy. Server is authenticated, and all data is completely encrypted.The `security` parameter, if specified, takes precedence over the `security_default` parameter. If neither of these parameters are specified, then `sys` is the default authentication.

For NFS server Kerberos security setup, review the relevant NFS server documentation. For Kerberos client setup, review the relevant operating system documentation.
- `dontroute`

Specifies that outgoing messages should not be routed by the operating system, but instead sent using the IP address to which they are bound.

Note

The `dontroute` option is a POSIX option, which sometimes does not work on Linux systems with multiple paths in the same subnet.

- **management**

Enables Direct NFS Client to use the management interface for SNMP queries. You can use this parameter if SNMP is running on separate management interfaces on the NFS server. The default value is the server parameter value.

- **community**

Specifies the community string for use in SNMP queries. Default value is `public`.

The following examples show three possible NFS server entries in `orafstab`. A single `orafstab` can have multiple NFS server entries.

Example 8-1 Using Local and Path NFS Server Entries

The following example uses both local and path. Because they are in different subnets, you do not have to specify `dontroute`.

```
server: MyDataServer1
local: 192.0.2.0
path: 192.0.2.1
local: 192.0.100.0
path: 192.0.100.1
export: /vol/oradata1 mount: /mnt/oradata1
```

Example 8-2 Using Local and Path in the Same Subnet, with dontroute

Local and path in the same subnet, where `dontroute` is specified:

```
server: MyDataServer2
local: 192.0.2.0
path: 192.0.2.128
local: 192.0.2.1
path: 192.0.2.129
dontroute
export: /vol/oradata2 mount: /mnt/oradata2
```

Example 8-3 Using Names in Place of IP Addresses, with Multiple Exports, management and community

```
server: MyDataServer3
local: LocalPath1
path: NfsPath1
local: LocalPath2
path: NfsPath2
local: LocalPath3
path: NfsPath3
local: LocalPath4
```

```
path: NfsPath4
dontroute
export: /vol/oradata3 mount: /mnt/oradata3
export: /vol/oradata4 mount: /mnt/oradata4
export: /vol/oradata5 mount: /mnt/oradata5
export: /vol/oradata6 mount: /mnt/oradata6
management: MgmtPath1
community: private
```

Example 8-4 Using Kerberos Authentication with Direct NFS Export

The security parameter overrides security_default:

```
server: nfsserver
local: 192.0.2.0
path: 192.0.2.2
local: 192.0.2.3
path: 192.0.2.4
export: /private/oracle1/logs mount: /logs security: krb5
export: /private/oracle1/data mount: /data security: krb5p
export: /private/oracle1/archive mount: /archive security: sys
export: /private/oracle1/data1 mount: /data1
security_default: krb5i
```

Related Topics

- [About Direct NFS Client Mounts to NFS Storage Devices](#)
Direct NFS (dNFS) Client integrates the NFS client functionality directly in the Oracle software to optimize the I/O path between Oracle and the NFS server. This integration can provide significant performance improvements.

Enabling and Disabling Direct NFS Client Control of NFS

By default, Direct NFS Client is installed in a disabled state with single-instance Oracle Database installations. Before enabling Direct NFS, you must configure an oranfstab file.

Use these procedures to enable or disable Direct NFS Client Oracle Disk Manager Control of NFS

Enabling Direct NFS Client Control of NFS

1. Change the directory to \$ORACLE_HOME/rdbms/lib.
2. Enter the following command:

```
make -f ins_rdbms.mk dnfs_on
```

Disabling Direct NFS Client Control of NFS

1. Log in as the Oracle software installation owner, and disable Direct NFS Client using the following commands:

```
cd $ORACLE_HOME/rdbms/lib
make -f ins_rdbms.mk dnfs_off
```

2. Remove the oranfstab file.

Note

If you remove an NFS path that an Oracle Database is using, then you must restart the database for the change to take effect.

Enabling Hybrid Columnar Compression on Direct NFS Client

Perform these steps to enable Hybrid Columnar Compression (HCC) on Direct NFS Client:

1. Ensure that SNMP is enabled on the ZFS storage server. For example:

```
$ snmpget -v1 -c public server_name .1.3.6.1.4.1.42.2.225.1.4.2.0  
SNMPv2-SMI::enterprises.42.2.225.1.4.2.0 = STRING: "Sun Storage 7410"
```

2. If SNMP is enabled on an interface other than the NFS server, then configure `orantstab` using the `management` parameter.
3. If SNMP is configured using a community string other than `public`, then configure `orantstab` file using the `community` parameter.
4. Ensure that `libnetsnmp.so` is installed by checking if `snmpget` is available.

Configuring Storage for Oracle Grid Infrastructure for a Standalone Server

Complete these procedures to use Oracle Grid Infrastructure for a standalone server, which includes Oracle Automatic Storage Management (Oracle ASM).

Oracle Grid Infrastructure for a standalone server, also known as Oracle Restart, provides system support for a single-instance Oracle Database. Oracle ASM is a volume manager and a file system for Oracle database files that supports single-instance Oracle Database and Oracle Real Application Clusters (Oracle RAC) configurations. Oracle Automatic Storage Management also supports a general purpose file system for your application needs, including Oracle Database binaries. Oracle Automatic Storage Management is Oracle's recommended storage management solution. It provides an alternative to conventional volume managers and file systems.

Note

- If you want to use Oracle ASM or Oracle Restart, then you must install Oracle Grid Infrastructure for a standalone server before you install and create the database. Otherwise, you must manually register the database with Oracle Restart.
- You can neither install Oracle Restart on an Oracle Grid Infrastructure cluster member node, nor add an Oracle Restart server to an Oracle Grid Infrastructure cluster member node. Oracle Restart supports single-instance databases on one server, while Oracle Grid Infrastructure for a Cluster supports single-instance or Oracle RAC databases on a cluster.

- [Configuring Storage for Oracle Automatic Storage Management](#)
Identify storage requirements and Oracle ASM disk group options.
- [Configuring Storage Device Path Persistence Using Oracle ASMFD](#)
Oracle ASM Filter Driver (Oracle ASMFD) maintains storage file path persistence and helps to protect files from accidental overwrites.
- [Configuring Disk Devices for Oracle ASM on Oracle Solaris](#)
Complete these tasks to configure disk devices for use with Oracle Automatic Storage Management (Oracle ASM).
- [Creating DAS or SAN Disk Partitions for Oracle Automatic Storage Management](#)
You can use direct-attached storage (DAS) and storage area network (SAN) disks with Oracle ASM.
- [Creating Directories for Oracle Database Files](#)
You can store Oracle Database and recovery files on a separate file system from the configuration files.
- [Creating Files on a NAS Device for Use with Oracle Automatic Storage Management](#)
If you have a certified NAS storage device, then you can create zero-padded files in an NFS mounted directory and use those files as disk devices in an Oracle ASM disk group.

Configuring Storage for Oracle Automatic Storage Management

Identify storage requirements and Oracle ASM disk group options.

Note

Oracle ASM Filter Driver (ASMFDD) is deprecated on both Linux and Oracle Solaris beginning with Oracle Database 19c. It will be desupported in a future release.

- [Identifying Storage Requirements for Oracle Automatic Storage Management](#)
To identify the storage requirements for using Oracle ASM, you must determine the number of devices and the amount of free disk space that you require.
- [Oracle ASM Disk Space Requirements](#)
Determine the total amount of Oracle Automatic Storage Management (Oracle ASM) disk space that you require for the database files and recovery files.
- [ASM Disk Group Options for Installation](#)
Plan how you want to configure Oracle ASM disk groups for deployment.
- [Using an Existing Oracle ASM Disk Group](#)
Use Oracle Enterprise Manager Cloud Control or the Oracle ASM command line tool (asmcmd) to identify existing disk groups, and to determine if sufficient space is available in the disk group.

Related Topics

- [Oracle Automatic Storage Management Filter Driver \(ASMFDD\) \(Doc ID 2806979.1\)](#)
- *Oracle Automatic Storage Management Administrator's Guide*

Identifying Storage Requirements for Oracle Automatic Storage Management

To identify the storage requirements for using Oracle ASM, you must determine the number of devices and the amount of free disk space that you require.

To complete this task, perform the following steps:

1. Determine whether you want to use Oracle ASM for Oracle Database files, recovery files, or both. Oracle Database files include data files, control files, redo log files, the server parameter file, and the password file.

During the database installation, you have the option to select either a file system or Oracle ASM as the storage mechanism for Oracle Database files. Similarly, you also have the option to select either a file system or Oracle ASM as the storage mechanism for your recovery files.

Note

You do not have to use the same storage mechanism for both Oracle Database files and recovery files. You can use a file system for one file type and Oracle ASM for the other.

If you select Oracle ASM as your storage option for Oracle Database files, then depending on your choice in the Specify Recovery Options screen, you have the following recovery options:

- If you select the Oracle ASM option for your recovery files, then Oracle Universal Installer provides you with only the option to use the same disk group for both Oracle Database files and recovery files.
 - If you decide not to enable recovery during the database installation, then, after the database installation, you can modify the `DB_RECOVERY_FILE_DEST` parameter to enable the fast recovery area.
2. Choose the Oracle ASM redundancy level to use for each Oracle ASM disk group that you create.

The redundancy level that you choose for the Oracle ASM disk group determines how Oracle ASM mirrors files in the disk group and determines the number of disks and amount of disk space that you require, as follows:

- **External redundancy**

This option does not allow Oracle ASM to mirror the contents of the disk group. Oracle recommends that you select this redundancy level either when the disk group contains devices, such as RAID devices, that provide their own data protection or when the database does not require uninterrupted access to data.

- **Normal redundancy**

To optimize performance and reliability in a normal redundancy disk group, Oracle ASM uses two-way mirroring for data files and three-way mirroring for control files, by default. In addition, you can choose the mirroring characteristics for individual files in a disk group.

A normal redundancy disk group requires a minimum of two failure groups (or two disk devices) if you are using two-way mirroring. The effective disk space in a normal redundancy disk group is half the sum of the disk space of all of its devices.

For most installations, Oracle recommends that you use normal redundancy disk groups. On Oracle Exadata, Oracle recommends that you use high redundancy disk groups for added protection against failure.

- **High redundancy**

The contents of the disk group are three-way mirrored by default. To create a disk group with high redundancy, you must specify at least three failure groups (a minimum of three devices).

Although high-redundancy disk groups provide a high level of data protection, you must consider the higher cost of additional storage devices before deciding to use this redundancy level.

- **Flex redundancy**

A flex redundancy disk group is a new disk group type with features such as flexible file redundancy, mirror splitting, and redundancy change. A flex disk group can consolidate files with different redundancy requirements into a single disk group. It also provides the capability for databases to change the redundancy of its files.

For database data, you can choose no mirroring (unprotected), two-way mirroring (mirrored), or three-way mirroring (high). A flex redundancy disk group requires a minimum of three disk devices (or three failure groups).

- **Extended redundancy**

Extended redundancy disk group has features similar to the flex redundancy disk group. Extended redundancy is available when you configure an Oracle Extended

Cluster. Extended redundancy extends Oracle ASM data protection to cover failure of sites by placing enough copies of data in different failure groups of each site.

3. Determine the total amount of disk space that you require for the database files and recovery files.

If an Oracle ASM instance is running on the system, then you can use an existing disk group to meet these storage requirements. If necessary, you can add disks to an existing disk group during the database installation.

See, "Oracle ASM Disk Space Requirements" in *Oracle Database Installation Guide* for the Oracle ASM disk space requirements.

Note

- The disk devices must be owned by the user performing the grid installation.
Check with your system administrator to determine if the disks used by Oracle ASM are mirrored at the storage level. If so, select External for the redundancy. If the disks are not mirrored at the storage level, then select Normal for the redundancy.
- Every Oracle ASM disk is divided into allocation units (AU). An allocation unit is the fundamental unit of allocation within a disk group. You can select the AU Size value from 1, 2, 4, 8, 16, 32 or 64 MB, depending on the specific disk group compatibility level. The default value is 4 MB for flex disk group and 1 MB for all other disk group types. On engineered systems, the default value is 4 MB.

4. Optionally, identify failure groups for the Oracle ASM disk group devices.

If you intend to use a normal, high or flex redundancy disk group, then you can further protect your database against hardware failure by associating a set of disk devices in a custom failure group. By default, each device comprises its own failure group. However, if two disk devices in a normal redundancy disk group are attached to the same Host Bus Adapter (HBA), then the disk group becomes unavailable if the controller fails. The controller in this example is a single point of failure.

To protect against failures of this type, use two HBAs, each with two disks, and define a failure group for the disks attached to each controller. This configuration enables the disk group to tolerate the failure of one HBA.

Consider the following guidelines while defining custom failure groups:

- You can specify custom failure groups in the **Create ASM Disk Group** screen during an Oracle Grid Infrastructure installation.
 - You can also define custom failure groups after installation, using the GUI tool ASMCA, the command line tool `asmcmd`, or SQL commands.
 - If you define custom failure groups, then for failure groups containing database files only, you must specify a minimum of two failure groups for normal redundancy disk groups and three failure groups for high redundancy disk groups.
5. If you are sure that a suitable disk group does not exist on the system, then install or identify appropriate disk devices to add to a new disk group.

Use the following guidelines when identifying appropriate disk devices:

- The disk devices must be owned by the user performing the Oracle Grid Infrastructure installation.

- All the devices in an Oracle ASM disk group must be the same size and have the same performance characteristics.
- Do not specify multiple partitions on a single physical disk as a disk group device. Oracle ASM expects each disk group device to be on a separate physical disk.
- Although you can specify a logical volume as a device in an Oracle ASM disk group, Oracle does not recommend their use because it adds a layer of complexity that is unnecessary with Oracle ASM. Oracle recommends that if you choose to use a logical volume manager, then use the logical volume manager to represent a single logical unit number (LUN) without striping or mirroring, so that you can minimize the effect on storage performance of the additional storage layer.

Related Topics

- [Oracle ASM Disk Space Requirements](#)
Determine the total amount of Oracle Automatic Storage Management (Oracle ASM) disk space that you require for the database files and recovery files.
- *Oracle Automatic Storage Management Administrator's Guide*

Oracle ASM Disk Space Requirements

Determine the total amount of Oracle Automatic Storage Management (Oracle ASM) disk space that you require for the database files and recovery files.

Table 9-1 Oracle ASM Disk Number and Minimum Space Requirements for an Oracle database (non-CDB)

Redundancy Level	Minimum Number of Disks	Data Files	Recovery Files	Both File Types
External	1	2.5 GB	7.5 GB	10 GB
Normal or Flex with two-way mirroring	2	5.2 GB	15.6 GB	20.8 GB
High or Flex with three-way mirroring	3	7.6 GB	22.8 GB	30.4 GB

Table 9-2 Oracle ASM Disk Number and Minimum Space Requirements for a multitenant container database (CDB) with one pluggable database (PDB)

Redundancy Level	Minimum Number of Disks	Data Files	Recovery Files	Both File Types
External	1	4 GB	12 GB	16 GB
Normal or Flex with two-way mirroring	2	8 GB	24 GB	32 GB
High or Flex with three-way mirroring	3	12 GB	36 GB	48 GB

Note

- If an Oracle ASM instance is running on the system, then you can use an existing disk group to meet these storage requirements. If necessary, you can add disks to an existing disk group during the database installation.
- The disk devices must be owned by the user performing the grid installation.

Check with your system administrator to determine if the disks used by Oracle ASM are mirrored at the storage level. If so, select External for the redundancy. If the disks are not mirrored at the storage level, then select Normal for the redundancy.

- Every Oracle ASM disk is divided into allocation units (AU). An allocation unit is the fundamental unit of allocation within a disk group. You can select the AU Size value from 1, 2, 4, 8, 16, 32 or 64 MB, depending on the specific disk group compatibility level. The default value is 4 MB for flex disk group and 1 MB for all other disk group types. On engineered systems, the default value is 4 MB.

ASM Disk Group Options for Installation

Plan how you want to configure Oracle ASM disk groups for deployment.

During Oracle Grid Infrastructure installation, you can create one Oracle ASM disk group. After the Oracle Grid Infrastructure installation, you can create additional disk groups using Oracle Automatic Storage Management Configuration Assistant (Oracle ASMCA), SQL*Plus, or Automatic Storage Management Command-Line Utility (ASMCMD).

Related Topics

- *Oracle Automatic Storage Management Administrator's Guide*

Using an Existing Oracle ASM Disk Group

Use Oracle Enterprise Manager Cloud Control or the Oracle ASM command line tool (asmcmd) to identify existing disk groups, and to determine if sufficient space is available in the disk group.

1. Connect to the Oracle ASM instance and start the instance if necessary:

```
$ $ORACLE_HOME/bin/asmcmd  
ASMCMD> startup
```

2. Enter one of the following commands to view the existing disk groups, their redundancy level, and the amount of free disk space in each one:

```
ASMCMD> lsdg
```

or

```
$ORACLE_HOME/bin/asmcmd -p lsdg
```

The `lsdg` command lists information about mounted disk groups only.

3. From the output, identify a disk group with the appropriate redundancy level and note the free space that it contains.
4. If necessary, install or identify the additional disk devices required to meet the storage requirements for your installation.

Note

If you are adding devices to an existing disk group, then Oracle recommends that you use devices that have the same size and performance characteristics as the existing devices in that disk group.

Configuring Storage Device Path Persistence Using Oracle ASMFD

Oracle ASM Filter Driver (Oracle ASMFD) maintains storage file path persistence and helps to protect files from accidental overwrites.

Note

Oracle ASM Filter Driver (ASMFD) is deprecated on both Linux and Oracle Solaris beginning with Oracle Database 19c. It will be desupported in a future release.

- [About Oracle ASM with Oracle ASM Filter Driver](#)
During Oracle Grid Infrastructure installation, you can choose to install and configure Oracle Automatic Storage Management Filter Driver (Oracle ASMFD). Oracle ASMFD helps prevent corruption in Oracle ASM disks and files within the disk group.
- [Guidelines for Installing Oracle ASMFD on Oracle Solaris](#)
Review these best practices for Oracle Automatic Storage Management Filter Driver (Oracle ASMFD).

Related Topics

- [Oracle Automatic Storage Management Filter Driver \(ASMFD\) \(Doc ID 2806979.1\)](#)

About Oracle ASM with Oracle ASM Filter Driver

During Oracle Grid Infrastructure installation, you can choose to install and configure Oracle Automatic Storage Management Filter Driver (Oracle ASMFD). Oracle ASMFD helps prevent corruption in Oracle ASM disks and files within the disk group.

Oracle ASM Filter Driver (Oracle ASMFD) rejects write I/O requests that are not issued by Oracle software. This write filter helps to prevent users with administrative privileges from inadvertently overwriting Oracle ASM disks, thus preventing corruption in Oracle ASM disks and files within the disk group. For disk partitions, the area protected is the area on the disk managed by Oracle ASMFD, assuming the partition table is left untouched by the user.

Oracle ASMFD simplifies the configuration and management of disk devices by eliminating the need to rebind disk devices used with Oracle ASM each time the system is restarted.

If Oracle ASMLIB exists on your Linux system, then deinstall Oracle ASMLIB before installing Oracle Grid Infrastructure, so that you can choose to install and configure Oracle ASMFD during an Oracle Grid Infrastructure installation.

 Caution

When you configure Oracle ASM, including Oracle ASMFD, do not modify or erase the contents of the Oracle ASM disks, or modify any files, including the configuration files.

Related Topics

- *Oracle Automatic Storage Management Administrator's Guide*

Guidelines for Installing Oracle ASMFD on Oracle Solaris

Review these best practices for Oracle Automatic Storage Management Filter Driver (Oracle ASMFD).

On Oracle Solaris systems, consider the following guidelines before you install Oracle ASMFD:

- Ensure that you label the disk as either SMI or Extensible Firmware Interface (EFI).
- Ensure that the disk has at least one slice that represents the entire disk. For example, Slice 2.
- Ensure that the slices on the disk do not overlap.

Configuring Disk Devices for Oracle ASM on Oracle Solaris

Complete these tasks to configure disk devices for use with Oracle Automatic Storage Management (Oracle ASM).

1. If necessary, install the disks that you intend to use for the disk group and restart the system.
2. Identify devices that are part of a logical volume manager (LVM) disk group:

This command displays information about VERITAS Volume Manager disks. If you use a different LVM, then refer to the appropriate documentation for information about determining which disk devices it is managing.

```
# vxdiskconfig
# /usr/sbin/vxdisk list
```

If this command displays disk group information associated with a disk device, then the disk is already part of an LVM disk group. Do not use disks that are part of an LVM disk group.

3. Create or identify the disk slices (partitions) that you want to include in the Oracle Automatic Storage Management disk group:
 - a. List the disks attached to the system:

```
# /usr/sbin/format
```

The output from this command is similar to the following:

```
AVAILABLE DISK SELECTIONS:
AVAILABLE DISK SELECTIONS:
  0. c0t0d0 < ST34321A cyl 8892 alt 2 hd 15 sec 63 >
    /pci@1f,0/pci@1,1/ide@3/dad@0,0
  1. c1t5d0 < SUN9.0G cyl 4924 alt 2 hd 27 sec 133 >
    /pci@1f,0/pci@1/scsi@1/sd@5,0
```

This command displays information about each disk attached to the system, including the device name. For example, cxytdz.

- b. Enter the number corresponding to the disk that you want to use.
- c. Use the `fdisk` command to create an Oracle Solaris partition on the disk if one does not already exist.

Oracle Solaris `fdisk` partitions must start at cylinder 1, not cylinder 0. If you create an `fdisk` partition, then you must label the disk before continuing.

- d. Enter the `partition` command, followed by the `print` command to display the partition table for the disk that you want to use.
- e. If necessary, create a single whole-disk slice, starting at cylinder 1.

Note

To prevent Automatic Storage Management from overwriting the partition table, you cannot use slices that start at cylinder 0 (for example, slice 2).

- f. Make a note of the number of the slice that you want to use.
 - g. If you modified a partition table or created a new one, then enter the `label` command to write the partition table and label to the disk.
 - h. Enter **q** to return to the **format** menu.
 - i. If you have finished creating slices, then enter **q** to quit the `format` utility. Else, enter the `disk` command to select a new disk and repeat steps *b* to *g* to create or identify the slices on that disks.
4. If you plan to use existing slices, then verify that they are not mounted as file systems:

```
# df -h /tmp
```

This command displays information about the slices on disk devices that are mounted as file systems. The device name for a slice includes the disk device name followed by the slice number, for example cxytdzsn, where sn is the slice number.

- 5. On every node, change the owner, group, and permissions on the file for each disk slice that you want to add to a disk group:

```
# chown grid:asmadmin /dev/rdisk/cxytdzs6
# chmod 660 /dev/rdisk/cxytdzs6
```

In this example, the device name specifies slice 6.

Note

If you are using a multi-pathing disk driver with Oracle Automatic Storage Management, then ensure that you set the permissions only on the correct logical device name for the disk.

Creating DAS or SAN Disk Partitions for Oracle Automatic Storage Management

You can use direct-attached storage (DAS) and storage area network (SAN) disks with Oracle ASM.

To use a DAS or SAN disk in Oracle ASM, Oracle recommends that the disk have a partition table. Oracle recommends creating exactly one partition for each disk.

Creating Directories for Oracle Database Files

You can store Oracle Database and recovery files on a separate file system from the configuration files.

Perform this procedure to place the Oracle Database or recovery files on a separate file system from the Oracle base directory:

1. Use the following command to determine the free disk space on each mounted file system:

```
# df -h
```

2. Identify the file systems to use, from the display:

Option	Description
Database Files	Select one of the following: • A single file system with at least 1.5 GB of free disk space • Two or more file systems with at least 3.5 GB of free disk space in total
Recovery Files	Choose a file system with at least 2 GB of free disk space

If you are using the same file system for multiple file types, then add the disk space requirements for each type to determine the total disk space requirement.

3. Note the names of the mount point directories for the file systems that you identified.
4. If the user performing installation has permissions to create directories on the disks where you plan to install Oracle Database, then Oracle DBCA creates the Oracle Database file directory, and the Recovery file directory. If the user performing installation does not have write access, then you must create these directories manually.

For example, given the user `oracle` and Oracle Inventory Group `oinstall`, and using the paths `/u03/oradata/wrk_area` for Oracle Database files, and `/u01/oradata/rcv_area` for the recovery area, these commands create the recommended subdirectories in each of the mount point directories and set the appropriate owner, group, and permissions on them:

- Database file directory:

```
# mkdir -p /u01/oradata/  
# chown oracle:oinstall /u01/oradata/  
# chmod 775 /u01/oradata
```

The default location for the database file directory is \$ORACLE_BASE/oradata.

- Recovery file directory (fast recovery area):

```
# mkdir -p /u01/oradata/rcv_area  
# chown oracle:oinstall /u01/oradata/rcv_area  
# chmod 775 /u01/oradata/rcv_area
```

The default fast recovery area is \$ORACLE_BASE/fast_recovery_area.

Oracle recommends that you keep the fast recovery area on a separate physical disk than that of the database file directory. This method enables you to use the fast recovery area to retrieve data if the disk containing oradata is unusable for any reason.

Creating Files on a NAS Device for Use with Oracle Automatic Storage Management

If you have a certified NAS storage device, then you can create zero-padded files in an NFS mounted directory and use those files as disk devices in an Oracle ASM disk group.

Ensure that you specify the ASM discovery path for Oracle ASM disks.

During installation of Oracle Grid Infrastructure, Oracle Universal Installer (OUI) can create files in the NFS mounted directory you specify. The following procedure explains how to manually create files in an NFS mounted directory to use as disk devices in an Oracle ASM disk group:

1. If necessary, create an exported directory for the disk group files on the NAS device.
2. Switch user to root.
3. Create a mount point directory on the local system.

For example:

```
# mkdir -p /mnt/oracleasm
```

4. To ensure that the NFS file system is mounted when the system restarts, add an entry for the file system in the mount file /etc/fstab.
5. Enter a command similar to the following to mount the NFS on the local system:

```
# mount /mnt/oracleasm
```

6. Choose a name for the disk group to create, and create a directory for the files on the NFS file system, using the disk group name as the directory name.

For example, if you want to set up a disk group for a sales database:

```
# mkdir /mnt/oracleasm/sales1
```

7. Use commands similar to the following to create the required number of zero-padded files in this directory:

```
# dd if=/dev/zero  
of=/mnt/oracleasm/sales1/disk1 bs=1024k  
count=1000
```

This example creates 1 GB files on the NFS file system. You must create one, two, or three files respectively to create an external, normal, or high redundancy disk group.

Note

Creating multiple zero-padded files on the same NAS device does not guard against NAS failure. Instead, create one file for each NAS device and mirror them using the Oracle ASM technology.

8. Enter commands similar to the following to change the owner, group, and permissions on the directory and files that you created:

```
# chown -R grid:asmadmin /mnt/oracleasm  
# chmod -R 660 /mnt/oracleasm
```

In this example, the installation owner is grid and the OSASM group is asmadmin.

9. During Oracle Database installations, edit the Oracle ASM disk discovery string to specify a regular expression that matches the file names you created.

For example:

```
/mnt/oracleasm/sales1/
```

Related Topics

- [My Oracle Support Note 359515.1](#)

Installing and Configuring Oracle Grid Infrastructure for a Standalone Server

Oracle Grid Infrastructure for a standalone server includes Oracle Restart and Oracle Automatic Storage Management.

If you install Oracle Grid Infrastructure for a standalone server and then create your database, then the database is automatically added to the Oracle Restart configuration. Oracle Restart automatically restarts the database when required.

If you install Oracle Grid Infrastructure for a standalone server on a host computer on which a database already exists, then you must manually add the database, the listener, the Oracle ASM instance, and other components to the Oracle Restart configuration before you are able to configure automatic database restarts.

Note

Oracle Grid Infrastructure for a standalone server can support multiple single-instance databases on a single host computer.

- [About Image-Based Oracle Grid Infrastructure Installation](#)
Installation and configuration of Oracle Grid Infrastructure software is simplified with image-based installation.
- [Setup Wizard Installation Options for Creating Images](#)
Before you start the setup wizards for your Oracle Database or Oracle Grid Infrastructure installation, decide if you want to use any of the available image-creation options.
- [Installing Oracle Grid Infrastructure for a Standalone Server](#)
Complete these steps to install Oracle Grid Infrastructure for a standalone server and then create a database that is managed by Oracle Restart.
- [Installing Oracle Grid Infrastructure for a Standalone Server for an Existing Database](#)
Follow the high-level instructions in this section to install Oracle Grid Infrastructure for a standalone server and configure it for an existing Oracle Database.
- [Installing Oracle Grid Infrastructure for a Standalone Server Using a Software-Only Installation](#)
A software-only installation only installs the Oracle Grid Infrastructure for a standalone server binaries at the specified location. You must complete a few manual configuration steps to enable Oracle Restart after you install the software.
- [Testing the Oracle Automatic Storage Management Installation](#)
After installing Oracle Grid Infrastructure for a single instance, use the ASMCMD command-line utility to test the Oracle ASM installation.
- [Relinking Oracle Restart and Oracle ASM Binaries](#)
Relink the Oracle Restart and Oracle ASM binaries every time you apply an operating system patch or after an operating system upgrade.

- [Configuring Oracle ASM Disk Groups Manually using Oracle ASMCA](#)
After installing Oracle Grid Infrastructure for a standalone server, you can also use Oracle ASMCA to create and configure disk groups, Oracle ADVM, and Oracle ACFS.
- [Enabling Oracle ACFS on Oracle Restart Configurations](#)
You must enable root access to use Oracle ACFS.
- [Applying Patches During an Oracle Grid Infrastructure Installation or Upgrade](#)
Starting with Oracle Grid Infrastructure 18c, you can download and apply Release Updates (RUs) and one-off patches during an Oracle Grid Infrastructure installation or upgrade.
- [Patching and Switching Oracle Grid Infrastructure Homes](#)
Perform an out-of-place Oracle Restart patching by switching from the current Oracle Grid Infrastructure home to a patched Oracle Grid Infrastructure home.

About Image-Based Oracle Grid Infrastructure Installation

Installation and configuration of Oracle Grid Infrastructure software is simplified with image-based installation.

To install Oracle Grid Infrastructure, create the new Grid home with the necessary user group permissions, and then extract the image file into the newly-created Grid home, and run the setup wizard to register the Oracle Grid Infrastructure product.

Using image-based installation, you can do the following:

- Configure Oracle Grid Infrastructure for a new cluster.
- Configure Oracle Grid Infrastructure for a standalone server (Oracle Restart).
- Upgrade Oracle Grid Infrastructure.
- Setup software only.
- Add or remove nodes from your existing cluster, if the Oracle Grid Infrastructure software is already installed or configured.

This installation feature streamlines the installation process and supports automation of large-scale custom deployments. You can also use this installation method for deployment of customized images, after you patch the base-release software with the necessary Release Updates (RUs) or Release Update Revisions (RURs).

Note

You must extract the image software into the directory where you want your Grid home to be located, and then run the `ORACLE_HOME\gridSetup.sh` script to start the Oracle Grid Infrastructure Setup Wizard. Ensure that the Grid home directory path you create is in compliance with the Oracle Optimal Flexible Architecture recommendations.

Setup Wizard Installation Options for Creating Images

Before you start the setup wizards for your Oracle Database or Oracle Grid Infrastructure installation, decide if you want to use any of the available image-creation options.

In image-based installations, you can start your Oracle Database installation or Oracle Grid Infrastructure installations by running the setup wizards `runInstaller` and `gridSetup.sh` respectively. Both these wizards come with the following image-creation options.

Table 10-1 Image-Creation Options for Setup Wizard

Option	Description
-createGoldImage	Creates a gold image from the current Oracle home.
-destinationLocation	Specify the complete path, or location, where the gold image will be created.
-exclFiles	Specify the complete paths to the files to be excluded from the newly created gold image.
-help	Displays help for all the available options.

For example:

```
cd $ORACLE_HOME
./runInstaller -createGoldImage -destinationLocation /tmp/my_db_images -
exclFiles /u01/app/oracle/product/19.0.0/dbhome_1/relnotes
```

```
cd Grid_home
./gridSetup.sh -createGoldImage -destinationLocation /tmp/my_grid_images -
exclFiles /u01/app/oracle/product/19.0.0/dbhome_1/relnotes
```

Where:

/tmp/my_db_images is a temporary file location where the image zip file is created.

/tmp/my_grid_images is a temporary file location where the image zip file is created.

/u01/app/oracle/product/19.0.0/dbhome_1/relnotes is the file to be excluded from the newly created gold image.

Refer to runInstaller -createGoldImage and gridSetup.sh -createGoldImage for more examples.

Installing Oracle Grid Infrastructure for a Standalone Server

Complete these steps to install Oracle Grid Infrastructure for a standalone server and then create a database that is managed by Oracle Restart.

Note

Oracle ASM Filter Driver (ASMFDD) is deprecated on both Linux and Oracle Solaris beginning with Oracle Database 19c. It will be desupported in a future release.

Install Oracle Grid Infrastructure for a standalone server, which installs Oracle Restart and Oracle ASM, and creates one disk group.

You should have your network information, storage information, and operating system users and groups available to you before you start the installation. You should also be prepared to run root scripts or provide information to automate root scripts.

1. Log in as the Oracle Restart software owner user (oracle).

2. If this is the first time you are installing Oracle software, then create the Oracle base and the Oracle inventory directories as per the Oracle Optimal Flexible Architecture (OFA) recommendations. Specify the correct owner, group, and permissions for these directories.

```
# mkdir -p /u01/app/oracle
# mkdir -p /u01/app/oraInventory
# chown -R oracle:oinstall /u01/app/oracle
# chown -R oracle:oinstall /u01/app/oraInventory
# chmod -R 775 /u01/app
```

3. Download the Oracle Grid Infrastructure for a standalone server installation image files, create the grid home directory, and extract the image files in this grid home directory.

For example:

```
$ mkdir -p /u01/app/oracle/product/19.0.0/grid
$ cd /u01/app/oracle/product/19.0.0/grid
$ unzip -q /tmp/grid_home.zip
```

Note

- Ensure that the Grid home directory path you create is in compliance with the Oracle Optimal Flexible Architecture recommendations. Also, unzip the installation image files only in this Grid home directory that you created.
- Oracle home or Oracle base cannot be symlinks, nor can any of their parent directories, all the way to up to the root directory.

4. Log in as the Oracle Restart software owner user and run `gridSetup.sh` to start the Oracle Grid Infrastructure setup wizard:

```
$ Grid_home/gridSetup.sh
```

Where *Grid_home* is the Oracle Grid Infrastructure home directory.

Note

You can use the `gridSetup.sh` command with the `-applyRU` and `-applyOneOffs` options to install Release Updates (RUs) and one-off patches during an Oracle Grid Infrastructure installation or upgrade.

5. In the Select Configuration Option screen, select the **Configure Oracle Grid Infrastructure for a Standalone Server (Oracle Restart)** option to install and configure Oracle Restart and Oracle ASM. Click **Next**.
6. During installation, disk paths mounted on Oracle ASM with the string `ORCL:*` are listed as default database storage candidate disks.
7. Configure Oracle ASM as needed with additional disk groups.
 - The default Disk Group Name is DATA. You can enter a new name for the disk group, or use the default name.
 - Any additional disk devices that you create must be owned by the user performing the grid installation.

8. Respond to the configuration prompts as needed to configure Oracle Grid Infrastructure. Click **Help** for information.
9. Provide information to automate root scripts, or run scripts as root when prompted by OUI.
If you configure automation for running root scripts, and a root script fails, then you can fix the problem manually, and click **Retry** to run the root script again.
10. Start the Oracle Database installation, and select Oracle ASM disk groups for Oracle Database files storage. For assistance during installation, click **Help** on the Oracle Universal Installer page where you need more information.

Related Topics

- *Oracle Automatic Storage Management Administrator's Guide*

Installing Oracle Grid Infrastructure for a Standalone Server for an Existing Database

Follow the high-level instructions in this section to install Oracle Grid Infrastructure for a standalone server and configure it for an existing Oracle Database.

Oracle Restart can manage resources from the same release and releases up to one version lower than Oracle Restart. For instance, you can install Oracle Grid Infrastructure for a standalone server 19c (Oracle Restart) to provide services for Oracle Database 19c and Oracle Database 18c. Earlier Oracle Database releases can coexist on the same server without being managed by Oracle Restart.

To install Oracle Grid Infrastructure for a standalone server for a database that is already installed:

1. Log in as the Oracle Database software owner user.
2. Set the ORACLE_HOME and ORACLE_SID environment variables.

Bourne, Bash or Korn shell:

```
$ ORACLE_SID=orcl
$ export ORACLE_SID
$ ORACLE_HOME=/u01/app/oracle/product/19.0.0/dbhome_1
$ export ORACLE_HOME
```

C shell:

```
% setenv ORACLE_SID orcl
% setenv ORACLE_HOME /u01/app/oracle/product/19.0.0/dbhome_1
```

3. Go to the Oracle home and stop any existing database listeners.

```
$ $ORACLE_HOME/lsnrctl stop listener_name
```

Note

If this command throws an error, reload the listener using `lsnrctl reload LISTENER`

4. On the same host computer as the database, install Oracle Grid Infrastructure for a standalone server, and select **Configure Oracle Grid Infrastructure for a Standalone Server (Oracle Restart)** as the installation option. See, “Installing Oracle Grid Infrastructure for a Standalone Server with a New Database Installation” in *Oracle Database Installation Guide*.

The Oracle Grid Infrastructure for a standalone server components are installed in an Oracle Grid Infrastructure Oracle home (Grid home), which is in a different location from existing Oracle Database homes.

5. If you have an existing Oracle Database, then register it for High Availability with Oracle Restart using the `srvctl` command:

```
$ cd $ORACLE_HOME/bin
$ srvctl add database -db dbname -o oracle_home_path
```

Related Topics

- [Installing Oracle Grid Infrastructure for a Standalone Server with a New Database Installation](#)
- *Oracle Database Administrator's Guide*

Installing Oracle Grid Infrastructure for a Standalone Server Using a Software-Only Installation

A software-only installation only installs the Oracle Grid Infrastructure for a standalone server binaries at the specified location. You must complete a few manual configuration steps to enable Oracle Restart after you install the software.

- [About Oracle Grid Infrastructure Software-Only Installations](#)
Manually installing and configuring the software binaries for Oracle Grid Infrastructure.
- [Installing Software Binaries for Oracle Grid Infrastructure for a Standalone Server](#)
Use this procedure to do a software-only installation of Oracle Grid Infrastructure for a standalone server.
- [Configuring Software Binaries for Oracle Grid Infrastructure for a Standalone Server](#)
Use these steps to configure and activate a software-only Oracle Grid Infrastructure for a standalone server installation for Oracle Restart.

About Oracle Grid Infrastructure Software-Only Installations

Manually installing and configuring the software binaries for Oracle Grid Infrastructure.

Oracle recommends that only advanced users perform software-only installations, because this installation method provides no validation of the installation, and this installation option requires manual postinstallation steps to enable the Oracle Grid Infrastructure for a standalone server software.

Performing a software-only installation requires the following steps:

1. Installing the software binaries.
2. Configuring the software binaries.

Installing Software Binaries for Oracle Grid Infrastructure for a Standalone Server

Use this procedure to do a software-only installation of Oracle Grid Infrastructure for a standalone server.

1. Log in as the Oracle Restart software owner user (oracle).
2. Download the Oracle Grid Infrastructure for a standalone server installation image files, create the Grid home directory, and extract the image files in this Grid home directory.

For example:

```
$ mkdir -p /u01/app/oracle/product/19.0.0/grid
$ chown oracle:oinstall /u01/app/oracle/product/19.0.0/grid
$ cd /u01/app/oracle/product/19.0.0/grid
$ unzip -q /tmp/grid_home.zip
```

Note

Ensure that the Grid home directory path you create is in compliance with the Oracle Optimal Flexible Architecture recommendations. Also, unzip the installation image files only in this Grid home directory that you created.

3. Ensure that you complete all the storage and server preinstallation requirements. Verify that your server meets the installation requirements using the following command:

```
$ cd /u01/app/oracle/product/19.0.0/grid
$ runcluvfy.sh stage -pre hacfg
```

4. Run `gridSetup.sh` to start the Oracle Grid Infrastructure setup wizard:

```
$ Grid_home/gridSetup.sh
```

Where, *Grid_home* is the Oracle Grid Infrastructure home directory.

Note

You must install Oracle Grid Infrastructure for a standalone server from the Oracle Grid Infrastructure media.

5. In the Select Configuration Option screen, select the **Set Up Software Only** option to perform a software-only installation of Oracle Grid Infrastructure for a standalone server. Click **Next**.
6. Respond to the prompts as needed to set up Oracle Grid Infrastructure. Click **Help** for information.
7. The Oracle Grid Infrastructure setup wizard prompts you to run the `root.sh` script and, if required, the `oraInstRoot.sh` script.
8. The `root.sh` script output provides information about how to proceed, depending on the configuration you plan to complete in this installation. Note this information.

Configuring Software Binaries for Oracle Grid Infrastructure for a Standalone Server

Use these steps to configure and activate a software-only Oracle Grid Infrastructure for a standalone server installation for Oracle Restart.

Configuring With Oracle Automatic Storage Management

1. Install the software binaries. See, “Installing Software Binaries for Oracle Grid Infrastructure for a Standalone Server” in *Oracle Database Installation Guide*.
2. Run `gridSetup.sh` to start the Oracle Grid Infrastructure setup wizard.
See, “Installing Oracle Grid Infrastructure for a Standalone Server with a New Database Installation” in *Oracle Database Installation Guide*.

Configuring Without Oracle Automatic Storage Management

1. Log in as root and run the `roothas.sh` script located in the `Grid_home` path, using the following syntax:

```
# cd Grid_home/crs/install
# ./roothas.sh
```

Where, `Grid_home` is the Oracle Grid Infrastructure home directory.

For example:

```
# cd /u01/app/oracle/product/19.0.0/grid/crs/install
# ./roothas.sh
```

2. Change directory to the path `Grid_home/oui/bin`.
3. Log in as the Oracle Restart software owner user and use the following command syntax, where `Grid_home` is the path of the Oracle Grid Infrastructure for a standalone server home.

```
$ ./runInstaller -updateNodeList ORACLE_HOME=Grid_home -defaultHomeName
CLUSTER_NODES= CRS=TRUE
```

For example:

```
$ ./runInstaller -updateNodeList ORACLE_HOME=/u01/app/oracle/product/
19.0.0/grid
-defaultHomeName CLUSTER_NODES= CRS=TRUE
```

4. Use the `SRVCTL` utility along with Oracle Network Configuration Assistant to add the listener to the Oracle Restart configuration.

Note

This procedure does not work for Oracle Restart upgrades from previous releases.

Related Topics

- [Installing Software Binaries for Oracle Grid Infrastructure for a Standalone Server](#)
Use this procedure to do a software-only installation of Oracle Grid Infrastructure for a standalone server.
- [Installing Oracle Grid Infrastructure for a Standalone Server](#)
Complete these steps to install Oracle Grid Infrastructure for a standalone server and then create a database that is managed by Oracle Restart.

Testing the Oracle Automatic Storage Management Installation

After installing Oracle Grid Infrastructure for a single instance, use the ASMCMD command-line utility to test the Oracle ASM installation.

1. Open a shell window, and temporarily set the ORACLE_SID and ORACLE_HOME environment variables to specify the appropriate values for the Oracle ASM instance to use.

For example, if the Oracle ASM SID is named +ASM and the Oracle home is located in the grid subdirectory of the ORACLE_BASE directory, then enter the following commands to create the required settings:

Bourne, Bash or Korn shell:

```
$ ORACLE_SID=+ASM
$ export ORACLE_SID
$ ORACLE_HOME=/u01/app/oracle/product/19.0.0/grid
$ export ORACLE_HOME
```

C shell:

```
% setenv ORACLE_SID +ASM
% setenv ORACLE_HOME /u01/app/oracle/product/19.0.0/grid
```

2. Use ASMCMD to list the disk groups for the Oracle ASM instance:

```
$ORACLE_HOME/bin/asmcmd lsdg
```

If the Oracle ASM instance is running, then ASMCMD connects by default as the SYS user with SYSASM privileges, and is available.

3. If the Oracle ASM instance is not running, then start the Oracle ASM instance using the following command:

```
$ORACLE_HOME/bin/srvctl start asm
```

Related Topics

- *Oracle Automatic Storage Management Administrator's Guide*

Relinking Oracle Restart and Oracle ASM Binaries

Relink the Oracle Restart and Oracle ASM binaries every time you apply an operating system patch or after an operating system upgrade.

 Caution

Before relinking executables, you must shut down all executables that run in the Oracle home directory that you are relinking. In addition, shut down applications linked with Oracle shared libraries.

1. Log in as root and unlock the grid home:

```
# cd Grid_home/crs/install
# roothas.sh -unlock
```

2. Log in as the grid user and relink the binaries:

```
$ export ORACLE_HOME=Grid_home
$ Grid_home/bin/relink
```

3. Log in as root again and perform the following steps:

```
# cd Grid_home/rdbms/install/
# ./rootadd_rdbms.sh
# cd Grid_home/crs/install
# roothas.sh -lock
```

Configuring Oracle ASM Disk Groups Manually using Oracle ASMCA

After installing Oracle Grid Infrastructure for a standalone server, you can also use Oracle ASMCA to create and configure disk groups, Oracle ADVM, and Oracle ACFS.

During Oracle Grid Infrastructure for a standalone server installation, Oracle Automatic Storage Management Configuration Assistant (Oracle ASMCA) utility creates a new Oracle Automatic Storage Management instance if there is no Oracle ASM instance currently configured on the computer. After installation, you can create and configure additional disk groups, and you can configure Oracle ADVM and Oracle ACFS.

To create disk groups or manually configure Oracle ASM disks, start Oracle ASMCA, where *Grid_home* is the path to the Oracle Grid Infrastructure home:

```
$ cd Grid_home/bin
$ ./asmca
```

Related Topics

- *Oracle Automatic Storage Management Administrator's Guide*

Enabling Oracle ACFS on Oracle Restart Configurations

You must enable root access to use Oracle ACFS.

To enable root access, log in as root, navigate to the path *Grid_home/crs/install*, and run the script *roothas.sh -lockacfs*.

Where, *Grid_home* is the Oracle Grid Infrastructure home directory path.

For example:

```
# cd /u01/app/oracle/product/19.0.0/grid/crs/install
# roothas.sh -lockacfs
```

Note

Starting with Oracle Database 12c Release 1 (12.1.0.2), the `roothas.sh` script replaces the `roothas.pl` script in the Oracle Grid Infrastructure home.

Applying Patches During an Oracle Grid Infrastructure Installation or Upgrade

Starting with Oracle Grid Infrastructure 18c, you can download and apply Release Updates (RUs) and one-off patches during an Oracle Grid Infrastructure installation or upgrade.

1. Download the patches you want to apply from My Oracle Support:

<https://support.oracle.com>

2. Select the **Patches and Updates** tab to locate the patch.

Oracle recommends that you select **Recommended Patch Advisor**, and enter the product group, release, and platform for your software.

3. Move the patches to an accessible directory like `/tmp`.

4. Change to the Oracle Grid Infrastructure home directory:

```
$ cd /u01/app/oracle/product/19.0.0/grid
```

5. Apply Release Updates (RUs) and any one-off patches during the installation or upgrade process:

```
$ ./gridSetup.sh -applyRU patch_directory_location -applyOneOffs
comma_seperated_list_of_patch_directory_locations
```

Note

You can apply RUs and one-off patches separately or together in the same command.

6. Complete the remaining steps in the Oracle Grid Infrastructure configuration wizard to complete the installation or upgrade.

Related Topics

- *Oracle Grid Infrastructure Upgrading Oracle Restart for Linux and Unix-Based Operating Systems*

Patching and Switching Oracle Grid Infrastructure Homes

Perform an out-of-place Oracle Restart patching by switching from the current Oracle Grid Infrastructure home to a patched Oracle Grid Infrastructure home.

1. Download the 19.3 Oracle Grid Infrastructure base release image files.

<https://www.oracle.com/database/technologies/oracle-database-software-downloads.html#19c>

2. As the grid user, extract the downloaded image files into a new Oracle Grid Infrastructure home directory.

```
$ mkdir -p /u01/app/oracle/product/19.17.0/grid
$ chown grid:oinstall /u01/app/oracle/product/19.17.0/grid
$ cd /u01/app/oracle/product/19.17.0/grid
$ unzip -q download_location/grid.zip
```

Here:

- /u01/app/oracle/product/19.17.0/grid is the new Grid home.
- /u01/app/oracle/product/19.16.0/grid is the old Grid home.

3. As the grid user, download and install the latest version of the OPatch utility in the new Grid home.

<https://updates.oracle.com/download/6880880.html>

```
$ mv /u01/app/oracle/product/19.17.0/grid/OPatch /u01/app/oracle/product/
19.17.0/grid/bak_OPatch
$ unzip latest_opatch.zip -d /u01/app/oracle/product/19.17.0/grid/
```

4. Download the Oracle Database RU version that you want to apply from [My Oracle Support](#). In this example, Oracle Database 19.17 RU.

For more information, see, [Downloading Release Update Patches](#)

5. Start the Oracle Grid Infrastructure installer to perform a software-only Oracle Restart installation. You can apply the optional -applyRU or -applyOneOff flags to apply Release Updates (RUs) during the installation.

```
$ /u01/app/oracle/product/19.17.0/grid/gridSetup.sh [-applyRU
patch_directory_location]
[-applyOneOffs comma_separated_list_of_patch_directory_locations]
```

6. Follow the steps in the configuration wizard to complete the Oracle Grid Infrastructure installation.
7. As the root user, run the following command to prepare the new home for the out-of-place patching:

```
# /u01/app/oracle/product/19.17.0/grid/crs/install/roothas.sh -prepatch -
dstcrshome
/u01/app/oracle/product/19.17.0/grid
```

This command does not shut down any services.

8. Run the following command to switch to the new Oracle Grid Infrastructure home and perform the out of place patching:

```
# /u01/app/oracle/product/19.17.0/grid/crs/install/roothas.sh -postpatch -  
dstcrshome  
/u01/app/oracle/product/19.17.0/grid
```

This command shuts down the old Oracle Grid Infrastructure home and starts resources from the new Oracle Grid Infrastructure home. All Oracle Grid Infrastructure services start running from the new Grid home.

9. Update the Oracle central inventory (oraInventory).

```
$ /u01/app/oracle/product/19.17.0/grid/oui/bin/runInstaller -  
updateNodeList ORACLE_HOME=/u01/app/oracle/product/19.17.0/grid CRS=TRUE  
$ /u01/app/oracle/product/19.16.0/grid/oui/bin/runInstaller -  
updateNodeList ORACLE_HOME=/u01/app/oracle/product/19.16.0/grid CRS=FALSE
```

10. If the patching fails, then perform the following steps to rollback the patch:

- a. As the root user, run the prepatch script.

```
# /u01/app/oracle/product/19.17.0/grid/crs/install/roothas.sh -prepatch  
-dstcrshome Old_GI_Home -rollback
```

- b. As the root user, run the postpatch script.

```
# /u01/app/oracle/product/19.17.0/grid/crs/install/roothas.sh -  
postpatch -dstcrshome Old_GI_Home -rollback
```

11. If you have successfully switched to the new Grid home on all nodes and want to switch back to the old Grid home, then perform the following steps:

- a. As the root user, run the prepatch script.

```
# Old_GI_Home/crs/install/roothas.sh -prepatch -dstcrshome Old_GI_Home
```

- b. As the grid user, run the postpatch script.

```
# Old_GI_Home/crs/install/roothas.sh -postpatch -dstcrshome Old_GI_Home
```

- c. Update the Oracle central inventory (oraInventory).

```
$ /u01/app/oracle/product/19.16.0/grid/oui/bin/runInstaller -  
updateNodeList ORACLE_HOME=/u01/app/oracle/product/19.16.0/grid CRS=TRUE  
$ /u01/app/oracle/product/19.17.0/grid/oui/bin/runInstaller -  
updateNodeList ORACLE_HOME=/u01/app/oracle/product/19.17.0/grid  
CRS=FALSE
```

Installing Oracle Database

Oracle Database and Oracle Grid Infrastructure installation software is available as image-based zip files and can be installed using several options.

You can download Oracle Database software from the Oracle website or the Oracle Software Delivery Cloud portal. In most cases, you use the graphical user interface (GUI) provided by Oracle Universal Installer to install the software. However, you can also run silent mode installations, without using the GUI. You can also use Oracle Fleet Patching and Provisioning for Oracle Database and Oracle Grid Infrastructure (clusterware) deployments.

Oracle Database software may be available on installation media on-demand.

Note

To install Oracle Database files on Oracle Automatic Storage Management (Oracle ASM), you must first complete an Oracle Grid Infrastructure for a standalone server installation. Oracle Grid Infrastructure for a standalone server includes Oracle Restart and Oracle ASM.

To upgrade an existing Oracle ASM installation, upgrade Oracle ASM by running an Oracle Grid Infrastructure upgrade. If you do not have Oracle ASM installed and you want to use Oracle ASM as your storage option, then you must complete an Oracle Grid Infrastructure for a standalone server installation before you start your Oracle Database installation.

You cannot use Oracle Universal Installer from an earlier Oracle release to install components from this release.

- [About Image-Based Oracle Database Installation](#)
Understand image-based installation to simplify installation and configuration of Oracle Database software.
- [About Deploying Oracle Database Using Oracle Fleet Patching and Provisioning](#)
You can use Oracle Fleet Patching and Provisioning (Oracle FPP) to provision Oracle Database software.
- [Downloading Oracle Software](#)
Select the method you want to use to download the software.
- [About Character Set Selection During Installation](#)
Before you create the database, decide the character set that you want to use.
- [About Automatic Memory Management Installation Options](#)
Decide if you want to configure Automatic Memory Management during installation.
- [Running the Installer in a Different Language](#)
Describes how to run the installer in other languages.
- [Installing the Oracle Database Software](#)
These topics explain how to run Oracle Universal Installer to perform most database installations.

- [Installing Standard Edition High Availability](#)
Learn how to Install high availability on Oracle Database Standard Edition 2.

About Image-Based Oracle Database Installation

Understand image-based installation to simplify installation and configuration of Oracle Database software.

To install Oracle Database, create the new Oracle home, extract the image file into the newly-created Oracle home, and run the setup wizard to register the Oracle Database product.

Using image-based installation, you can install and upgrade Oracle Database for single-instance and cluster configurations. If you install or clone an Oracle Database image, then all Oracle Database options such as Oracle OLAP (o1ap) and Oracle Real Application Testing (rat) are enabled by default.

This installation feature streamlines the installation process and supports automation of large-scale custom deployments. You can also use this installation method for deployment of customized images, after you patch the base-release software with the necessary Release Updates (Updates) or Monthly Recommended Patches (MRPs).

Note

You must extract the image software (db_home.zip) into the directory where you want your Oracle Database home to be located, and then run the Oracle Database Setup Wizard to start the Oracle Database installation and configuration. Oracle recommends that the Oracle home directory path you create is in compliance with the Oracle Optimal Flexible Architecture recommendations.

About Deploying Oracle Database Using Oracle Fleet Patching and Provisioning

You can use Oracle Fleet Patching and Provisioning (Oracle FPP) to provision Oracle Database software.

Starting with Oracle Database 19c, Rapid Home Provisioning is renamed to Oracle Fleet Patching and Provisioning (Oracle FPP).

With Oracle Fleet Patching and Provisioning, you create, store, and manage templates of Oracle homes as images (called gold images) of Oracle software, such as databases, middleware, and applications. You can make a working copy of any gold image and then you can provision that working copy to any node in the data center or cloud computing environment.

You can use Oracle Fleet Patching and Provisioning to provision, patch, and upgrade single-instance databases running on Oracle Restart, on clusters, or on single, standalone nodes. These may be on nodes with or without Oracle Grid Infrastructure installed.

Note

Oracle Fleet Patching and Provisioning is not supported on Microsoft Windows and HP-UX Itanium systems.

Related Topics

- *Oracle Fleet Patching and Provisioning Administrator's Guide*

Downloading Oracle Software

Select the method you want to use to download the software.

You can download Oracle Database software from the Oracle website or the Oracle Software Delivery Cloud portal and extract them to the Oracle home. Ensure that you review and understand the terms of the license.

- [Downloading the Software from Oracle Software Delivery Cloud Portal](#)
You can download the software from Oracle Software Delivery Cloud.
- [Downloading the Installation Archive Files from Oracle Website](#)
Download installation archive files from the Oracle website.

Downloading the Software from Oracle Software Delivery Cloud Portal

You can download the software from Oracle Software Delivery Cloud.

1. Use a browser to access the Oracle Software Delivery Cloud portal:
<https://edelivery.oracle.com/>
2. Click **Sign In** and enter your Oracle account username and password.
3. Type **Oracle Database** in the search bar. Click the **Add to Cart** button corresponding to the Oracle Database version that you want to download.
4. In the Checkout page, click **Checkout** and deselect any products that you do not want to download.
5. Select the operating system platform on which you want to install the software from the **Platform/Languages** column.
6. Click **Continue**.
7. Review the license agreement.
8. Select the **I reviewed and accept the Oracle License Agreement** checkbox. Click **Continue**.
9. Click **Download** to start downloading the software.
10. After you download the files, click **View Digest** to verify that the checksum matches the value listed on the download page.

Downloading the Installation Archive Files from Oracle Website

Download installation archive files from the Oracle website.

1. Use any browser to access the software download page on the Oracle website:
<http://www.oracle.com/technetwork/indexes/downloads/index.html>
2. Go to the download page for the product to install.
3. On the download page, identify the required disk space by adding the file sizes for each required file.

The file sizes are listed next to the file names.

4. Select a file system with enough free space to store and expand the archive files.
In most cases, the available disk space must be at least twice the size of all of the archive files.
5. On the file system, create a parent directory for each product (for example, OraDB19c) to hold the installation directories.
6. Download all of the installation archive files to the directory you created for the product.

Note

For Oracle Database Client installations, there are two installation archive files available for download. The first file is the client installation binary and the second file is a client gold image file. Download the appropriate zip file based on the type of installation you want to perform.

7. Verify that the files you downloaded are the same size as the corresponding files on the Oracle website. Also verify the checksums are the same as noted on the Oracle website using a command similar to the following, where *filename* is the name of the file you downloaded:

```
cksum filename.zip
```

8. Extract the files in each directory that you just created.

About Character Set Selection During Installation

Before you create the database, decide the character set that you want to use.

After a database is created, changing its character set is usually very expensive in terms of time and resources. Such operations may require converting all character data by exporting the whole database and importing it back. Therefore, it is important that you carefully select the database character set at installation time.

Oracle Database uses character sets for the following:

- Data stored in SQL character data types (CHAR, VARCHAR2, CLOB, and LONG).
- Identifiers such as table names, column names, and PL/SQL variables.
- Stored SQL and PL/SQL source code, including text literals embedded in this code.

Starting with Oracle Database 12c Release 2 (12.2), the default database character set of a database created from the General Purpose/Transaction Processing or the Data Warehousing template is Unicode AL32UTF8.

Unicode is the universal character set that supports most of the currently spoken languages of the world. It also supports many historical scripts (alphabets). Unicode is the native encoding of many technologies, including Java, XML, XHTML, ECMAScript, and LDAP. Unicode is ideally suited for databases supporting the Internet and the global economy.

Because AL32UTF8 is a multibyte character set, database operations on character data may be slightly slower when compared to single-byte database character sets, such as WE8ISO8859P1 or WE8MSWIN1252. Storage space requirements for text in most languages that use characters outside of the ASCII repertoire are higher in AL32UTF8 compared to legacy character sets supporting the language. English data may require more space only if stored in CLOB (character large object) columns. Storage for non-character data types, such as NUMBER or DATE, does not

depend on a character set. The universality and flexibility of Unicode usually outweighs these additional costs.

Consider legacy character sets only when the database needs to support a single group of languages and the use of a legacy character set is critical for fulfilling compatibility, storage, or performance requirements. The database character set to be selected in this case is the character set of most clients connecting to this database.

The database character set of a multitenant container database (CDB) determines which databases can be plugged in later. Ensure that the character set you choose for the CDB is compatible with the database character sets of the databases to be plugged into this CDB. If you use Unicode AL32UTF8 as your CDB character set, then you can plug in a pluggable database (PDB) in any database character set supported by Oracle Database (with the exception of EBCDIC-based character sets).

See Also

Oracle Database Globalization Support Guide for more information about choosing a database character set for a multitenant container database (CDB)

About Automatic Memory Management Installation Options

Decide if you want to configure Automatic Memory Management during installation.

During a Typical installation, you create your database with Database Configuration Assistant (DBCA), and automatic memory management is enabled. If you choose advanced installation, then you can either specify memory allocation manually, or enable automatic memory management.

With automatic memory management, the Oracle Database instances automatically manage and tune memory for you. With automatic memory management, you choose a memory target, and the instance automatically distributes memory between the system global area (SGA) and the instance program global area (instance PGA). As memory requirements change, the instance dynamically redistributes memory between the SGA and instance PGA.

You can enable automatic memory management either during, or after the database installation. Enabling automatic memory management after installation involves a shutdown and restart of the database.

Starting with 12c, Oracle Database uses the Optimized Shared Memory (OSM) model of Oracle Solaris on Oracle Solaris 10 1/13 or later and Oracle Solaris 11 SRU 7.5 or later systems to implement Automatic Memory Management. However, Oracle Database 12c continues to use Intimate Shared Memory (ISM) or Dynamic Intimate Shared Memory (DISM) on systems where OSM is not available.

Related Topics

- *Oracle Database Administrator's Reference for Linux and UNIX-Based Operating Systems*
- *Oracle Database Administrator's Guide*

Running the Installer in a Different Language

Describes how to run the installer in other languages.

Your operating system locale determines the language in which the database installer runs. You can run the installer in one of these languages:

- Brazilian Portuguese (pt_BR)
- French (fr)
- German (de)
- Italian (it)
- Japanese (ja)
- Korean (ko)
- Simplified Chinese (zh_CN)
- Spanish (es)
- Traditional Chinese (zh_TW)

To run the database installer in a supported language, change the locale in which your operating system session is running before you start the installer.

If the selected language is not one of the supported languages, then the installer runs in English.

Installing the Oracle Database Software

These topics explain how to run Oracle Universal Installer to perform most database installations.

Note

- If you plan to use Oracle Restart or Oracle ASM, then you must install Oracle Grid Infrastructure for a standalone server before you install and create the database. Otherwise, you must manually register the database with Oracle Restart.
- You can install Oracle Database by using the silent or response file installation method, without the GUI. This method is useful to perform multiple installations of Oracle Database.

- [Setup Wizard Installation Options for Creating Images](#)
- [Applying Patches During an Oracle Database Installation or Upgrade](#)
Starting with Oracle Database 18c, you can download and apply Release Updates (RUs) during an Oracle Database installation or upgrade.
- [Running Oracle Database Setup Wizard to Install Oracle Database](#)
Extract the database image files and use the `runInstaller` command to start the installation.

Setup Wizard Installation Options for Creating Images

Before you start the setup wizards for your Oracle Database or Oracle Grid Infrastructure installation, decide if you want to use any of the available image-creation options.

In image-based installations, you can start your Oracle Database installation or Oracle Grid Infrastructure installations by running the setup wizards `runInstaller` and `gridSetup.sh` respectively. Both these wizards come with the following image-creation options.

Table 11-1 Image-Creation Options for Setup Wizard

Option	Description
<code>-createGoldImage</code>	Creates a gold image from the current Oracle home.
<code>-destinationLocation</code>	Specify the complete path, or location, where the gold image will be created.
<code>-exclFiles</code>	Specify the complete paths to the files to be excluded from the newly created gold image.
<code>-help</code>	Displays help for all the available options.

For example:

```
cd $ORACLE_HOME
./runInstaller -createGoldImage -destinationLocation /tmp/my_db_images -
exclFiles /u01/app/oracle/product/19.0.0/dbhome_1/relnotes
```

```
cd Grid_home
./gridSetup.sh -createGoldImage -destinationLocation /tmp/my_grid_images -
exclFiles /u01/app/oracle/product/19.0.0/dbhome_1/relnotes
```

Where:

`/tmp/my_db_images` is a temporary file location where the image zip file is created.

`/tmp/my_grid_images` is a temporary file location where the image zip file is created.

`/u01/app/oracle/product/19.0.0/dbhome_1/relnotes` is the file to be excluded from the newly created gold image.

Refer to `runInstaller -createGoldImage` and `gridSetup.sh -createGoldImage` for more examples.

Applying Patches During an Oracle Database Installation or Upgrade

Starting with Oracle Database 18c, you can download and apply Release Updates (RUs) during an Oracle Database installation or upgrade.

1. Download the patches you want to apply from My Oracle Support:

<https://support.oracle.com>

2. Select the **Patches and Updates** tab to locate the patch.

Oracle recommends that you select **Recommended Patch Advisor**, and enter the product group, release, and platform for your software.

3. Move the patches to an accessible directory like /tmp.
4. Change to the Oracle Database home directory:

```
$ cd $ORACLE_HOME
```

5. Apply Release Updates (RUs) during the installation or upgrade process:

```
$ ./runInstaller -applyRU patch_directory_location
```

6. Complete the remaining steps in the Oracle Database configuration wizard to complete the installation or upgrade.

Running Oracle Database Setup Wizard to Install Oracle Database

Extract the database image files and use the `runInstaller` command to start the installation.

Have all the information you need to provide regarding users groups, and storage paths before you start installation. Oracle recommends that you have your My Oracle Support credentials available during installation. You should also be prepared to run root scripts or provide information to automate root scripts.

1. Log in as the Oracle installation owner user account (`oracle`) that you want to own the software binaries.
2. If this is the first time you are installing Oracle software, then create the Oracle base and the Oracle inventory directories as per the Oracle Optimal Flexible Architecture (OFA) recommendations. Specify the correct owner, group, and permissions for these directories.

```
# mkdir -p /u01/app/oracle
# mkdir -p /u01/app/oraInventory
# chown -R oracle:oinstall /u01/app/oracle
# chown -R oracle:oinstall /u01/app/oraInventory
# chmod -R 775 /u01/app
```

3. Download the Oracle Database installation image files (`db_home.zip`) to a directory of your choice. For example, you can download the image files to the /tmp directory.
4. Create an OFA-compliant Oracle home directory and extract the image files that you have downloaded in to this Oracle home directory. For example:

```
$ mkdir -p /u01/app/oracle/product/19.0.0/dbhome_1
$ cd /u01/app/oracle/product/19.0.0/dbhome_1
$ unzip -q /tmp/db_home.zip
```

Note

- Ensure that the Oracle home directory path you create is in compliance with the Oracle Optimal Flexible Architecture recommendations. Also, unzip the installation image files only in this Oracle home directory that you created.
- Oracle home or Oracle base cannot be symlinks, nor can any of their parent directories, all the way up to the root directory.

5. From the Oracle home directory, run the `runInstaller` command to start the Oracle Database Setup Wizard.

```
$ cd /u01/app/oracle/product/19.0.0/dbhome_1
$ ./runInstaller
```

Note

- Run the `runInstaller` command from the Oracle home directory only. Do not use the `runInstaller` command that resides at `$ORACLE_HOME/oui/bin/`, or any other location, to install Oracle Database, Oracle Database Client, or Oracle Grid Infrastructure.
- Use the `runInstaller` command with the `-applyRU` option to install Release Updates (RUs) during an Oracle Database installation or upgrade.

6. In the Select Configuration Option screen, select **Create and configure a single instance database**.

7. Select your installation type.

Installation screens vary depending on the installation option you select. Respond to the configuration prompts as needed.

8. Provide information to automate root scripts, or run scripts as root when prompted by the setup wizard.

If you configure automation for running root scripts, and a root script fails, then you can fix the problem manually, and click **Retry** to run the root script again.

Note

Click **Help** if you have any questions about the information you are asked to submit during installation.

Installing Standard Edition High Availability

Learn how to Install high availability on Oracle Database Standard Edition 2.

- [About Standard Edition High Availability](#)
Starting with Oracle Database 19c, you can install Oracle Database Standard Edition 2 in high availability mode.
- [Requirements for Installing Standard Edition High Availability](#)
Review these requirements before you install and deploy the Standard Edition High Availability feature.
- [Deploying Standard Edition High Availability](#)
Learn the process and options to deploy high availability on Oracle Database Standard Edition 2.

About Standard Edition High Availability

Starting with Oracle Database 19c, you can install Oracle Database Standard Edition 2 in high availability mode.

Standard Edition High Availability provides cluster-based failover for single-instance Standard Edition Oracle Databases using Oracle Clusterware.

Oracle Standard Edition High Availability benefits from the cluster capabilities and storage solutions that are already part of Oracle Grid Infrastructure, such as Oracle Clusterware, Oracle Automatic Storage Management (Oracle ASM) and Oracle ASM Cluster File System (Oracle ACFS).

Using integrated, shared, and concurrently mounted storage, such as Oracle ASM and Oracle ACFS for database files as well as for unstructured data, enables Oracle Grid Infrastructure to restart an Oracle Database on a failover node much faster than any cluster solution that relies on failing over and remounting volumes and file systems.

Starting with Oracle Database 19c Release Update (19.7), Standard Edition High Availability is supported on Linux x86-64, Oracle Solaris on SPARC (64-bit), and Microsoft Windows.

Starting with Oracle Database 19c Release Update (19.13), Standard Edition High Availability is supported on IBM AIX on POWER Systems (64-bit) and HP-UX Itanium.

Note

This section is specific to Standard Edition High Availability, which provides cluster-based database failover for Standard Edition Oracle Databases 19c. For more information about high availability options for Oracle Database, see *Oracle Clusterware Administration and Deployment Guide*.

Requirements for Installing Standard Edition High Availability

Review these requirements before you install and deploy the Standard Edition High Availability feature.

- You must configure Standard Edition High Availability using at least two nodes of a cluster running Oracle Grid Infrastructure 19.7 or later for Standalone Cluster.
- You must configure Standard Edition High Availability using an Oracle Database home of version 19.7 or later.

Note

When updating your Oracle Database home using Release Update (RU) 19.7 or later, ensure that you apply the Oracle Clusterware (OCW) RU of the same version to the Oracle Database home.

- Ensure that all cluster nodes on which you plan to install Oracle Database have the same operating system users, groups, and resource limits for the database software owner.
- You must store the Oracle Database binaries only on local storage or Oracle Automatic Storage Management Cluster File System (Oracle ACFS).
- You must store the Oracle Database data files only on Oracle ASM or Oracle ACFS.

- If you are using Oracle ACFS for storing data files, then you must register Oracle ACFS as a clusterware resource. The Oracle Database software owner user (oracle) must be the mount owner of the Oracle ACFS volume. When using an Oracle home on Oracle ACFS, Oracle recommends that you have Oracle base on the local file system.
- If you are using a local file system, then you must have the same single-instance Standard Edition 2 installation and updates on each node. You must also use the same Oracle base and Oracle home directory structure on each node.
- You must perform the same Oracle home operations on all the nodes.
- You must use an SPFILE for the database instance initialization parameters and a database password file stored for the database instance initialization parameters in Oracle ASM or Oracle ACFS. This approach ensures that the parameters are consistent across all nodes and the password file is available after a failover or a relocation.
- You must register the Standard Edition 2 single-instance database with SCAN listeners as remote listener and node listeners as local listener.

Deploying Standard Edition High Availability

Learn the process and options to deploy high availability on Oracle Database Standard Edition 2.

After installing Oracle Clusterware, as described in *Oracle Grid Infrastructure Installation and Upgrade Guide* for your platform, install single instance Standard Edition 2 Oracle Database software on the cluster nodes on which you want to configure Standard Edition High Availability.

- [Installing Standard Edition High Availability Database Software on Local File System](#)
You can install Oracle Database software binaries on a local file system to enable the Oracle Database Standard Edition high availability feature.
- [Installing Standard Edition High Availability Database Software on Oracle ACFS](#)
You can install Oracle Database software binaries on an Oracle ASM Cluster File System (Oracle ACFS) volume to enable the Oracle Database Standard Edition High Availability feature.

Related Topics

- *Oracle Grid Infrastructure Installation and Upgrade Guide for Linux*

Installing Standard Edition High Availability Database Software on Local File System

You can install Oracle Database software binaries on a local file system to enable the Oracle Database Standard Edition high availability feature.

Ensure that all the cluster nodes, on which you plan to configure Standard Edition High Availability, have the same operating system configuration, users, groups, resource limits, and SSH equivalence for the Oracle Database software owner user (oracle).

Before you start the installation, have all the information you need about users, groups, and storage paths. You should also be prepared to run root scripts or provide information to automate root scripts.

1. As the root user, log into the first cluster node on which you want to configure Standard Edition High Availability and create the Oracle base directory on the local file system.

Follow the Oracle Optimal Flexible Architecture (OFA) recommendations and specify the correct owner, group, and permissions for this directory.

```
# mkdir -p /u01/app/oracle
# chown oracle:oinstall /u01/app/oracle
```

2. Log in to the first cluster node as the Oracle Database software owner user (oracle).
3. Download the Oracle Database 19c release 19.3 installation image file (db_home.zip) from Oracle Software Delivery Cloud website to a directory of your choice.

<https://edelivery.oracle.com/>

4. Download the Oracle Database Release Update 19.7 or later patch from My Oracle Support to a directory of your choice and unzip it.

<https://support.oracle.com/>

5. Create an OFA-compliant Oracle home directory on the local file system and extract the image files that you have downloaded in to this Oracle home directory. For example:

```
$ mkdir -p /u01/app/oracle/product/19.0.0/dbhome_1
$ cd /u01/app/oracle/product/19.0.0/dbhome_1
$ unzip -q /tmp/db_home.zip
```

6. From the Oracle home directory, run the runInstaller command with the -applyRU flag to start the Oracle Database Setup Wizard and apply the Oracle Database Release Update 19.7 or later patch during installation.

```
$ ./runInstaller -applyRU patch_directory_location/patch_ID
```

Note

Run the runInstaller command from the Oracle home directory only. Do not use the runInstaller command that resides at \$ORACLE_HOME/oui/bin/, or any other location, to install Oracle Database.

7. In the Select Configuration Option screen, select **Set Up Software Only**.
8. In the Select Database Installation Option screen, select **Single instance database installation**.
9. In the Select Database Edition screen, select **Standard Edition 2**.
10. Respond to the configuration prompts as needed.
11. Provide information to automate root scripts, or run scripts as root when prompted by the setup wizard.

If you configure automation for running root scripts, and a root script fails, then you can fix the problem manually, and click **Retry** to run the root script again.

Note

Click **Help** if you have any questions about the information you are asked to submit during installation.

12. Optional: As the `oracle` user, enable read-only Oracle home on the first cluster node.

```
$ $ORACLE_HOME/bin/roohctl -enable
```

13. As the root user, create the Oracle base directory on all of the other cluster nodes on which you want to configure Standard Edition High Availability.

```
# mkdir -p /u01/app/oracle
# chown oracle:oinstall /u01/app/oracle
```

14. As the `oracle` user, run the `addnode.sh` script from the first node to perform the following operations on the other nodes on which you want to configure Standard Edition High Availability:

- Copy the Oracle home directory from the first node to the other nodes.
- Setup Oracle base and Oracle inventory directories on the other nodes.

```
$ $ORACLE_HOME/addnode/addnode.sh -silent
CLUSTER_NEW_NODES=comma_separated_list_of_other_nodes
```

15. As the root user, run the `root.sh` script on all the other cluster nodes on which you are configuring Standard Edition High Availability.

```
# /u01/app/oracle/product/19.0.0/dbhome_1/root.sh
```

After the Oracle Database software installation is complete, use Oracle Database Configuration Assistant (Oracle DBCA), in either interactive or silent mode, to create a Standard Edition database on the first cluster node on which you installed the Oracle Database software.

For more information about the requirements for creating a database, and the procedure for enabling and configuring Standard Edition High Availability for Oracle Databases, refer to *Oracle Database Administrator's Guide*.

Related Topics

- [Response Files](#)
Review the following topics to install and configure Oracle Database, Oracle Grid Infrastructure, and other Oracle products using response files.
- *Oracle Grid Infrastructure Installation and Upgrade Guide for Linux*
- *Creating a Database with Oracle DBCA*

Installing Standard Edition High Availability Database Software on Oracle ACFS

You can install Oracle Database software binaries on an Oracle ASM Cluster File System (Oracle ACFS) volume to enable the Oracle Database Standard Edition High Availability feature.

Ensure that all the cluster nodes, on which you plan to configure Standard Edition High Availability, have the same operating system configuration, users, groups, resource limits, and SSH equivalence for the Oracle Database software owner user (`oracle`) between the nodes.

Before you start the installation, have all the information you need about users groups, and storage paths. You should also be prepared to run root scripts or provide information to automate root scripts.

1. As the root user, register Oracle ACFS as an Oracle Clusterware resource specifying the Oracle Database software owner user (oracle) as the mount owner.

```
# Grid_home/bin/srvctl add filesystem -volume acfs_volume_name -diskgroup  
diskgroup_name -path mount_point -fstype ACFS -autostart ALWAYS -user  
oracle
```

Note

Mount Oracle ACFS to a mount point where you plan to create Oracle base and Oracle home directories for this installation. For example, /u01/app/oracle.

2. As the oracle user, mount the Oracle ACFS file system on all of the cluster nodes on which you want to configure Standard Edition High Availability.

```
$ Grid_home/bin/srvctl start filesystem -volume acfs_volume_name -  
diskgroup diskgroup_name
```

3. As the root user, create the Oracle base directory in an Oracle ACFS volume on the first cluster node on which you want to configure Standard Edition High Availability. Follow the Oracle Optimal Flexible Architecture (OFA) recommendations and specify the correct owner, group, and permissions for this directory.

```
# mkdir -p /u01/app/oracle  
# chown oracle:oinstall /u01/app/oracle
```

Note

If you have configured the Oracle ACFS file system with the oracle user as the mount owner and the /u01/app/oracle directory as the mount point, then this step is not required.

4. Log in to the first cluster node as the Oracle Database software owner user (oracle).
5. Download the Oracle Database 19c release 19.3 installation image file (db_home.zip) from Oracle Software Delivery Cloud website to a directory of your choice.

<https://edelivery.oracle.com/>

6. Download the Oracle Database Release Update 19.7 or later patch from My Oracle Support to a directory of your choice and unzip it.

<https://support.oracle.com/>

7. Create an OFA-compliant Oracle home directory in an Oracle ACFS volume and extract the image files that you have downloaded in to this Oracle home directory. For example:

```
$ mkdir -p /u01/app/oracle/product/19.0.0/dbhome_1  
$ cd /u01/app/oracle/product/19.0.0/dbhome_1  
$ unzip -q /tmp/db_home.zip
```

8. From the Oracle home directory, run the `runInstaller` command with the `-applyRU` flag to start the Oracle Database Setup Wizard and apply the Oracle Database Release Update 19.7 or later patch during installation.

```
$ ./runInstaller -applyRU patch_directory_location/patch_ID
```

Note

Run the `runInstaller` command from the Oracle home directory only. Do not use the `runInstaller` command that resides at `$ORACLE_HOME/oui/bin/`, or any other location, to install Oracle Database.

9. In the Select Configuration Option screen, select **Set Up Software Only**.
10. In the Select Database Installation Option screen, select **Single instance database installation**.
11. In the Select Database Edition screen, select **Standard Edition 2**.
12. Respond to the configuration prompts as needed.
13. Provide information to automate root scripts, or run scripts as root when prompted by the setup wizard.

If you configure automation for running root scripts, and a root script fails, then you can fix the problem manually, and click **Retry** to run the root script again.

Note

Click **Help** if you have any questions about the information you are asked to submit during installation.

14. Optional: As the `oracle` user, enable read-only Oracle home on the first cluster node.

```
$ $ORACLE_HOME/bin/roohctl -enable
```

15. Attach the Oracle home on the first node to the other cluster nodes on which you want to configure Standard Edition High Availability.

```
$ $ORACLE_HOME/addnode/addnode.sh -silent  
CLUSTER_NEW_NODES=comma_separated_list_of_other_nodes
```

16. As the root user, run the `root.sh` script on all the other cluster nodes on which you are configuring Standard Edition High Availability.

```
# /u01/app/oracle/product/19.0.0/dbhome_1/root.sh
```

After the Oracle Database software installation is complete, use Oracle Database Configuration Assistant (Oracle DBCA), in either interactive or silent mode, to create a Standard Edition database on the first cluster node on which you installed the Oracle Database software.

For more information about the requirements for creating a database, and the procedure for enabling and configuring Standard Edition High Availability for Oracle Databases, refer to *Oracle Database Administrator's Guide*.

Related Topics

- *Oracle Grid Infrastructure Installation and Upgrade Guide for Linux*
- *Configuring Oracle Automatic Storage Management Cluster File System*
- *Creating a Database with Oracle DBCA*

Oracle Database Postinstallation Tasks

Complete configuration tasks after you install Oracle Database.

You are required to complete some configuration tasks after Oracle Database is installed. In addition, Oracle recommends that you complete additional tasks immediately after installation. You must also complete product-specific configuration tasks before you use those products.

Note

This chapter describes basic configuration only. Refer to product-specific administration and tuning guides for more detailed configuration and tuning information.

- [Required Postinstallation Tasks](#)
Download and apply required patches for your software release after completing your initial installation.
- [Recommended Postinstallation Tasks](#)
Oracle recommends that you complete these tasks after installation.
- [About Changes in Default SGA Permissions for Oracle Database](#)
Starting with Oracle Database 12c Release 2 (12.2.0.1), by default, permissions to read and write to the System Global Area (SGA) are limited to the Oracle software installation owner.
- [Checking Installed Oracle Database Contents and Directory Location](#)
Use these steps to check the contents and directory location of an Oracle Database installation:
- [Enabling and Disabling Oracle Database Options After Installation](#)
The chopt tool changes your database options after installation.
- [Starting Oracle Enterprise Manager Database Express](#)
Use these steps to log in to Oracle Enterprise Manager Database Express (EM Express).
- [Creating a Fast Recovery Area](#)
During an Oracle Restart installation, you can create only one disk group. During an Oracle Clusterware installation, you can create multiple disk groups. If you plan to add an Oracle Database for a standalone server or an Oracle RAC database, then you should create the fast recovery area for database files.
- [Cloning an Oracle Database Home](#)
Cloning an Oracle home involves creating a copy of the Oracle home and then configuring it for a new environment.

Required Postinstallation Tasks

Download and apply required patches for your software release after completing your initial installation.

- [Downloading Release Update Patches](#)
Download and install Release Updates (RU) and Monthly Recommended Patches (MRPs) patches for your Oracle software after you complete installation.
- [Unlocking and Resetting Oracle Database User Passwords](#)
Passwords for all Oracle system administration accounts except SYS, SYSTEM, and DBSMP are revoked after installation. Before you use a locked account, you must unlock it and reset its password.

Downloading Release Update Patches

Download and install Release Updates (RU) and Monthly Recommended Patches (MRPs) patches for your Oracle software after you complete installation.

Oracle provides quarterly updates in the form of Release Updates (RU) and Monthly Recommended Patches (MRPs). Oracle no longer releases patch sets. For more information, see [My Oracle Support Note 2285040.1](#).

Check the My Oracle Support website for required updates for your installation.

1. Use a web browser to view the My Oracle Support website:
<https://support.oracle.com>
2. In the **sign in to other Oracle Support portals** section, click **Support for Oracle Hardware, Software, and Managed Cloud services** to log in to My Oracle Support.

Note

If you are not a My Oracle Support registered user, then click **Register for My Oracle Support** and register.

3. On the main My Oracle Support page, click **Patches & Updates**.
4. In the **Patch Search** region, select **Product or Family (Advanced)**.
5. On the **Product or Family (Advanced)** display, provide information about the product, release, and platform for which you want to obtain patches, and click **Search**.
The Patch Search pane opens, displaying the results of your search.
6. Select the patch number and click **ReadMe**.
The README page is displayed. It contains information about the patch and how to apply the patches to your installation.
7. Uncompress the Oracle patch updates that you downloaded from My Oracle Support.

Related Topics

- [My Oracle Support note 888.1](#)
- [Patch Delivery Methods for Oracle Database](#)

Unlocking and Resetting Oracle Database User Passwords

Passwords for all Oracle system administration accounts except SYS, SYSTEM, and DBSMP are revoked after installation. Before you use a locked account, you must unlock it and reset its password.

If you created a preconfigured database during the installation, but you did not unlock accounts required to use the database, then you must unlock and reset those accounts using these procedures.

- [Requirements for Database Passwords](#)
To secure your database, use passwords that satisfy the Oracle recommended password requirements, even the passwords for predefined user accounts.
- [Oracle Database System Privileges Accounts and Passwords](#)
Review these system privileges accounts after installation in preparation for unlocking accounts and changing passwords.
- [Guidelines for Changing System Privileges Account Passwords](#)
Follow these rules for changing Oracle Database system privileges account passwords.
- [Locking and Unlocking User Accounts](#)
- [Using SQL*Plus to Unlock Accounts and Reset Passwords](#)
Use this SQL*Plus procedure to unlock and reset user account passwords.

Requirements for Database Passwords

To secure your database, use passwords that satisfy the Oracle recommended password requirements, even the passwords for predefined user accounts.

Oracle Database provides a set of predefined user accounts. Create passwords in a secure fashion. If you have default passwords, change these passwords to secure passwords.

You can manage the security for Oracle Database users in various ways:

- Enforce restrictions on the way that passwords are created
- Create user profiles
- Use user resource limits to further secure user accounts

Related Topics

- *Oracle Database Security Guide*

Oracle Database System Privileges Accounts and Passwords

Review these system privileges accounts after installation in preparation for unlocking accounts and changing passwords.

All databases created by the Database Configuration Assistant (DBCA) include the SYS, SYSTEM, and DBSNMP database accounts. In addition, Oracle Database provides several other administrative accounts. Before using these accounts, you must unlock them and reset their passwords.

Starting with Oracle Database 12c Release 2 (12.2), only the HR sample schema is automatically installed after a database installation. All sample schemas, including HR, are distributed on GitHub:

<https://github.com/oracle/db-sample-schemas>

Note

This list contains some of the important system privileges user accounts, but it is not complete. Use Oracle Enterprise Manager Database Express 12c to view the complete list of database accounts.

Table 12-1 Partial List of Oracle Database System Privileges Accounts Locked After Installation

User Name	Description	For More Information
ANONYMOUS	Enables HTTP access to Oracle XML DB.	<i>Oracle XML DB Developer's Guide</i>
APEX_050100	The account that owns the Oracle Application Express schema and metadata.	<i>Oracle Application Express App Builder User's Guide</i>
APEX_PUBLIC_USER	The minimally privileged account used for Oracle Application Express configuration with Oracle Application Express Listener or Oracle HTTP Server and mod_plsql.	<i>Oracle Application Express App Builder User's Guide</i>
AUDSYS	The account where the unified audit data trail resides.	<i>Oracle Database Security Guide</i>
CTXSYS	The Oracle Text account.	<i>Oracle Text Application Developer's Guide</i>
DBSFUSER	The account used to run the DBMS_SFW_ACL_ADMIN package.	<i>Oracle Database PL/SQL Packages and Types Reference</i>
DBSNMP	The account used by the Management Agent component of Oracle Enterprise Manager to monitor and manage the database.	<i>Oracle Enterprise Manager Cloud Control Administrator's Guide</i>
DIP	The account used by the Directory Integration Platform (DIP) to synchronize the changes in Oracle Internet Directory with the applications in the database.	None
DVSYS	There are two roles associated with this account. The Database Vault owner role manages the Database Vault roles and configurations. The Database Vault Account Manager is used to manage database user accounts. Note: Part of Oracle Database Vault user interface text is stored in database tables in the DVSYS schema. By default, only the English language is loaded into these tables. You can use the DVSYS.DBMS_MACADM.ADD-NLS_DATA procedure to add other languages to Oracle Database Vault.	<i>Oracle Database Vault Administrator's Guide</i>
DVF	The account owned by Database Vault that contains public functions to retrieve the Database Vault Factor values.	<i>Oracle Database Vault Administrator's Guide</i>
FLows_FILES	The account owns the Oracle Application Express uploaded files.	<i>Oracle Application Express App Builder User's Guide</i>

Table 12-1 (Cont.) Partial List of Oracle Database System Privileges Accounts Locked After Installation

User Name	Description	For More Information
GGSYS	The internal account used by Oracle GoldenGate. It should not be unlocked or used for a database login.	None
GSMADMIN_INTERN AL	The internal account that owns the Global Data Services schema. It should not be unlocked or used for a database login.	<i>Oracle Database Global Data Services Concepts and Administration Guide</i>
GSMCATUSER	The account used by Global Service Manager to connect to the Global Data Services catalog.	<i>Oracle Database Global Data Services Concepts and Administration Guide</i>
GSMUSER	The account used by Global Service Manager to connect to the database.	<i>Oracle Database Global Data Services Concepts and Administration Guide</i>
HR	The account that owns the Human Resources schema included in the Oracle Sample Schemas.	<i>Oracle Database Sample Schemas</i>
LBACSYS	The Oracle Label Security administrator account. Starting with Oracle Database 18c, the LBACSYS user account is created as a schema-only account.	<i>Oracle Label Security Administrator's Guide</i>
MDDATA	The schema used by Oracle Spatial and Graph for storing geocoder and router data.	<i>Oracle Spatial and Graph Developer's Guide</i>
MDSYS	The Oracle Spatial and Graph administrator account.	<i>Oracle Spatial and Graph Developer's Guide</i>
OUTLN	The account that supports plan stability. Plan stability enables you to maintain the same execution plans for the same SQL statements. OUTLN acts as a role to centrally manage metadata associated with stored outlines.	None
ORACLE_OCM	This account contains the instrumentation for configuration collection used by the Oracle Configuration Manager.	None
REMOTE_SCHEDUL ER_AGENT	The account to disable remote jobs on a database. This account is created during the remote scheduler agent configuration. You can disable the capability of a database to run remote jobs by dropping this user.	<i>Oracle Database Administrator's Guide</i>
SYS	The account used to perform database administration tasks.	<i>Oracle Database Administrator's Guide</i>
SYSTEM	Another account used to perform database administration tasks.	<i>Oracle Database Administrator's Guide</i>
SYSBACKUP	The account used to perform backup and recovery tasks.	<i>Oracle Database Administrator's Guide</i>
SYSKM	The account used to perform encryption key management.	<i>Oracle Database Administrator's Guide</i>
SYSDBG	The account used to administer and monitor Oracle Data Guard.	<i>Oracle Database Administrator's Guide</i>
SYSRAC	The account used to administer Oracle Real Application Clusters (RAC).	<i>Oracle Database Administrator's Guide</i>

Table 12-1 (Cont.) Partial List of Oracle Database System Privileges Accounts Locked After Installation

User Name	Description	For More Information
SYS\$UMF	The account used to administer Remote Management Framework, including the remote Automatic Workload Repository (AWR).	<i>Oracle Database Performance Tuning Guide</i>
WMSYS	The account used to store the metadata information for Oracle Workspace Manager.	<i>Oracle Database Workspace Manager Developer's Guide</i>
XDB	The account used for storing Oracle XML DB data and metadata.	<i>Oracle XML DB Developer's Guide</i>
XS\$NULL	The internal account that represents the absence of a database schema user in a session, and indicates an application user session is in use. XS\$NULL cannot be authenticated to a database, nor can it own any database schema objects, or possess any database privileges.	<i>Oracle Database Real Application Security Administrator's and Developer's Guide</i>

Except for the accounts provided with the Oracle Sample Schemas, most of these database accounts are locked by default and created without passwords as schema only. This prevents malicious users from logging into these accounts using the default password set during catalog creation. To find the status of an account, query the `AUTHENTICATION_TYPE` column of the `DBA_USERS` data dictionary view. If `AUTHENTICATION_TYPE` is schema only, then the status is `NONE`.

Many of these accounts are automatically created when you run standard scripts such as the various `cat*.sql` scripts. To find user accounts that are created and maintained by Oracle, query the `USERNAME` and `ORACLE_MAINTAINED` columns of the `ALL_USERS` data dictionary view. If the output for `ORACLE_MAINTAINED` is `Y`, then you must not modify the user account except by running the script that was used to create it.

Related Topics

- *Oracle Database Security Guide*
- *Oracle Database Sample Schemas*

Guidelines for Changing System Privileges Account Passwords

Follow these rules for changing Oracle Database system privileges account passwords.

Before you use a locked account, you must unlock it and reset its password. Passwords for all Oracle system administration accounts except `SYS`, `SYSTEM`, and `DBSNMP` are revoked after installation. If you created a starter database during the installation, Oracle Database Configuration Assistant displays a screen with your database information and the Password Management button. Use the Password Management button to unlock only the user names you use.

For more information about how to create a secure password, see:

Oracle Database Security Guide

Locking and Unlocking User Accounts

To temporarily deny access to the database for a particular user account, you can lock the user account. If the user then attempts to connect, then the database displays an error message and does not allow the connection. You can unlock the user account when you want to permit database access again for that user. You can use Oracle Enterprise Manager Database Express (EM Express) to lock and unlock user accounts.

To lock or unlock a user account:

1. In EM Express, go to the Users page.
2. Click the desired user account.
3. From the **Actions** menu, select **Alter Account**.
The Alter Account page appears.
4. Do one of the following:
 - To lock the account, enable the **Account Locked** option, and then click **OK**.
 - To unlock the account, disable the **Account Locked** option, and then click **OK**.

Using SQL*Plus to Unlock Accounts and Reset Passwords

Use this SQL*Plus procedure to unlock and reset user account passwords.

1. Log in as the Oracle Database software owner user.
2. Set the ORACLE_HOME and ORACLE_SID environment variables.
Bourne, Bash or Korn shell:

```
$ ORACLE_SID=orcl
$ export ORACLE_SID
$ ORACLE_HOME=/u01/app/oracle/product/19.0.0/dbhome_1
$ export ORACLE_HOME
```

C shell:

```
% setenv ORACLE_SID orcl
% setenv ORACLE_HOME /u01/app/oracle/product/19.0.0/dbhome_1
```

3. Start SQL*Plus and log in as the SYS user, connecting as SYSDBA:

```
$ $ORACLE_HOME/bin/sqlplus
SQL> CONNECT SYS as SYSDBA
Enter password: sys_password
```

4. To unlock an account:

```
ALTER USER account ACCOUNT UNLOCK;
```

5. To reset the password:

```
ALTER USER user_name IDENTIFIED BY new_password;
```

Note

If you unlock an account but do not reset the password, then the password remains expired. The first time someone connects as that user, they must change the user's password.

Related Topics

- *Oracle Database Administrator's Guide*

Recommended Postinstallation Tasks

Oracle recommends that you complete these tasks after installation.

- [Creating a Backup of the root.sh Script](#)
Oracle recommends that you back up the `root . sh` script after you complete an installation.
- [Setting Language and Locale Preferences for Client Connections](#)
Configure client applications connecting to an Oracle Database according to your locale preferences and your I/O device character set.
- [Recompile Invalid Objects in the Database](#)
After you install, patch, or upgrade a database, recompile invalid objects on the CDB and PDBs with a recompilation driver script.
- [About Installing Oracle Autonomous Health Framework](#)
Install the latest version of Oracle Autonomous Health Framework to perform proactive health checks and collect diagnostics data for the Oracle software stack.
- [Enabling Data Analytics Accelerators on SPARC for Oracle Database](#)
The microprocessors for the SPARC M7 and T7 series servers include Data Analytics Accelerator (DAX) coprocessors. These coprocessors perform query-related operations directly through the hardware, which improves Oracle Database performance.

Creating a Backup of the root.sh Script

Oracle recommends that you back up the `root . sh` script after you complete an installation.

If you install other products in the same Oracle home directory subsequent to this installation, then Oracle Universal Installer updates the contents of the existing `root . sh` script during the installation. If you require information contained in the original `root . sh` script, then you can recover it from the backed up `root . sh` file.

Setting Language and Locale Preferences for Client Connections

Configure client applications connecting to an Oracle Database according to your locale preferences and your I/O device character set.

You must configure client applications connecting to an Oracle Database according to your locale preferences and your I/O device character set. If your applications do not have their own specific methods to configure locale preferences, then the method you use to configure an Oracle database client connection depends on the access API you use to connect to the database. Check your application documentation, before you configure locale preferences for your applications.

For applications that connect to Oracle Databases using Oracle Call Interface (OCI) use NLS_LANG and other client settings with names that start with NLS_ to set the locale conventions and client character set for Oracle Database sessions. It is important that you set the character set part of the NLS_LANG value properly. The character set you set must correspond to the character set used by your I/O devices, which in case of Microsoft Windows is either the ANSI Code Page (for GUI applications), such as WE8MSWIN1252, or the OEM Code Page (for Console mode applications), such as US8PC437. By doing this, the OCI API is notified about the character set of data that it receives from the application. OCI can then convert this data correctly to and from the database character set.

NLS_LANG and the other NLS settings can be specified either as environment variables or as Windows Registry settings. Environment variable values take precedence over Registry values.

Oracle Universal Installer sets a default value for the NLS_LANG setting in Registry when it creates a new Oracle home on Microsoft Windows. The NLS_LANG value is based on the language of the Windows user interface, which is the language of Windows menu items and dialog box labels.

 Caution

Failure to set the client character set correctly can cause data loss.

Java applications that connect to Oracle Databases by using Oracle JDBC do not use NLS_LANG. Instead, Oracle JDBC maps the default locale of the Java VM in which the application runs to the Oracle Database language and territory settings. Oracle JDBC then configures the connected database session using these settings. Because Java works internally in Unicode, the client character set is always set to Unicode. Unless an application explicitly changes it, the default locale of the Java VM is set based on the locale of the user operating system on which the Java VM runs. Check your Java VM documentation for information about configuring the Java VM default locale.

 Note

In 3-tier architecture deployments, application servers that are database clients can have settings in their configuration files that specify the NLS_LANG value or the Java VM locale. Check the documentation accompanying these servers.

 See Also

Oracle Database Globalization Support Guide for more information about configuring user locale preferences

Recompile Invalid Objects in the Database

After you install, patch, or upgrade a database, recompile invalid objects on the CDB and PDBs with a recompilation driver script.

By default, AutoUpgrade performs a recompilation of invalid Oracle objects, which is controlled by the configuration file `run_utlrp` local parameter (default: `prefix.run_utlrp=yes`). In addition,

Oracle provides the recompilation scripts `utlrp.sql` and `utlprp.sql`. These scripts are located in the `Oracle_home/rdbms/admin` directory.

Note

Starting with AutoUpgrade 23.1, when you run the AutoUpgrade utility, AutoUpgrade runs the `utlprpom.sql` script, and does not run `utlrp.sql`. When AutoUpgrade is used for upgrades to Oracle Database 12c Release 2 (12.2.0.1) and later releases, AutoUpgrade only recompiles invalid objects owned by Oracle-maintained schemas. Because database upgrades do not need to touch user objects, AutoUpgrade maintains this policy when it recompiles invalid objects.

After installing a database, recompile all invalid objects;

1. Change directory to `Oracle_home/rdbms/admin`. For example

```
$ cd $ORACLE_HOME/rdbms/admin
```

2. Use the `catcon.pl` script in the Oracle home to run `utlrp.sql`. For example:

```
$ORACLE_HOME/perl/bin/perl catcon.pl --n 1 --e --b utlrp --d '','',''  
utlrp.sql
```

Note the following conditions of this use case:

- `--n` parameter: is set to 1, so the script runs each PDB recompilation in sequence.
- `--e` parameter: turns echo on.
- `--b` parameter: Sets the log file base name. It is set to `utlrp`.

Expect a time delay for the serial recompilation of PDBs to complete. Depending on the number of PDBs that you are upgrading, the recompilation can extend significantly beyond the time required for the upgrade scripts to complete.

The `utlrp.sql` script automatically recompiles invalid objects in either serial or parallel recompilation, based on both the number of invalid objects, and on the number of CPUs available. CPUs are calculated using the number of CPUs (`cpu_count`) multiplied by the number of threads for each CPU (`parallel_threads_per_cpu`). On Oracle Real Application Clusters (Oracle RAC), this number is added across all Oracle RAC nodes.

After patching or upgrading a database, there is more than one approach you can use to recompile invalid Oracle-owned and user-owned objects:

Recompile all invalid objects (the invalid objects in both Oracle and user schemas) by using `utlrp.sql` or `utlprp.sql`.

If time is a factor and the type of invalid objects is predominately application owned, then you can recompile Oracle-owned invalid objects first, and defer recompiling application-owned invalid objects to a later time. To recompile invalid objects in Oracle schemas, use `utlprpom.sql`. To recompile the remaining invalid objects, use `utlrp.sql` or `utlprp.sql`.

Note

When you use either `utlprp.sql` or `utlprpom.sql`, note that both scripts require you to define the degree of parallelism that the script should use, or determine the number of parallel recompile jobs to use.

The script uses syntax as follows, where *base* is the base name you want to have given to log files, *N* is the number of PDBs on which you want to run recompilation jobs in parallel (degrees of parallelism), *script.sql* is the Oracle recompilation script you chose to use, and *P* is the number of PDBs on which you want to run in parallel:

```
$ORACLE_HOME/perl/bin/perl $ORACLE_HOME/rdbms/admin/catcon.pl -b base -  
d $ORACLE_HOME/rdbms/admin  
-n N -l /tmp script.sql '--pP'
```

Suppose you are running recompilation in a CDB using the log file base name `recomp`, with a degrees of parallelism setting of 3 jobs per PDB container, the script you choose to use is `utlprp.sql`, and you want to recompile across at most 10 PDBs at a time. In that case, the syntax you use to run the recompile operation is similar to the following,

```
$ORACLE_HOME/perl/bin/perl $ORACLE_HOME/rdbms/admin/catcon.pl -b recomp -  
d $ORACLE_HOME/rdbms/admin -n 10 -l /tmp utlprp.sql '--p3'
```

Related Topics

- Syntax and Parameters for `catcon.pl`

About Installing Oracle Autonomous Health Framework

Install the latest version of Oracle Autonomous Health Framework to perform proactive health checks and collect diagnostics data for the Oracle software stack.

Oracle Autonomous Health Framework includes the functionality from Oracle ORAchk, Oracle EXAchk, and Oracle Trace File Analyzer (TFA). Oracle Autonomous Health Framework extends health check coverage to the entire Oracle software stack, based on critical and reoccurring problems. Oracle Autonomous Health Framework proactively scans for known problems with Oracle products and deployments, including the following:

- Standalone Oracle Database
- Oracle Grid Infrastructure
- Oracle Real Application Clusters
- Maximum Availability Architecture (MAA) Validation
- Upgrade Readiness Validations
- Oracle GoldenGate

Oracle Autonomous Health Framework is pre-installed with Oracle Database. However, Oracle recommends that you update to the latest version of Oracle Autonomous Health Framework by downloading and installing it from My Oracle Support Note 2550798.1.

<https://support.oracle.com/epmos/faces/DocContentDisplay?id=2550798.1&parent=DOCUMENTATION&sourceId=USERGUIDE>

Enabling Data Analytics Accelerators on SPARC for Oracle Database

The microprocessors for the SPARC M7 and T7 series servers include Data Analytics Accelerator (DAX) coprocessors. These coprocessors perform query-related operations directly through the hardware, which improves Oracle Database performance.

Starting with Oracle Database 12c Release 2 (12.2) and later releases, the DAX feature is enabled by default for Oracle Database on Oracle Solaris on SPARC (64-bit) systems. The `dax_access` privilege is available by default to the Oracle database installation owner (`oracle`) and all other users and processes as part of the basic Oracle Solaris operating system privilege set.

To use the DAX hardware with Oracle Database, you must install Oracle Database 12c Enterprise Edition, and enable the In-Memory option for your application.

For information about configuring the Oracle Database 12c in-memory feature, see *Oracle Database In-Memory Guide*.

Note

The DAX feature is not supported for use with Kernel Zones at this time.

Related Topics

- *Oracle Database In-Memory Guide*

About Changes in Default SGA Permissions for Oracle Database

Starting with Oracle Database 12c Release 2 (12.2.0.1), by default, permissions to read and write to the System Global Area (SGA) are limited to the Oracle software installation owner.

In previous releases, both the Oracle installation owner account and members of the OSDBA group had access to shared memory. The change in Oracle Database 12c Release 2 (12.2) and later releases to restrict access by default to the Oracle installation owner account provides greater security than previous configurations. However, this change may prevent DBAs who do not have access to the Oracle installation owner account from administering the database.

The Oracle Database initialization parameter `ALLOW_GROUP_ACCESS_TO_SGA` determines if the Oracle Database installation owner account (`oracle` in Oracle documentation examples) is the only user who can read and write to the database System Global Area (SGA), or if members of the OSDBA group can read the SGA. In Oracle Database 12c Release 2 (12.2) and later releases, the default value for this parameter is `FALSE`, so that only the Oracle Database installation owner has read and write permissions to the SGA. Group access to the SGA is removed by default. This change affects all Linux and UNIX platforms.

If members of the OSDBA group require read access to the SGA, then you can change the initialization parameter `ALLOW_GROUP_ACCESS_TO_SGA` setting from `FALSE` to `TRUE`. Oracle strongly recommends that you accept the default permissions that limit access to the SGA to the `oracle` user account.

Related Topics

- *Oracle Database Reference*

Checking Installed Oracle Database Contents and Directory Location

Use these steps to check the contents and directory location of an Oracle Database installation:

1. Go to \$ORACLE_HOME/oui/bin.
2. Start Oracle Universal Installer.

\$./runInstaller
3. Click **Installed Products** to display the Inventory dialog box on the Welcome screen.
4. Select an Oracle Database product from the list to check the installed contents.
5. Click **Details** to find additional information about an installed product.
6. Click **Close** to close the Inventory dialog box.
7. Click **Cancel** to close Oracle Universal Installer, and then click **Yes** to confirm.

Enabling and Disabling Oracle Database Options After Installation

The chopt tool changes your database options after installation.

When you install Oracle Database, some options are enabled and others are disabled. To enable or disable a particular database feature for an Oracle home, shut down the database and use the chopt tool.

- [Chopt Tool](#)
Use the chopt tool after installation to add or remove Oracle Database options.

Chopt Tool

Use the chopt tool after installation to add or remove Oracle Database options.

Purpose

The chopt tool is a command-line utility that enables and disables database options.

Note

If you install or clone an Oracle Database image, then all Oracle Database options are enabled by default.

Prerequisites

You must complete installation before you can use the chopt tool.

File Path

The tool is located in the `ORACLE_HOME/bin` directory

Syntax

`chopt [enable | disable] db_option`

Options

Command Option	Description
<code>olap</code>	Oracle OLAP
<code>partitioning</code>	Oracle Partitioning
<code>rat</code>	Oracle Real Application Testing

Note

- When you enable or disable OLAP, you must run the `SYS.XOQ_VALIDATE` and `SYS.APS_VALIDATE` procedures to update the database registry. When you disable OLAP, its status in the database registry should be `OPTION OFF` and when you enable OLAP, its status in the database registry should be `VALID`.
- The Oracle Advanced Analytics (OAA) feature is enabled by default for Oracle Database. You cannot disable it using the `chopt` tool.

Examples

To use the `chopt` tool to modify your Oracle Database, you must shut down the database before you run the `chopt` tool, and then start up the database after you add or remove database options.

Example 12-1 Enabling Oracle Real Application Testing Using `chopt`

The following example shows how to use the `chopt` tool to enable the Oracle Real Application Testing option in an Oracle Database called Sales:

```
cd $ORACLE_HOME/bin
srvctl stop database -d Sales
chopt enable rat
srvctl start database -d Sales
```

Starting Oracle Enterprise Manager Database Express

Use these steps to log in to Oracle Enterprise Manager Database Express (EM Express).

To start Oracle Enterprise Manager Database Express, use the EM Express URL provided by Oracle Database Configuration Assistant (Oracle DBCA) during the database installation and creation. For information about logging in to Oracle Enterprise Manager Database Express see *Oracle Database 2 Day DBA*.

If Oracle DBCA did not provide the EM Express URL during the database installation and creation, or if you need to change the EM Express port later on, then see *Oracle Database 2 Day DBA*.

Related Topics

- *Oracle Database 2 Day DBA*
- *Configuring the HTTPS Port for EM Express*

Creating a Fast Recovery Area

During an Oracle Restart installation, you can create only one disk group. During an Oracle Clusterware installation, you can create multiple disk groups. If you plan to add an Oracle Database for a standalone server or an Oracle RAC database, then you should create the fast recovery area for database files.

- [About the Fast Recovery Area and the Fast Recovery Area Disk Group](#)
The fast recovery area is a unified storage location for all Oracle Database files related to recovery. Enabling rapid backups for recent data can reduce requests to system administrators to retrieve backup tapes for recovery operations.
- [Creating the Fast Recovery Area Disk Group](#)
Procedure to create the fast recovery area disk group.

About the Fast Recovery Area and the Fast Recovery Area Disk Group

The fast recovery area is a unified storage location for all Oracle Database files related to recovery. Enabling rapid backups for recent data can reduce requests to system administrators to retrieve backup tapes for recovery operations.

Database administrators can define the `DB_RECOVERY_FILE_DEST` parameter to the path for the fast recovery area to enable on disk backups and rapid recovery of data. When you enable fast recovery in the `init.ora` file, Oracle Database writes all RMAN backups, archive logs, control file automatic backups, and database copies to the fast recovery area. RMAN automatically manages files in the fast recovery area by deleting obsolete backups and archiving files no longer required for recovery.

Oracle recommends that you create a fast recovery area disk group. Oracle Clusterware files and Oracle Database files can be placed on the same disk group, and you can also place fast recovery files in the same disk group. However, Oracle recommends that you create a separate fast recovery disk group to reduce storage device contention.

The fast recovery area is enabled by setting the `DB_RECOVERY_FILE_DEST` parameter. The size of the fast recovery area is set with `DB_RECOVERY_FILE_DEST_SIZE`. As a general rule, the larger the fast recovery area, the more useful it becomes. For ease of use, Oracle recommends that you create a fast recovery area disk group on storage devices that can contain at least three days of recovery information. Ideally, the fast recovery area is large enough to hold a copy of all of your data files and control files, the online redo logs, and the archived redo log files needed to recover your database using the data file backups kept under your retention policy.

Multiple databases can use the same fast recovery area. For example, assume you have created a fast recovery area disk group on disks with 150 GB of storage, shared by 3 different databases. You can set the size of the fast recovery for each database depending on the importance of each database. For example, if `database1` is your least important database, `database2` is of greater importance, and `database3` is of greatest importance, then you can set different `DB_RECOVERY_FILE_DEST_SIZE` settings for each database to meet your retention

target for each database: 30 GB for database1, 50 GB for database2, and 70 GB for database3.

Creating the Fast Recovery Area Disk Group

Procedure to create the fast recovery area disk group.

1. Go to the Oracle Grid Infrastructure home bin directory, and start Oracle ASM Configuration Assistant (ASMCA).

For example:

```
$ cd /u01/app/oracle/product/19.0.0/grid/bin
$ ./asmca
```

ASMCA opens the home window.

2. Click **Disk Groups** in the left panel to open the Disk Groups tab.
3. Click **Create** to create a new disk group.

The Create Disk Groups window opens.

4. Provide configuration information for the fast recovery area as prompted:

In the Disk Group Name field, enter a descriptive name for the fast recovery area group. For example: FRA.

In the Redundancy section, select the level of redundancy you want to use. For example: Normal

In the Select Member Disks field, select eligible disks you want to add to the fast recovery area, and click **OK**.

5. When the Fast Recovery Area disk group creation is complete, click **Exit** and click **Yes** to confirm closing the ASMCA application.

Cloning an Oracle Database Home

Cloning an Oracle home involves creating a copy of the Oracle home and then configuring it for a new environment.

If you are performing multiple Oracle Database installations, then you may want to use cloning to create each Oracle home, because copying files from an existing Oracle Database installation takes less time than creating a new version of them. This method is also useful if the Oracle home that you are cloning has had patches applied to it. When you clone the Oracle home, the new Oracle home has the patch updates.

Perform these steps to clone an Oracle home.

Note

During cloning, Oracle Universal Installer (OUI) prompts you to run scripts that require root privileges.

1. Verify that the installation of Oracle Database that you want clone is successful.

You can do this by reviewing the `installActionsdate_time.log` file for the installation session, which is typically located in the `/u01/app/oracle/oraInventory/logs` directory.

If you install patches, then check their status using the following:

```
$ cd $ORACLE_HOME/OPatch
```

```
Include $ORACLE_HOME/OPatch in $PATH
```

```
$ opatch lsinventory
```

2. Stop all processes related to the Oracle home.
3. Create a ZIP or TAR file with the Oracle home (but not the Oracle base) directory.

For example, if the source Oracle installation is in the path `/u01/app/oracle/product/19.0.0/dbhome_1`, then you zip the `dbhome_1` directory by using the following command:

```
# zip -r dbhome_1.zip /u01/app/oracle/product/19.0.0/dbhome_1
```

You can also use the TAR command. For example:

```
# tar -cvf dbhome_1.tar /u01/app/oracle/product/19.0.0/dbhome_1
```

Do not include the `admin`, `fast_recovery_area`, and `oradata` directories that are under the Oracle base directory. These directories are created in the target installation later, when you create a new database there.

4. Copy the ZIP or TAR file to the root directory of the target computer. If you use File Transfer Protocol (FTP), then transfer the ZIP or TAR file in binary mode only.
5. Extract the ZIP or TAR file content using the following command:

```
# unzip -d / dbhome_1.zip  
# tar -xvf dbhome_1.tar
```

6. If necessary, change the ownership of the Oracle Database home to the Oracle Database installation owner user (`oracle`) belonging to the Oracle Inventory group (`oinstall`).

```
# chown -R oracle:oinstall /u01/app/oracle/product/19.0.0/dbhome_1
```

7. On the target computer, change the directory to the unzipped Oracle home directory, and remove all the `.ora` (`*.ora`) files present in the unzipped `$ORACLE_HOME/network/admin` directory.
8. Delete unnecessary files from the unzipped Oracle home directory.

The unzipped Oracle home directory contains files that are relevant only to the source Oracle home. The following example shows how to remove these unnecessary files from the unzipped Oracle home directory:

Remove the .ora files from the network/admin directory, and remove the old database entries from the dbs directory.

```
# cd $ORACLE_HOME
# rm -rf network/admin/*.ora
# rm dbs/old_database_entries
```

9. From the \$ORACLE_HOME/clone/bin directory, run the clone.pl file for the unzipped Oracle home.

Use the following syntax (you can also include one or more of the extended Oracle Database groups in the syntax):

```
$ORACLE_HOME/perl/bin/perl $ORACLE_HOME/clone/bin/clone.pl
ORACLE_BASE="target_oracle_base" ORACLE_HOME="target_oracle_home"
OSDBA_GROUP=OSDBA_privileged_group
OSOPER_GROUP=OSOPER_privileged_group
OSBACKUPDBA_GROUP=OSBACKUPDBA_privileged_group
OSDGDBA_GROUP=OSDGDBA_privileged_group
OSKMDBA_GROUP=OSKMDBA_privileged_group
OSRACDBA_GROUP=OSRACDBA_privileged_group -defaultHomeName
```

For example:

```
$ORACLE_HOME/perl/bin/perl $ORACLE_HOME/clone/bin/clone.pl
ORACLE_BASE="/u01/app/oracle" ORACLE_HOME="/u01/app/oracle/product/19.0.0/
dbhome_1"
OSDBA_GROUP=dba OSOPER_GROUP=oper OSBACKUPDBA_GROUP=backupdba
OSDGDBA_GROUP=dgdba OSKMDBA_GROUP=kmdba OSRACDBA_GROUP=racdba -
defaultHomeName
```

Note

- In this command, if you do not provide the parameters for the operating system groups, then clone.pl uses the operating system group values from the source home.
- Run the \$ORACLE_HOME/perl/bin/perl \$ORACLE_HOME/clone/bin/clone.pl -help command for more information about the command option flags.
- The clone.pl script is deprecated in Oracle Database 19c and can be removed in a future release. Hence, Oracle recommends that you use the software-only installation option, available in the database installer, instead of clone.pl to clone your database.

OUI starts, and then records the cloning actions in the cloneActionstimestamp.log file. This log file is typically located in /u01/app/oracle/orainventory/logs directory.

10. Use the following commands to run Net Configuration Assistant to configure the connection information for the new database:

```
$ cd $ORACLE_HOME/bin
$ ./netca
```

11. Use the following commands to run Database Configuration Assistant to create a new Oracle Database for the newly-cloned oracle home:

```
$ cd $ORACLE_HOME/bin  
$ ./dbca
```

Related Topics

- *Oracle Clusterware Administration and Deployment Guide*
- *Oracle Database Administrator's Guide*

Removing Oracle Database Software

These topics describe how to remove Oracle software and configuration files.

Use the `deinstall` command that is included in Oracle homes to remove Oracle software. Oracle does not support the removal of individual products or components.

Caution

If you have a standalone database on a node in a cluster, and if you have multiple databases with the same global database name (GDN), then you cannot use the `deinstall` command to remove one database only.

- [About Oracle Deinstallation Options](#)
You can stop and remove Oracle Database software and components in an Oracle Database home with the `deinstall` command.
- [Oracle Deinstallation \(Deinstall\)](#)
You can run the `deinstall` command from an Oracle home directory after installation.
- [Deinstallation Examples for Oracle Database](#)
Use these examples to help you understand how to run the `deinstall` command.
- [Downgrading Oracle Restart](#)
Use this procedure to deconfigure and downgrade Oracle Restart, or to troubleshoot Oracle Restart installation errors.
- [Deinstalling Previous Release Grid Home](#)
Use this procedure to deinstall the previous release Grid home.

About Oracle Deinstallation Options

You can stop and remove Oracle Database software and components in an Oracle Database home with the `deinstall` command.

You can remove the following software using `deinstall` :

- Oracle Database
- Oracle Grid Infrastructure, which includes Oracle Clusterware and Oracle Automatic Storage Management (Oracle ASM)
- Oracle Real Application Clusters (Oracle RAC)
- Oracle Database Client

The `deinstall` command is available in Oracle home directories after installation. It is located in the `$ORACLE_HOME/deinstall` directory.

`deinstall` creates a response file by using information in the Oracle home and using the information you provide. You can use a response file that you generated previously by running

the `deinstall` command using the `-checkonly` option. You can also edit the response file template.

If you run `deinstall` to remove an Oracle Grid Infrastructure installation, then the deinstaller prompts you to run the `deinstall` command as the root user. For Oracle Grid Infrastructure for a cluster, the script is `rootcrs.sh`, and for Oracle Grid Infrastructure for a standalone server (Oracle Restart), the script is `roothas.sh`.

Note

- You must run the `deinstall` command from the same release to remove Oracle software. Do not run the `deinstall` command from a later release to remove Oracle software from an earlier release. For example, do not run the `deinstall` command from the 19c Oracle home to remove Oracle software from an existing 11.2.0.4 Oracle home.
- Starting with Oracle Database 12c Release 1 (12.1.0.2), the `roothas.sh` script replaces the `roothas.pl` script in the Oracle Grid Infrastructure home for Oracle Restart, and the `rootcrs.sh` script replaces the `rootcrs.pl` script in the Grid home for Oracle Grid Infrastructure for a cluster.

If the software in the Oracle home is not running (for example, after an unsuccessful installation), then `deinstall` cannot determine the configuration, and you must provide all the configuration details either interactively or in a response file.

In addition, before you run `deinstall` for Oracle Grid Infrastructure installations:

- Dismount Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and disable Oracle Automatic Storage Management Dynamic Volume Manager (Oracle ADVM).
- If Grid Naming Service (GNS) is in use, then notify your DNS administrator to delete the subdomain entry from the DNS.

Files Deleted by deinstall

When you run `deinstall`, if the central inventory (`oraInventory`) contains no other registered homes besides the home that you are deconfiguring and removing, then `deinstall` removes the following files and directory contents in the Oracle base directory of the Oracle Database installation owner:

- `admin`
- `cfgtoollogs`
- `checkpoints`
- `diag`
- `oradata`
- `fast_recovery_area`

Oracle strongly recommends that you configure your installations using an Optimal Flexible Architecture (OFA) configuration, and that you reserve Oracle base and Oracle home paths for exclusive use of Oracle software. If you have any user data in these locations in the Oracle base that is owned by the user account that owns the Oracle software, then `deinstall` deletes this data.

 Caution

deinstall deletes Oracle Database configuration files, user data, and fast recovery area (FRA) files even if they are located outside of the Oracle base directory path.

Oracle Deinstallation (Deinstall)

You can run the deinstall command from an Oracle home directory after installation.

Purpose

deinstall stops Oracle software, and removes Oracle software and configuration files on the operating system for a specific Oracle home.

Syntax

The deinstall command uses the following syntax:

```
(./deinstall [-silent] [-checkonly] [-paramfile complete path of input response file]  
[-params name1=value name2=value . . .]  
[-o complete path of directory for saving files]  
[-tmpdir complete path of temporary directory to use]  
[-logdir complete path of log directory to use] [-local] [-  
skipLocalHomeDeletion] [-skipRemoteHomeDeletion] [-help]
```

Parameters

Parameter	Description
-silent	Use this flag to run deinstall in noninteractive mode. This option requires one of the following: • A working system that it can access to determine the installation and configuration information. The -silent flag does not work with failed installations. • A response file that contains the configuration values for the Oracle home that is being deinstalled or deconfigured. You can generate a response file to use or modify by running deinstall with the -checkonly flag. deinstall then discovers information from the Oracle home to deinstall and deconfigure. It generates the response file that you can then use with the -silent option. You can also modify the template file deinstall.rsp.tmpl, located in the \$ORACLE_HOME/deinstall/response directory.

Parameter	Description
-checkonly	Use this flag to check the status of the Oracle software home configuration. Running <code>deinstall</code> with the <code>-checkonly</code> flag does not remove the Oracle configuration. The <code>-checkonly</code> flag generates a response file that you can then use with the <code>deinstall</code> command and <code>-silent</code> option.
-paramfile <i>complete path of input response file</i>	Use this flag to run <code>deinstall</code> with a response file in a location other than the default. When you use this flag, provide the complete path where the response file is located. The default location of the response file is <code>\$ORACLE_HOME/deinstall/response</code> .
-params <i>[name1=value name2=value name3=value . . .]</i>	Use this flag with a response file to override one or more values to change in a response file you have created.
-o <i>complete path of directory for saving response files</i>	Use this flag to provide a path other than the default location where the response file (<code>deinstall.rsp.tpl</code>) is saved. The default location of the response file is <code>\$ORACLE_HOME/deinstall/response</code> .
-tmpdir <i>complete path of temporary directory to use</i>	Use this flag to specify a non-default location where <code>deinstall</code> writes the temporary files for the deinstallation.
-logdir <i>complete path of log directory to use</i>	Use this flag to specify a non-default location where <code>deinstall</code> writes the log files for the deinstallation.
-local	Use this flag on a multinode environment to deinstall Oracle software in a cluster. When you run <code>deinstall</code> with this flag, it deconfigures and deinstalls the Oracle software on the local node (the node where <code>deinstall</code> is run). On remote nodes, it deconfigures Oracle software, but does not deinstall the Oracle software.
-skipLocalHomeDeletion	Use this flag in Oracle Grid Infrastructure installations on a multinode environment to deconfigure a local Grid home without deleting the Grid home.
-skipRemoteHomeDeletion	Use this flag in Oracle Grid Infrastructure installations on a multinode environment to deconfigure a remote Grid home without deleting the Grid home.
-help	Use this option to obtain additional information about the command option flags.

Deinstallation Examples for Oracle Database

Use these examples to help you understand how to run the `deinstall` command.

Run `deinstall` from the `$ORACLE_HOME/deinstall` directory. The deinstallation starts without prompting you for the Oracle home path.

```
$ ./deinstall
```

You can generate a deinstallation response file by running `deinstall` with the `-checkonly` flag. Alternatively, you can use the response file template located at `$ORACLE_HOME/deinstall/response/deinstall.rsp.tmpl`. If you have a response file, then use the optional flag `-paramfile` to provide a path to the response file.

In the following example, the `deinstall` command is in the path `/u01/app/oracle/product/19.0.0/dbhome_1/deinstall`. It uses a response file called `my_db_paramfile.tmpl` in the software owner location `/home/usr/oracle`:

```
$ cd /u01/app/oracle/product/19.0.0/dbhome_1/deinstall
$ ./deinstall -paramfile /home/usr/oracle/my_db_paramfile.tmpl
```

To remove the Oracle Grid Infrastructure home, use the `deinstall` command in the Oracle Grid Infrastructure home.

In this example, the Oracle Grid Infrastructure home is `/u01/app/oracle/product/19.0.0/grid`

```
$ cd /u01/app/oracle/product/19.0.0/grid/deinstall
$ ./deinstall -paramfile /home/usr/oracle/my_grid_paramfile.tmpl
```

Downgrading Oracle Restart

Use this procedure to deconfigure and downgrade Oracle Restart, or to troubleshoot Oracle Restart installation errors.

Running `roothas.sh` with the command flags `-deconfig -force` enables you to deconfigure Oracle Restart without removing the installed binaries. This feature is useful if you encounter an error during an Oracle Grid Infrastructure for a standalone server installation. For example, when you run the `root.sh` command, and find a missing operating system package. By running `roothas.sh -deconfig -force`, you can deconfigure Oracle Restart, correct the cause of the error, and then run `root.sh` again.

1. As the `oracle` user, create a backup of the SPFILE to a PFILE.

```
CREATE PFILE='/u01/app/oracle/product/19.0.0/dbhome_1/dbs/test_init.ora'
FROM SPFILE='/u01/oracle/dbs/test_spfile.ora';
```

2. List all the Oracle Databases on the server with their version, unique name of the database, and Oracle home information.

```
$ srvctl config database -home
```

3. Downgrade Oracle Database. Refer to *Oracle Database Upgrade Guide* for more information about required pre-downgrade tasks, downgrade tasks, post-downgrade tasks, and compatibility information.

Note

Downgrade Oracle Database only if the Oracle Database version is higher than the Oracle Restart version to which you are downgrading Oracle Restart.

4. As the oracle user, downgrade the Oracle Restart resources corresponding to the Oracle Database, only if you have downgraded your Oracle Database.

```
$ srvctl downgrade database -d db_unique_name -oraclehome $ORACLE_HOME -t  
to_version
```

5. Inspect the Oracle Restart configuration of each database, service, and listener.

```
$ srvctl config database -db db_unique_name  
$ srvctl config service -db db_unique_name  
$ srvctl config listener -listener listener_name
```

Make a note of the configuration information and use this information when adding the components back to Oracle Restart.

6. Stop all databases and listeners that are running before you deconfigure or downgrade Oracle Restart.

```
$ srvctl stop database -db db_unique_name  
$ srvctl stop listener [-listener listener_name]
```

7. As the root user, run `roothas.sh` with the `-deconfig -force` flags to deconfigure Oracle Restart.

```
# /u01/app/oracle/product/19.0.0/grid/crs/install/roothas.sh -deconfig -  
force
```

8. As the grid user, update the Oracle central inventory (oraInventory).

```
$ /u01/app/oracle/product/19.0.0/grid/oui/bin/runInstaller -updateNodeList  
-silent ORACLE_HOME=upgraded_Grid_home -local CRS=false
```

9. As the root user, run `roothas.sh` with the `-unlock` flag to unlock the previous release Oracle Restart home.

```
# /u01/app/oracle/product/18.0.0/grid/crs/install/roothas.sh -unlock -  
dstcrshome previous_release_Grid_home
```

10. As the grid user, reconfigure the previous release Oracle Restart home using the `gridSetup.sh` command.

```
$ /u01/app/oracle/product/18.0.0/grid/gridSetup.sh
```

11. As the oracle user, add the components back to Oracle Restart with the same attributes that you noted down before deconfiguring Oracle Restart.

- a. Add Oracle Database to the Oracle Restart configuration.

```
$ srvctl add database -db db_unique_name -oraclehome Oracle_home
```

- b. Add the listener to the Oracle Restart configuration.

```
$ srvctl add listener -listener listener_name -oraclehome Oracle_home
```

For the `-oraclehome` parameter, provide the Oracle home path from which the listener was running before the downgrade.

- c. Add each service to the database, using the `srvctl add service` command.

```
$ srvctl add service -db db_unique_name -service service_name_list
```

Related Topics

- [Oracle Database Upgrade Guide](#)

Deinstalling Previous Release Grid Home

Use this procedure to deinstall the previous release Grid home.

For upgrades from previous releases, if you want to deinstall the previous release Grid home, then perform the following steps:

1. Log in as the root user.
2. Manually change the permissions of the previous release Grid home.

```
# chmod -R 775 /u01/app/oracle/product/18.0.0/grid
# chown -R oracle:oinstall /u01/app/oracle/product/18.0.0/grid
# chown oracle /u01/app/oracle/product/18.0.0
```

In this example:

- `/u01/app/oracle/product/18.0.0/grid` is the previous release Oracle Grid Infrastructure for a standalone server home
 - `oracle` is the Oracle Grid Infrastructure installation owner user
 - `oinstall` is the name of the Oracle Inventory group (OINSTALL group)
 - `/u01/app/oracle/product/18.0.0` is the parent directory of the previous Grid home.
3. Log in as the Oracle Grid Infrastructure software owner user (`oracle`) and run the `deinstall` command.

Completing Preinstallation Tasks Manually

You can complete the preinstallation configuration tasks manually.

Oracle recommends that you use Oracle Universal Installer and Cluster Verification Utility fixup scripts to complete minimal configuration settings. If you cannot use fixup scripts, then complete minimum system settings manually.

- [Configuring Kernel Parameters on Oracle Solaris](#)
These topics explain how to configure kernel parameters manually for Oracle Solaris if you cannot complete them using the fixup scripts.
- [Configuring Shell Limits for Oracle Solaris](#)
For each installation software owner user account, check the shell limits for installation.

Configuring Kernel Parameters on Oracle Solaris

These topics explain how to configure kernel parameters manually for Oracle Solaris if you cannot complete them using the fixup scripts.

- [Minimum Parameter Settings for Installation](#)
Use this table to set parameters manually if you cannot use the fixup scripts.
- [Checking Shared Memory Resource Controls](#)
Use the `prctl` command to make runtime interrogations of and modifications to the resource controls associated with an active process, task, or project on the system.
- [Configuring Additional Shared Memory Identifiers Limit](#)
Starting with 18c, on Oracle Solaris systems, Oracle Database uses a new method of sharing memory among a group of processes, also known as Managed Global Areas (MGA). The operating system memory allocation mechanism for this new method is Optimized Shared Memory (OSM).
- [Displaying and Changing Kernel Parameter Values](#)
Use these procedures to display the current value specified for resource controls and to change them if necessary:
- [Setting UDP and TCP Kernel Parameters Manually](#)
If you do not use a Fixup script or CVU to set ephemeral ports, then set TCP/IP ephemeral port range parameters to provide enough ephemeral ports for the anticipated server workload.

Minimum Parameter Settings for Installation

Use this table to set parameters manually if you cannot use the fixup scripts.

Table 14-1 Minimum Oracle Solaris Resource Control Parameter Settings

Resource Control	Minimum Value
<code>project.max-sem-ids</code>	100
<code>process.max-sem-nsems</code>	256

Table 14-1 (Cont.) Minimum Oracle Solaris Resource Control Parameter Settings

Resource Control	Minimum Value
<code>project.max-shm-memory</code>	This value varies according to the RAM size. See section “Requirements for Shared Memory Resources” for minimum values.
<code>project.max-shm-ids</code>	100 See section “Configuring Additional Shared Memory Identifiers Limit”.
<code>tcp_smallest_anon_port</code>	9000
<code>tcp_largest_anon_port</code>	65500
<code>udp_smallest_anon_port</code>	9000
<code>udp_largest_anon_port</code>	65500

Guidelines for Setting Resource Control Parameters

- Unless otherwise specified, the kernel parameter and shell limit values in the preceding table are minimum values only. Verify that the kernel parameters shown in the preceding table are set to values greater than or equal to the minimum value shown. For production database systems, Oracle recommends that you tune these values to optimize the performance of the system. See your operating system documentation for more information about kernel resource management.
- If the current value for any parameter is greater than the value listed in the preceding table, then the Fixup scripts do not change the value of that parameter.
- The `project.max-shm-memory` resource control value assumes that no other application is using the shared memory segment from this project other than the Oracle instances. If applications, other than the Oracle instances are using the shared memory segment, then you must add that shared memory usage to the `project.max-shm-memory` resource control value.
- `project.max-shm-memory` resource control = the cumulative sum of all shared memory allocated on each Oracle database instance started under the corresponding project.
- Ensure that `memory_target` or `max_sga_size` does not exceed `process.max-address-space` and `project.max-shm-memory`. For more information, see My Oracle Support Note [1370537.1](#).

Requirements for Shared Memory Resources `project.max-shm-memory`**Table 14-2 Requirement for Resource Control `project.max-shm-memory`**

RAM	<code>project.max-shm-memory</code> setting
1 GB to 16 GB	Half the size of physical memory
Greater than 16 GB	At least 8 GB

Related Topics

- [Checking TCP Network Protocol Buffer for Direct NFS Client](#)
Check your TCP network buffer size to ensure that it is adequate for the speed of your servers.

Related Topics

- [Configuring Additional Shared Memory Identifiers Limit](#)
Starting with 18c, on Oracle Solaris systems, Oracle Database uses a new method of sharing memory among a group of processes, also known as Managed Global Areas (MGA). The operating system memory allocation mechanism for this new method is Optimized Shared Memory (OSM).

Checking Shared Memory Resource Controls

Use the `prctl` command to make runtime interrogations of and modifications to the resource controls associated with an active process, task, or project on the system.

To view the current value of `project.max-shm-memory` set for a project and system-wide:

```
# prctl -n project.max-shm-memory -i project default
```

default is the project ID obtained by running the `id -p` command.

```
project.max-shm-memory
```

```
prctl -n project.max-shm-memory -v 6gb -r -i project default
```

Related Topics

- [Administering Oracle Solaris 11](#)

Configuring Additional Shared Memory Identifiers Limit

Starting with 18c, on Oracle Solaris systems, Oracle Database uses a new method of sharing memory among a group of processes, also known as Managed Global Areas (MGA). The operating system memory allocation mechanism for this new method is Optimized Shared Memory (OSM).

Oracle recommends that you configure the additional shared memory identifiers and increase the shared memory limits as follows:

Additional number of MGA segments (`project.max-shm-ids`) = MGA memory size / granule size

Where:

MGA memory size is the cumulative MGA memory requirement size for all processes in an Oracle Database instance. Typically, 2MB per Oracle process.

Minimum value of granule size is 4 MB. For large SGA, granule size may increase automatically upto 512 MB.

Refer to the following table to determine the approximate granule size.

Table 14-3 Granule Size for SGA Values

SGA	Granule Size
Less than 1 GB	4 MB
Between 1 GB and 8 GB	16 MB
Between 8 GB and 16 GB	32 MB

Table 14-3 (Cont.) Granule Size for SGA Values

SGA	Granule Size
Between 16 GB and 32 GB	64 MB
Between 32 GB and 64 GB	128 MB
Between 64 GB and 128 GB	256 MB
Greater than 128 GB	512 MB

Displaying and Changing Kernel Parameter Values

Use these procedures to display the current value specified for resource controls and to change them if necessary:

Displaying Resource Control Values

1. To display the current values of the resource control:

```
$ id -p // to verify the project id
uid=100(oracle) gid=100(dba) projid=1 (group.dba)
$ prctl -n project.max-shm-memory -i project group.dba
$ prctl -n project.max-sem-ids -i project group.dba
```

2. To change the current values use the `prctl` command. For example:

- To modify the value of `max-shm-memory` to 6 GB:


```
# prctl -n project.max-shm-memory -v 6gb -r -i project group.dba
```
- To modify the value of `max-sem-ids` to 256:


```
# prctl -n project.max-sem-ids -v 256 -r -i project group.dba
```

Note

When you use the `prctl` command (Resource Control) to change system parameters, you do not have to restart the system for these parameter changes to take effect. However, the changed parameters do not persist after a system restart.

Modifying Resource Control Values

Use the following procedure to modify the resource control project settings, so that they persist after a system restart:

1. By default, Oracle instances are run as the `oracle` user of the `dba` group. A project with the name `group.dba` is created to serve as the default project for the `oracle` user. Run the `id` command to verify the default project for the `oracle` user:

```
# su - oracle
$ id -p
```

```
uid=100(oracle) gid=100(dba) projid=100(group.dba)
$ exit
```

2. To set the maximum shared memory size to 2 GB, run the `projmod` command:

```
# projmod -sK "project.max-shm-memory=(privileged,2G,deny)" group.dba
```

Alternatively, add the resource control value `project.max-shm-memory=(privileged,2147483648,deny)` to the last field of the project entries for the Oracle project.

3. Check the values for the `/etc/project` file:

```
# cat /etc/project
```

The output is similar to the following:

```
system:0:::
user.root:1:::
noproject:2:::
default:3:::
group.staff:10:::
group.dba:100:Oracle default project :: project.max-shm-
memory=(privileged,2147483648,deny)
```

4. To verify that the resource control is active, check process ownership, and run the commands `id` and `prctl`:

```
# su - oracle
$ id -p
uid=100(oracle) gid=100(dba) projid=100(group.dba)
$ prctl -n project.max-shm-memory -i process $$
process: 5754: -bash
NAME                                PRIVILEGE    VALUE    FLAG    ACTION
RECIPIENT
project.max-shm-memory privileged    2.00GB    -       deny
```

Note

The value for the maximum shared memory depends on the SGA requirements and should be set to a value greater than the SGA size.

Related Topics

- [Oracle Solaris Tunable Parameters Reference Manual](#)

Setting UDP and TCP Kernel Parameters Manually

If you do not use a Fixup script or CVU to set ephemeral ports, then set TCP/IP ephemeral port range parameters to provide enough ephemeral ports for the anticipated server workload.

Ensure that the lower range is set to at least 9000 or higher, to avoid Well Known ports, and to avoid ports in the Registered Ports range commonly used by Oracle and other server ports.

Set the port range high enough to avoid reserved ports for any applications you may intend to use. If the lower value of the range you have is greater than 9000, and the range is large enough for your anticipated workload, then you can ignore Oracle Universal Installer warnings regarding the ephemeral port range.

On Oracle Solaris 11, use the `ipadm` command to check your current range for ephemeral ports:

```
# ipadm show-prop -p smallest_anon_port,largest_anon_port tcp
```

PROTO	PROPERTY	PERM	CURRENT	PERSISTENT	DEFAULT	POSSIBLE
tcp	smallest_anon_port	rw	32768	--	32768	1024-65535
tcp	largest_anon_port	rw	65500	--	65535	32768-65535

In the preceding examples, the ephemeral ports are set to the default range (32768-65535).

If necessary for your anticipated workload or number of servers, update the UDP and TCP ephemeral port range to a broader range. For example:

On Oracle Solaris 11:

```
# ipadm set-prop -p smallest_anon_port=9000 tcp
# ipadm set-prop -p largest_anon_port=65500 tcp
# ipadm set-prop -p smallest_anon_port=9000 udp
# ipadm set-prop -p largest_anon_port=65500 udp
```

Oracle recommends that you make these settings permanent. Refer to your system administration documentation for information about how to automate this ephemeral port range alteration on system restarts.

Configuring Shell Limits for Oracle Solaris

For each installation software owner user account, check the shell limits for installation.

Note

The shell limit values in this section are minimum values only. For production database systems, Oracle recommends that you tune these values to optimize the performance of the system. See your operating system documentation for more information about configuring shell limits.

The `ulimit` settings determine process memory related resource limits. Verify that the following shell limits are set to the values shown:

Table 14-4 Oracle Solaris Shell Limit Recommended Ranges

Resource Shell Limit	Description	Soft Limit	Hard Limit
STACK	Size (KB) of the stack segment of the process	at least 10240	at most 32768
NOFILES	Open file descriptors	at least 1024	at least 65536
MAXUPRC or MAXPROC	Maximum user processes	at least 2047	at least 16384

To display the current value specified for these shell limits:

```
ulimit -s  
ulimit -n
```

A

Response Files

Review the following topics to install and configure Oracle Database, Oracle Grid Infrastructure, and other Oracle products using response files.

- [How Response Files Work](#)
Response files can assist you with installing an Oracle product multiple times on multiple computers.
- [Reasons for Using Silent Mode or Response File Mode](#)
Review this section for use cases for running the installer in silent mode or response file mode.
- [Using Response Files](#)
Review this information to use response files.
- [Preparing Response Files](#)
Review this information to prepare response files for use during silent mode or response file mode installations.
- [Running Oracle Universal Installer Using a Response File](#)
After creating the response file, run Oracle Universal Installer at the command line, specifying the response file you created, to perform the installation.
- [Running Configuration Assistants Using Response Files](#)
You can run configuration assistants in response file or silent mode to configure and start Oracle software after it is installed on the system. To run configuration assistants in response file or silent mode, you must copy and edit a response file template.
- [Postinstallation Configuration Using Response File Created During Installation](#)
Use response files to configure Oracle software after installation. You can use the same response file created during installation to also complete postinstallation configuration.
- [Postinstallation Configuration Using the ConfigToolAllCommands Script](#)
You can create and run a response file configuration after installing Oracle software. The configToolAllCommands script requires users to create a second response file, of a different format than the one used for installing the product.

How Response Files Work

Response files can assist you with installing an Oracle product multiple times on multiple computers.

When you start Oracle Universal Installer (OUI), you can use a response file to automate the installation and configuration of Oracle software, either fully or partially. OUI uses the values contained in the response file to provide answers to some or all installation prompts.

Typically, the installer runs in interactive mode, which means that it prompts you to provide information in graphical user interface (GUI) screens. When you use response files to provide this information, you run the installer from a command prompt using either of the following modes:

- Silent mode

If you include responses for all of the prompts in the response file and specify the `-silent` option when starting the installer, then it runs in silent mode. During a silent mode installation, the installer does not display any screens. Instead, it displays progress information in the terminal that you used to start it.

- Response file mode

If you include responses for some or all of the prompts in the response file and omit the `-silent` option, then the installer runs in response file mode. During a response file mode installation, the installer displays all the screens, screens for which you specify information in the response file, and also screens for which you did not specify the required information in the response file.

You define the settings for a silent or response file installation by entering values for the variables listed in the response file. For example, to specify the Oracle home name, provide the Oracle home path for the `ORACLE_HOME` environment variable:

```
ORACLE_HOME=/u01/app/oracle/product/19.0.0/dbhome_1
```

Reasons for Using Silent Mode or Response File Mode

Review this section for use cases for running the installer in silent mode or response file mode.

Mode	Uses
Silent	Use silent mode for the following installations: • Complete an unattended installation, which you schedule using operating system utilities such as <code>at</code>. • Complete several similar installations on multiple systems without user interaction. • Install the software on a system that does not have X Window System software installed on it. The installer displays progress information on the terminal that you used to start it, but it does not display any of the installer screens.
Response file	Use response file mode to complete similar Oracle software installations on more than one system, providing default answers to some, but not all of the installer prompts.

Using Response Files

Review this information to use response files.

Use the following general steps to install and configure Oracle products using the installer in silent or response file mode:

Note

You must complete all required preinstallation tasks on a system before running the installer in silent or response file mode.

1. Prepare a response file.
2. Run the installer in silent or response file mode.

3. Run the root scripts as prompted by Oracle Universal Installer.
4. If you completed a software-only installation, then run Net Configuration Assistant and Oracle DBCA in silent or response file mode to create the database listener and an Oracle Database instance respectively.

Preparing Response Files

Review this information to prepare response files for use during silent mode or response file mode installations.

- [Editing a Response File Template](#)
Oracle provides response file templates for each product and each configuration tool.
- [Recording Response Files](#)
You can use OUI in interactive mode to record response files, which you can then edit and use to complete silent mode or response file mode installations. This method is useful for Advanced or software-only installations.

Editing a Response File Template

Oracle provides response file templates for each product and each configuration tool.

About Response File Templates

For Oracle Database, the response file templates are located in the `$ORACLE_HOME/install/response` directory. For Oracle Grid Infrastructure, the response file templates are located in the `Grid_home/install/response` directory.

Where, *Grid_home* is the Oracle Grid Infrastructure home directory path.

Note

If you copied the software to a hard disk, then the response files are located in the `$ORACLE_HOME/install/response` directory.

All response file templates contain comment entries, sample formats, examples, and other useful instructions. Read the response file instructions to understand how to specify values for the response file variables, so that you can customize your installation.

The following table lists the response files provided with this software:

Table A-1 Response Files for Oracle Database and Oracle Grid Infrastructure

Response File	Description
db_install.rsp	Silent installation of Oracle Database.
dbca.rsp	Silent creation and configuration of Oracle Database using Oracle DBCA.
netca.rsp	Silent configuration of Oracle Net using Oracle NETCA.
gridsetup.rsp	Silent configuration of Oracle Grid Infrastructure installations.

 Caution

When you modify a response file template and save a file for use, the response file may contain plain text passwords. Ownership of the response file should be given to the Oracle software installation owner only, and permissions on the response file should be changed to 600. Oracle strongly recommends that database administrators or other administrators delete or secure response files when they are not in use.

To copy and modify a response file:

1. Copy the response file from the response file directory to a directory on your system:

For example, for Oracle Database:

```
$ cp $ORACLE_HOME/install/response/db_install.rsp local_directory
```

2. Open the response file in a text editor:

```
$ vi /local_directory/db_install.rsp
```

3. Follow the instructions in the file to edit it.

 Note

The installer or configuration assistant fails if you do not correctly configure the response file. Also, ensure that your response file name has the `.rsp` suffix.

4. Secure the response file by changing the permissions on the file to 600:

```
$ chmod 600 /local_dir/db_install.rsp
```

Ensure that only the Oracle software owner user can view or modify response files or consider deleting them after the installation succeeds.

 Note

A fully-specified response file for an Oracle Database installation contains the passwords for database administrative accounts and for a user who is a member of the OSDBA group (required for automated backups).

Recording Response Files

You can use OUI in interactive mode to record response files, which you can then edit and use to complete silent mode or response file mode installations. This method is useful for Advanced or software-only installations.

You can save all the installation steps into a response file during installation by clicking **Save Response File** on the Summary page. You can use the generated response file for a silent installation later.

When you record the response file, you can either complete the installation, or you can exit from the installer on the Summary page, before OUI starts to set up the software to the system.

If you use record mode during a response file mode installation, then the installer records the variable values that were specified in the original source response file into the new response file.

Note

OUI does not save passwords while recording the response file.

To record a response file:

1. Complete preinstallation tasks as for a standard installation.
When you run the installer to record a response file, it checks the system to verify that it meets the requirements to install the software. For this reason, Oracle recommends that you complete all of the required preinstallation tasks and record the response file while completing an installation.
2. Ensure that the Oracle software owner user (typically `oracle`) has permissions to create or write to the Oracle home path that you specify when you run the installer.
3. On each installation screen, specify the required information.
4. When the installer displays the Summary screen, perform the following steps:
 - a. Click **Save Response File**. In the window, specify a file name and location for the new response file. Click **Save** to write the responses you entered to the response file.
 - b. Click **Finish** to continue with the installation.
Click **Cancel** if you do not want to continue with the installation. The installation stops, but the recorded response file is retained.

Note

Ensure that your response file name has the `.rsp` suffix.

5. Before you use the saved response file on another system, edit the file and make any required changes. Use the instructions in the file as a guide when editing it.

Running Oracle Universal Installer Using a Response File

After creating the response file, run Oracle Universal Installer at the command line, specifying the response file you created, to perform the installation.

Run Oracle Universal Installer at the command line, specifying the response file you created. The Oracle Universal Installer executables, `runInstaller` and `gridSetup.sh`, provide several options. For help information on the full set of these options, run the `gridSetup.sh` or `runInstaller` command with the `-help` option. For example:

- For Oracle Database:

```
$ $ORACLE_HOME/runInstaller -help
```

- For Oracle Grid Infrastructure:

```
$ /u01/app/oracle/product/19.0.0/grid/gridSetup.sh -help
```

The help information appears in a window after some time.

To run the installer using a response file:

1. Complete the preinstallation tasks for a normal installation.
2. Log in as the software installation owner user.
3. If you are completing a response file mode installation, then set the operating system DISPLAY environment variable for the user running the installation.

Note

You do not have to set the DISPLAY environment variable if you are completing a silent mode installation.

4. To start the installer in silent or response file mode, enter a command similar to the following:

- For Oracle Database:

```
$ $ORACLE_HOME/runInstaller [-silent] \  
-responseFile responsefilename
```

- For Oracle Grid Infrastructure:

```
$ /u01/app/oracle/product/19.0.0/grid/gridSetup.sh [-silent] \  
-responseFile responsefilename
```

Note

Do not specify a relative path to the response file. If you specify a relative path, then the installer fails.

In this example:

- `-silent` runs the installer in silent mode.
 - `responsefilename` is the full path and file name of the installation response file that you configured.
5. If this is the first time you are installing Oracle software on your system, then Oracle Universal Installer prompts you to run the `oraInstRoot.sh` script.

Log in as the root user and run the `oraInstRoot.sh` script:

```
$ su root  
password:  
# /u01/app/oraInventory/oraInstRoot.sh
```

Note

You do not have to manually create the `oraInst.loc` file. Running the `oraInstRoot.sh` script is sufficient as it specifies the location of the Oracle Inventory directory.

6. When the installation completes, log in as the root user and run the `root.sh` script. For example:

```
$ su root
password:
# $ORACLE_HOME/root.sh
```

Running Configuration Assistants Using Response Files

You can run configuration assistants in response file or silent mode to configure and start Oracle software after it is installed on the system. To run configuration assistants in response file or silent mode, you must copy and edit a response file template.

Note

If you copied the software to a hard disk, then the response file template is located in the `$ORACLE_HOME/install/response` directory.

- [Running Net Configuration Assistant Using Response Files](#)
You can run NETCA in silent mode to configure and start an Oracle Net Listener on the system, and configure naming methods and Oracle Net service names.
- [Running Oracle DBCA Using Response Files](#)
You can run Oracle Database Configuration Assistant (Oracle DBCA) in response file mode to configure and start an Oracle database on the system.

Running Net Configuration Assistant Using Response Files

You can run NETCA in silent mode to configure and start an Oracle Net Listener on the system, and configure naming methods and Oracle Net service names.

To run Oracle Net Configuration Assistant (NETCA) in silent mode, you must copy and edit a response file template. Oracle provides a response file template named `netca.rsp` in the `$ORACLE_HOME/assistants/netca` directory.

To run Net Configuration Assistant using a response file:

1. Copy the `netca.rsp` response file template from the response file directory to a directory on your system:

```
$ cp $ORACLE_HOME/assistants/netca/netca.rsp local_directory
```

2. Open the response file in a text editor:

```
$ vi /local_directory/netca.rsp
```

3. Follow the instructions in the file to edit it.

Note

Net Configuration Assistant fails if you do not correctly configure the response file.

4. Log in as the Oracle software owner user, and set the `ORACLE_HOME` environment variable to specify the correct Oracle home directory.
5. Enter a command similar to the following to run Net Configuration Assistant in silent mode:

```
$ $ORACLE_HOME/bin/netca -silent -responsefile /local_directory/netca.rsp
```

In this command:

- The `/silent` option indicates to run Net Configuration Assistant in silent mode.
- `local_directory` is the full path of the directory where you copied the `netca.rsp` response file template.

Running Oracle DBCA Using Response Files

You can run Oracle Database Configuration Assistant (Oracle DBCA) in response file mode to configure and start an Oracle database on the system.

To run Oracle DBCA in response file mode, you must copy and edit a response file template. Oracle provides a response file template named `dbca.rsp` in the `ORACLE_HOME/assistants/dbca` directory. To run Oracle DBCA in response file mode, you must use the `-responseFile` flag in combination with the `-silent` flag. You must also use a graphical display and set the `DISPLAY` environment variable.

To run Oracle DBCA in response file mode:

1. Copy the `dbca.rsp` response file template from the response file directory to a directory on your system:

```
$ cp /directory_path/assistants/dbca/dbca.rsp local_directory
```

In this example, `directory_path` is the path of the directory where you have copied the installation binaries.

As an alternative to editing the response file template, you can also create a database by specifying all required information as command line options when you run Oracle DBCA. For information about the list of options supported, enter the following command:

```
$ $ORACLE_HOME/bin/dbca -help
```

2. Open the response file in a text editor:

```
$ vi /local_dir/dbca.rsp
```

3. Follow the instructions in the file to edit the file.

Note

Oracle DBCA fails if you do not correctly configure the response file.

4. Log in as the Oracle software owner user, and set the ORACLE_HOME environment variable to specify the correct Oracle home directory.
5. To run Oracle DBCA in response file mode, set the DISPLAY environment variable.
6. Use the following command syntax to run Oracle DBCA in silent or response file mode using a response file:

```
$ORACLE_HOME/bin/dbca [-silent] -createDatabase -responseFile /local_dir/  
dbca.rsp
```

In this example:

- `-silent` option indicates that Oracle DBCA runs in silent mode.
- `local_dir` is the full path of the directory where you copied the `dbca.rsp` response file template.

During configuration, Oracle DBCA displays a window that contains the status messages and a progress bar.

Postinstallation Configuration Using Response File Created During Installation

Use response files to configure Oracle software after installation. You can use the same response file created during installation to also complete postinstallation configuration.

- [Using the Installation Response File for Postinstallation Configuration](#)
Starting with Oracle Database 12c release 2 (12.2), you can use the response file created during installation to also complete postinstallation configuration.
- [Running Postinstallation Configuration Using Response File](#)
You can use a response file to complete postinstallation tasks on one or more servers simultaneously.

Using the Installation Response File for Postinstallation Configuration

Starting with Oracle Database 12c release 2 (12.2), you can use the response file created during installation to also complete postinstallation configuration.

Run the installer with the `-executeConfigTools` option to configure configuration assistants after installing Oracle Grid Infrastructure or Oracle Database. You can use the response file located at `$ORACLE_HOME/install/response/product_timestamp.rsp` to obtain the passwords required to run the configuration tools. You must update the response file with the required passwords before running the `-executeConfigTools` command.

Oracle strongly recommends that you maintain security with a password response file:

- Permissions on the response file should be set to 600.
- The owner of the response file should be the installation owner user, with the group set to the central inventory (`oraInventory`) group.

Example A-1 Response File Passwords for Oracle Grid Infrastructure (grid user)

```
grid.install.crs.config.ipmi.bmcPassword=password  
grid.install.asm.SYSASMPassword=password
```

```
grid.install.asm.monitorPassword=password  
grid.install.config.emAdminPassword=password
```

If you do not have a BMC card, or you do not want to enable IPMI, then leave the `ipmi.bmcPassword` input field blank.

If you do not want to enable Oracle Enterprise Manager for management, then leave the `emAdminPassword` password field blank.

Example A-2 Response File Passwords for Oracle Grid Infrastructure for a Standalone Server (oracle user)

```
oracle.install.asm.SYSASMPassword=password  
oracle.install.asm.monitorPassword=password  
oracle.install.config.emAdminPassword=password
```

If you do not want to enable Oracle Enterprise Manager for management, then leave the `emAdminPassword` password field blank.

Example A-3 Response File Passwords for Oracle Database (oracle user)

This example illustrates the passwords to specify for use with the database configuration assistants.

```
oracle.install.db.config.starterdb.password.SYS=password  
oracle.install.db.config.starterdb.password.SYSTEM=password  
oracle.install.db.config.starterdb.password.DBSNMP=password  
oracle.install.db.config.starterdb.password.PDBADMIN=password  
oracle.install.db.config.starterdb.emAdminPassword=password  
oracle.install.db.config.asm.ASMSNMPPassword=password
```

You can also specify `oracle.install.db.config.starterdb.password.ALL=password` to use the same password for all database users.

The database configuration assistants require the SYS, SYSTEM, and DBSNMP passwords for use with Oracle DBCA. You must specify the following passwords, depending on your system configuration:

- If the database uses Oracle Automatic Storage Management (Oracle ASM) for storage, then you must specify a password for the `ASMSNMPPassword` variable. If you are not using Oracle ASM, then leave the value for this password variable blank.
- If you create a multitenant container database (CDB) with one or more pluggable databases (PDBs), then you must specify a password for the `PDBADMIN` variable. If you are not using Oracle ASM, then leave the value for this password variable blank.

Running Postinstallation Configuration Using Response File

You can use a response file to complete postinstallation tasks on one or more servers simultaneously.

Complete this procedure to run configuration assistants with the `executeConfigTools` command and a response file.

1. Edit the response file and specify the required passwords for your configuration. You can use the response file created during installation, located at `$ORACLE_HOME/install/response/product_timestamp.rsp`. For example:

For Oracle Database (oracle user)

```
oracle.install.asm.SYSASMPassword=password  
oracle.install.config.emAdminPassword=password
```

For Oracle Grid Infrastructure (grid user)

```
grid.install.asm.SYSASMPassword=password  
grid.install.config.emAdminPassword=password
```

2. Change directory to the Oracle home containing the installation software. For example:

For Oracle Grid Infrastructure:

```
cd Grid_home
```

Where, *Grid_home* is the path to the Oracle Grid Infrastructure home directory /u01/app/oracle/product/19.0.0/grid

For Oracle Database:

```
cd $ORACLE_HOME
```

3. Run the configuration script using the following syntax:

For Oracle Grid Infrastructure:

```
$ ./gridSetup.sh -executeConfigTools -responseFile Grid_home/install/  
response/product_timestamp.rsp
```

For Oracle Database:

```
$ ./runInstaller -executeConfigTools -responseFile $ORACLE_HOME/install/  
response/product_timestamp.rsp
```

For Oracle Database, you can also run the response file located in the directory \$ORACLE_HOME/inventory/response/:

```
$ ./runInstaller -executeConfigTools -responseFile $ORACLE_HOME/inventory/  
response/db_install.rsp
```

The postinstallation configuration tool runs the installer in the graphical user interface mode, displaying the progress of the postinstallation configuration. Specify the [-silent] option to run the postinstallation configuration in the silent mode.

For example, for Oracle Grid Infrastructure:

```
$ ./gridSetup.sh -executeConfigTools -responseFile /u01/app/oracle/product/  
19.0.0/grid/response/grid_2016-01-09_01-03-36PM.rsp [-silent]
```

For Oracle Database:

```
$ ./runInstaller -executeConfigTools -responseFile /u01/app/oracle/product/  
19.0.0/dbhome_1/inventory/response/db_2016-01-09_01-03-36PM.rsp [-silent]
```

Postinstallation Configuration Using the ConfigToolAllCommands Script

You can create and run a response file configuration after installing Oracle software. The configToolAllCommands script requires users to create a second response file, of a different format than the one used for installing the product.

Starting with Oracle Database 12c Release 2 (12.2), the configToolAllCommands script is deprecated and may be desupported in a future release.

- [About the Postinstallation Configuration File](#)
When you run a silent or response file installation, you provide information about your servers in a response file that you otherwise provide manually during a graphical user interface installation.
- [Creating a Password Response File](#)
You can create a password response file and use it with configuration assistants to perform silent installation.
- [Running Postinstallation Configuration Using a Password Response File](#)
Complete this procedure to run configuration assistants with the configToolAllCommands script.

About the Postinstallation Configuration File

When you run a silent or response file installation, you provide information about your servers in a response file that you otherwise provide manually during a graphical user interface installation.

However, the response file does not contain passwords for user accounts that configuration assistants require after software installation is complete. The configuration assistants are started with a script called configToolAllCommands. You can run this script in response file mode by using a password response file. The script uses the passwords to run the configuration tools in succession to complete configuration.

If you keep the password file to use for clone installations, then Oracle strongly recommends that you store the password file in a secure location. In addition, if you have to stop an installation to fix an error, then you can run the configuration assistants using configToolAllCommands and a password response file.

The configToolAllCommands password response file has the following syntax options:

- `oracle.crs` for Oracle Grid Infrastructure components or `oracle.server` for Oracle Database components that the configuration assistants configure
- `variable_name` is the name of the configuration file variable
- `value` is the desired value to use for configuration.

The command syntax is as follows:

```
internal_component_name|variable_name=value
```

For example:

```
oracle.crs|S_ASMPASSWORD=Password
```

The database configuration assistants require the SYS, SYSTEM, and DBSNMP passwords for use with Oracle DBCA. You may need to specify the following additional passwords, depending on your system configuration:

- If the database is using Oracle Automatic Storage Management (Oracle ASM) for storage, then you must specify a password for the S_ASMSNMPPASSWORD variable. If you are not using Oracle ASM, then leave the value for this password variable blank.
- If you create a multitenant container database (CDB) with one or more pluggable databases (PDBs), then you must specify a password for the S_PDBADMINPASSWORD variable. If you are not using Oracle ASM, then leave the value for this password variable blank.

Oracle strongly recommends that you maintain security with a password response file:

- Permissions on the response file should be set to 600.
- The owner of the response file should be the installation owner user, with the group set to the central inventory (oraInventory) group.

Creating a Password Response File

You can create a password response file and use it with configuration assistants to perform silent installation.

Perform the following steps to create a password response file:

1. Create a response file that has a name of the format *filename.properties*, for example:

```
$ touch pwdrsp.properties
```

2. Open the file with a text editor, and cut and paste the sample password file contents, as shown in the examples, modifying as needed.
3. Change permissions to secure the password response file. For example:

```
$ ls -al pwdrsp.properties
-rw----- 1 oracle oinstall 0 Apr 30 17:30 pwdrsp.properties
```

Example A-4 Password response file for Oracle Grid Infrastructure (grid user)

```
grid.crs|S_ASMPASSWORD=password
grid.crs|S_OMSPASSWORD=password
grid.crs|S_BMCPASSWORD=password
grid.crs|S_ASMMONITORPASSWORD=password
```

If you do not have a BMC card, or you do not want to enable IPMI, then leave the S_BMCPASSWORD input field blank.

Example A-5 Password response file for Oracle Grid Infrastructure for a Standalone Server (oracle user)

```
oracle.crs|S_ASMPASSWORD=password  
oracle.crs|S_OMSPASSWORD=password  
oracle.crs|S_ASMMONITORPASSWORD=password
```

Example A-6 Password response file for Oracle Database (oracle user)

This example provides a template for a password response file to use with the database configuration assistants.

```
oracle.server|S_SYSPASSWORD=password  
oracle.server|S_SYSTEMPASSWORD=password  
oracle.server|S_EMADMINPASSWORD=password  
oracle.server|S_DBSNMPPASSWORD=password  
oracle.server|S_ASMSNMPPASSWORD=password  
oracle.server|S_PDBADMINPASSWORD=password
```

If you do not want to enable Oracle Enterprise Manager for management, then leave those password fields blank.

Running Postinstallation Configuration Using a Password Response File

Complete this procedure to run configuration assistants with the configToolAllCommands script.

1. Create a password response file as described in *Creating a Password File*.
2. Change directory to \$ORACLE_HOME/cfgtoollogs.
3. Run the configuration script using the following syntax:

```
configToolAllCommands RESPONSE_FILE=/path/name.properties
```

For example:

```
$ ./configToolAllCommands RESPONSE_FILE=/home/oracle/pwdrsp.properties
```

B

Optimal Flexible Architecture

Oracle Optimal Flexible Architecture (OFA) rules are a set of configuration guidelines created to ensure well-organized Oracle installations, which simplifies administration, support and maintenance.

- [About the Optimal Flexible Architecture Standard](#)
Oracle Optimal Flexible Architecture (OFA) rules help you to organize database software and configure databases to allow multiple databases, of different versions, owned by different users to coexist.
- [About Multiple Oracle Homes Support](#)
Oracle Database supports multiple Oracle homes. You can install this release or earlier releases of the software more than once on the same system, in different Oracle home directories.
- [About the Oracle Inventory Directory and Installation](#)
The directory that you designate as the Oracle Inventory directory (oraInventory) stores an inventory of all software installed on the system.
- [Oracle Base Directory Naming Convention](#)
The Oracle Base directory is the database home directory for Oracle Database installation owners, and the log file location for Oracle Grid Infrastructure owners.
- [Oracle Home Directory Naming Convention](#)
By default, Oracle Universal Installer configures Oracle home directories using these Oracle Optimal Flexible Architecture conventions.
- [Optimal Flexible Architecture File Path Examples](#)
Review examples of hierarchical file mappings of an Optimal Flexible Architecture-compliant installation.

About the Optimal Flexible Architecture Standard

Oracle Optimal Flexible Architecture (OFA) rules help you to organize database software and configure databases to allow multiple databases, of different versions, owned by different users to coexist.

In earlier Oracle Database releases, the OFA rules provided optimal system performance by isolating fragmentation and minimizing contention. In current releases, OFA rules provide consistency in database management and support, and simplifies expanding or adding databases, or adding additional hardware.

By default, Oracle Universal Installer places Oracle Database components in directory locations and with permissions in compliance with OFA rules. Oracle recommends that you configure all Oracle components in accordance with OFA guidelines.

Oracle recommends that you accept the OFA default. Following OFA rules is especially of value if the database is large, or if you plan to have multiple databases.

Note

OFA assists in identification of an ORACLE_BASE with its Automatic Diagnostic Repository (ADR) diagnostic data to properly collect incidents.

About Multiple Oracle Homes Support

Oracle Database supports multiple Oracle homes. You can install this release or earlier releases of the software more than once on the same system, in different Oracle home directories.

Careful selection of mount point names can make Oracle software easier to administer. Configuring multiple Oracle homes in compliance with Optimal Flexible Architecture (OFA) rules provides the following advantages:

- You can install this release, or earlier releases of the software, more than once on the same system, in different Oracle home directories. However, you cannot install products from one release of Oracle Database into an Oracle home directory of a different release.
- Multiple databases, of different versions, owned by different users can coexist concurrently.
- To install Oracle Database software in multiple Oracle homes, you must extract the image file in each Oracle home, and then run the setup wizard from the respective Oracle home.
- You must install a new Oracle Database release in a new Oracle home that is separate from earlier releases of Oracle Database.

You cannot install multiple releases in one Oracle home. Oracle recommends that you create a separate Oracle Database Oracle home for each release, in accordance with the Optimal Flexible Architecture (OFA) guidelines.

- In production, the Oracle Database server software release is the release number in the format of major and RU release number. For example, with the release number 19.3.0.0.0, the major release is 19 and the RU release number is 3.
- Later Oracle Database releases can access earlier Oracle Database releases. However, this access is only for upgrades. For example, Oracle Database 19c can access an Oracle Database 18c if the 18c database is started up in upgrade mode.
- Structured organization of directories and files, and consistent naming for database files simplify database administration.
- Login home directories are not at risk when database administrators add, move, or delete Oracle home directories.
- For information about release support timelines, refer to My Oracle Support Doc ID 742060.1

Related Topics

- [My Oracle Support Note 742060.1](#)

About the Oracle Inventory Directory and Installation

The directory that you designate as the Oracle Inventory directory (oraInventory) stores an inventory of all software installed on the system.

All Oracle software installation owners on a server are granted the OINSTALL privileges to read and write to this directory. If you have previous Oracle software installations on a server,

then additional Oracle software installations detect this directory from the `/var/opt/oracle/oraInst.loc` file, and continue to use that Oracle Inventory. Ensure that the group designated as the OINSTALL group is available as a primary group for all planned Oracle software installation owners.

If you are installing Oracle software for the first time, then OUI creates an Oracle base and central inventory, and creates an Oracle inventory using information in the following priority:

- In the path indicated in the `ORACLE_BASE` environment variable set for the installation owner user account
- In an Optimal Flexible Architecture (OFA) path (`u[01–99]/app/owner` where *owner* is the name of the user account running the installation), and that user account has permissions to write to that path
- In the user home directory, in the path `/app/owner`, where *owner* is the name of the user account running the installation

For example:

If you are performing an Oracle Database installation, and you set `ORACLE_BASE` for user `oracle` to the path `/u01/app/oracle` before installation, and grant 755 permissions to `oracle` for that path, then Oracle Universal Installer creates the Oracle Inventory directory one level above the `ORACLE_BASE` in the path `ORACLE_BASE/..oraInventory`, so the Oracle Inventory path is `/u01/app/oraInventory`. Oracle Universal Installer installs the software in the `ORACLE_BASE` path. If you are performing an Oracle Grid Infrastructure for a Cluster installation, then the Grid installation path is changed to root ownership after installation, and the Grid home software location should be in a different path from the Grid user Oracle base.

If you create the OFA path `/u01`, and grant `oracle` 755 permissions to write to that path, then the Oracle Inventory directory is created in the path `/u01/app/oraInventory`, and Oracle Universal Installer creates the path `/u01/app/oracle`, and configures the `ORACLE_BASE` environment variable for the Oracle user to that path. If you are performing an Oracle Database installation, then the Oracle home is installed under the Oracle base. However, if you are installing Oracle Grid Infrastructure for a cluster, then be aware that ownership of the path for the Grid home is changed to root after installation and the Grid base and Grid home should be in different locations, such as `/u01/app/19.0.0/grid` for the Grid home path, and `/u01/app/grid` for the Grid base. For example:

```
/u01/app/oraInventory, owned by grid:oinstall
/u01/app/oracle, owned by oracle:oinstall
/u01/app/oracle/product/19.0.0/dbhome_1/, owned by oracle:oinstall
/u01/app/grid, owned by grid:oinstall
/u01/app/19.0.0/grid, owned by root
```

If you have neither set `ORACLE_BASE`, nor created an OFA-compliant path, then the Oracle Inventory directory is placed in the home directory of the user that is performing the installation, and the Oracle software is installed in the path `/app/owner`, where *owner* is the Oracle software installation owner. For example:

```
/home/oracle/oraInventory
/home/oracle/app/oracle/product/19.0.0/dbhome_1
```

Oracle Base Directory Naming Convention

The Oracle Base directory is the database home directory for Oracle Database installation owners, and the log file location for Oracle Grid Infrastructure owners.

Name Oracle base directories using the syntax `/pm/h/u`, where *pm* is a string mount point name, *h* is selected from a small set of standard directory names, and *u* is the name of the owner of the directory.

You can use the same Oracle base directory for multiple installations. If different operating system users install Oracle software on the same system, then you must create a separate Oracle base directory for each installation owner. For ease of administration, Oracle recommends that you create a unique owner for each Oracle software installation owner, to separate log files.

Because all Oracle installation owners write to the central Oracle inventory file, and that file mountpoint is in the same mount point path as the initial Oracle installation, Oracle recommends that you use the same `/pm/h` path for all Oracle installation owners.

Table B-1 Examples of OFA-Compliant Oracle Base Directory Names

Example	Description
<code>/u01/app/oracle</code>	Oracle Database Oracle base, where the Oracle Database software installation owner name is <code>oracle</code> . The Oracle Database binary home is located underneath the Oracle base path.
<code>/u01/app/grid</code>	Oracle Grid Infrastructure Oracle base, where the Oracle Grid Infrastructure software installation owner name is <code>grid</code> . Caution: The Oracle Grid Infrastructure Oracle base should not contain the Oracle Grid Infrastructure binaries for an Oracle Grid Infrastructure installation. Permissions for the file path to the Oracle Grid Infrastructure binary home are changed to <code>root</code> during installation.

Note

Oracle home or Oracle base cannot be symlinks, nor can any of their parent directories, all the way to up to the root directory.

Oracle Home Directory Naming Convention

By default, Oracle Universal Installer configures Oracle home directories using these Oracle Optimal Flexible Architecture conventions.

The directory pattern syntax for Oracle homes is `/pm/s/u/product/v/type_[n]`. The following table describes the variables used in this syntax:

Variable	Description
<i>pm</i>	A mount point name.
<i>s</i>	A standard directory name.

Variable	Description
<i>u</i>	The name of the owner of the directory.
<i>v</i>	The version of the software.
<i>type</i>	The type of installation. For example: Database (dbhome), Client (client), or Oracle Grid Infrastructure (grid)
<i>n</i>	An optional counter, which enables you to install the same product more than once in the same Oracle base directory. For example: Database 1 and Database 2 (dbhome_1, dbhome_2)

For example, the following path is typical for the first installation of Oracle Database on this system:

/u01/app/oracle/product/19.0.0/dbhome_1

Note

Oracle home or Oracle base cannot be symlinks, nor can any of their parent directories, all the way up to the root directory.

Optimal Flexible Architecture File Path Examples

Review examples of hierarchical file mappings of an Optimal Flexible Architecture-compliant installation.

/u02/u03 /u04

Note

- The Grid homes are examples of Grid homes used for an Oracle Grid Infrastructure for a standalone server deployment (Oracle Restart), or a Grid home used for an Oracle Grid Infrastructure for a cluster deployment (Oracle Clusterware). You can have either an Oracle Restart deployment, or an Oracle Clusterware deployment. You cannot have both options deployed at the same time.
- Oracle Automatic Storage Management (Oracle ASM) is included as part of an Oracle Grid Infrastructure installation. Oracle recommends that you use Oracle ASM to provide greater redundancy and throughput.

Caution

Refrain from placing any private or application files under the Optimal Flexible Architecture file directories such as Oracle base, Oracle home, and OraInventory.

Table B-2 Optimal Flexible Architecture Hierarchical File Path Examples

Directory	Description
/	Root directory
/u01/	User data mount point 1
/u01/app/	Subtree for application software
/u01/app/oraInventory	Central OraInventory directory, which maintains information about Oracle installations on a server. Members of the group designated as the OINSTALL group have permissions to write to the central inventory. All Oracle software installation owners must have the OINSTALL group as their primary group, and be able to write to this group.
/u01/app/oracle/	Oracle base directory for user oracle. There can be many Oracle Database installations on a server, and many Oracle Database software installation owners. Oracle software homes that an Oracle installation owner owns should be located in the Oracle base directory for the Oracle software installation owner, unless that Oracle software is Oracle Grid Infrastructure deployed for a cluster.
/u01/app/grid	Oracle base directory for user grid. The Oracle home (Grid home) for Oracle Grid Infrastructure installation is located outside of the Grid user. There can be only one Grid home on a server, and only one Grid software installation owner. The Grid home contains log files and other administrative files.
/u01/app/oracle/admin/	Subtree for database administration files
/u01/app/oracle/admin/TAR	Subtree for support log files
/u01/app/oracle/admin/db_sales/	Admin subtree for database named "sales"
/u01/app/oracle/admin/db_dwh/	Admin subtree for database named "dwh"

Table B-2 (Cont.) Optimal Flexible Architecture Hierarchical File Path Examples

Directory	Description
/u01/app/oracle/ fast_recovery_ are a/	Subtree for recovery files
/u01/app/oracle/ fast_recovery_ are a/db_sales	Recovery files for database named "sales"
/u01/app/oracle/ fast_recovery_ are a/db_dwh	Recovery files for database named "dwh"
/u02/app/oracle/ oradata /u03/app/oracle/ oradata /u04/app/oracle/ oradata	Oracle data file directories
/u01/app/oracle/ product/	Common path for Oracle software products other than Oracle Grid Infrastructure for a cluster
/u01/app/oracle/ product/19.0.0/ dbhome_1	Oracle home directory for Oracle Database 1, owned by Oracle Database installation owner account orac le
/u01/app/oracle/ product/19.0.0/ dbhome_2	Oracle home directory for Oracle Database 2, owned by Oracle Database installation owner account orac le
/u01/app/oracle2/ product/19.0.0/ dbhome_2	Oracle home directory for Oracle Database 2, owned by Oracle Database installation owner account orac le2

Table B-2 (Cont.) Optimal Flexible Architecture Hierarchical File Path Examples

Directory	Description
/u01/app/oracle/ product/19.0.0/ grid	Oracle home directory for Oracle Grid Infrastructure for a standalone server, owned by Oracle Database and Oracle Grid Infrastructure installation owner oracle.
/u01/app/19.0.0/ grid	Oracle home directory for Oracle Grid Infrastructure for a cluster (Grid home), owned by user grid before installation, and owned by root after installation.
/u01/app/oracle/ product/19.0.0/ client_1	Oracle home directory for Oracle Database Client 1, owned by Oracle Database installation owner account oracle

C

Configuring Read-Only Oracle Homes

Understand how read-only Oracle homes work and how you can configure read-only Oracle homes.

- [Evolution of Oracle Homes](#)
Learn about read-only Oracle home concepts like Oracle base home, Oracle base config, and orabasetab.
- [Enabling a Read-Only Oracle Home](#)
Configure your Oracle home as a read-only Oracle home after you have performed a software-only Oracle Database installation.
- [Copying demo Directories to Oracle Base Home](#)
In a read-only mode ORACLE_HOME, you must copy the demo directories listed in this topic from ORACLE_HOME to ORACLE_BASE_HOME.
- [Determining if an Oracle Home is Read-Only](#)
Run the orabasehome command to determine if your Oracle home is a read/write or read-only Oracle home.
- [File Path and Directory Changes in Read-Only Oracle Homes](#)
Examples of hierarchical file mappings in a read-only Oracle home as compared to a read/write Oracle home.

Evolution of Oracle Homes

Learn about read-only Oracle home concepts like Oracle base home, Oracle base config, and orabasetab.

- [About Read-Only Oracle Homes](#)
Starting with Oracle Database 18c, you can configure an Oracle home in read-only mode. A read-only Oracle home does not imply that the file system and the Oracle home is in read-only mode. The file system and mount point should always be in read/write mode.
- [About Oracle Base Homes](#)
Both, in a read-only ORACLE_HOME and read/write ORACLE_HOME, the user-specific files, instance-specific files, and log files reside in a location known as the ORACLE_BASE_HOME.
- [About Oracle Base Config](#)
Both, in a read-only ORACLE_HOME and read/write ORACLE_HOME, the configuration files reside in a location known as ORACLE_BASE_CONFIG.
- [About orabasetab](#)
The orabasetab file is used to define fundamental directories based on \$ORACLE_HOME, ORACLE_BASE, ORACLE_BASE_HOME and ORACLE_BASE_CONFIG.

About Read-Only Oracle Homes

Starting with Oracle Database 18c, you can configure an Oracle home in read-only mode. A read-only Oracle home does not imply that the file system and the Oracle home is in read-only mode. The file system and mount point should always be in read/write mode.

A read-only Oracle home simplifies provisioning by implementing separation of installation and configuration.

In a read-only Oracle home, the configuration data and log files reside outside of the read-only Oracle home. See, File Path and Directory Changes in Read-Only Oracle Homes.

Apart from the traditional ORACLE_BASE and ORACLE_HOME directories, the following directories contain files that used to be in ORACLE_HOME:

- ORACLE_BASE_HOME
- ORACLE_BASE_CONFIG

Ensure that you enable read-only Oracle home before you create the database or listener.

When you run the `roohctl` script to enable a read-only Oracle home, the Oracle home file system will continue to be a read/write file system.

Note

This feature does not affect how database administrators monitor, diagnose, and tune their system performance.

About Oracle Base Homes

Both, in a read-only ORACLE_HOME and read/write ORACLE_HOME, the user-specific files, instance-specific files, and log files reside in a location known as the ORACLE_BASE_HOME.

In a read/write ORACLE_HOME, the ORACLE_BASE_HOME path is the same as the ORACLE_HOME directory. However, in a read-only ORACLE_HOME, the ORACLE_BASE_HOME directory is not co-located with ORACLE_HOME but is located at ORACLE_BASE/homes/HOME_NAME.

Where, HOME_NAME is the internal name for ORACLE_HOME.

For example, the networking directories `network/admin`, `network/trace`, and `network/log` are located in the ORACLE_BASE_HOME directory. In a read/write ORACLE_HOME the networking directories appear to be in ORACLE_HOME because ORACLE_BASE_HOME is co-located with ORACLE_HOME, whereas in a read-only ORACLE_HOME the networking directories are located in ORACLE_BASE/homes/HOME_NAME.

To print the ORACLE_BASE_HOME path, run the `orabasehome` command from the `$ORACLE_HOME/bin` directory:

```
$ setenv ORACLE_HOME /u01/app/oracle/product/19.0.0/dbhome_1
$ cd $ORACLE_HOME/bin
$ ./orabasehome
```

For example:

```
$ ./orabasehome  
/u01/app/oracle/homes/OraDB19Home1
```

Where, /u01/app/oracle is ORACLE_BASE and OraDB19Home1 is HOME_NAME

About Oracle Base Config

Both, in a read-only ORACLE_HOME and read/write ORACLE_HOME, the configuration files reside in a location known as ORACLE_BASE_CONFIG.

In a read/write ORACLE_HOME, the ORACLE_BASE_CONFIG path is the same as the ORACLE_HOME path because it is located at \$ORACLE_HOME. However, in a read-only ORACLE_HOME, the ORACLE_BASE_CONFIG path is the same as ORACLE_BASE.

ORACLE_BASE_CONFIG/dbs contains the configuration files for ORACLE_HOME. Each file in the dbs directory contains \$ORACLE_SID so that the directory can be shared by many different ORACLE_SIDs.

To print the ORACLE_BASE_CONFIG path, run the orabaseconfig command from the \$ORACLE_HOME/bin directory:

```
$ setenv ORACLE_HOME /u01/app/oracle/product/19.0.0/dbhome_1  
$ cd $ORACLE_HOME/bin  
$ ./orabaseconfig
```

For example:

```
$ ./orabaseconfig  
/u01/app/oracle
```

Where, /u01/app/oracle is ORACLE_BASE.

About orabasetab

The orabasetab file is used to define fundamental directories based on \$ORACLE_HOME, ORACLE_BASE, ORACLE_BASE_HOME and ORACLE_BASE_CONFIG.

The orabasetab file resides in ORACLE_HOME/install/orabasetab. To determine if an ORACLE_HOME is read-only or read/write, you can check for the presence of the orabasetab file. The orabasetab file also defines the ORACLE_BASE and the HOME_NAME of the Oracle home. HOME_NAME is the internal name for ORACLE_HOME.

The last line in the orabasetab file, which starts with \$ORACLE_HOME, defines the directories for \$ORACLE_HOME. The last line consists of four fields, each separate by a colon delimiter(:).

1. The first field matches the current \$ORACLE_HOME.
2. The second field defines the ORACLE_BASE for the current ORACLE_HOME.
3. The third field defines the HOME_NAME which is used in constructing the ORACLE_BASE_HOME path in a read-only ORACLE_HOME.

4. The fourth field displays N in a read/write ORACLE_HOME and Y in a read-only ORACLE_HOME.

In a read-only ORACLE_HOME, the ORACLE_BASE_HOME path is ORACLE_BASE/homes/HOME_NAME and ORACLE_BASE_CONFIG is the same as ORACLE_BASE.

In a read/write ORACLE_HOME, ORACLE_HOME, ORACLE_BASE_HOME and ORACLE_BASE_CONFIG are all the same.

Viewing an orabasetab File

1. Log in as the Oracle installation owner user account (oracle).
2. Go to the \$ORACLE_HOME/install directory.

```
$ cd /u01/app/oracle/product/19.0.0/dbhome_1/install
```

3. View the contents of the orabasetab file.

```
$ cat orabasetab
/u01/app/oracle/product/19.0.0/dbhome_1:/u01/app/oracle:oraDB19Home1:Y:
```

In this example, a Y in the fourth field at the end of the line indicates you have a read-only Oracle home.

Enabling a Read-Only Oracle Home

Configure your Oracle home as a read-only Oracle home after you have performed a software-only Oracle Database installation.

To enable a read-only Oracle home:

1. Perform a software-only Oracle Database installation.
2. Run the `roohctl -enable` script.
3. Run Oracle Database Configuration Assistant (Oracle DBCA) to create a database.

Note

Ensure that you enable read-only Oracle home before you create the database or listener.

Software-Only Database Installation

1. Log in as the Oracle installation owner user account (oracle) that you want to own the software binaries.
2. Download the Oracle Database installation image files (db_home.zip) to a directory of your choice. For example, you can download the image files to the /tmp directory.
3. Create the Oracle home directory and extract the image files that you have downloaded in to this Oracle home directory. For example:

```
$ mkdir -p /u01/app/oracle/product/19.0.0/dbhome_1
$ chown oracle:oinstall /u01/app/oracle/product/19.0.0/dbhome_1
```

```
$ cd /u01/app/oracle/product/19.0.0/dbhome_1
$ unzip -q /tmp/db_home.zip
```

Note

Ensure that the Oracle home directory path you create is in compliance with the Oracle Optimal Flexible Architecture recommendations. Also, unzip the installation image files only in this Oracle home directory that you created.

4. From the Oracle home directory, run the `runInstaller` command to start the Oracle Database installer.
5. In the Select Configuration Option screen, select **Set Up Software Only**.
6. Select your installation type. Installation screens vary depending on the installation option you select. Respond to the configuration prompts as needed.

Note

Click **Help** if you have any questions about the information you are asked to submit during installation.

Run the roohctl Script

1. Go to the `bin` directory

```
$ cd /u01/app/oracle/product/19.0.0/dbhome_1/bin
```

2. Run the `roohctl` script to enable read-only Oracle home.

```
$ ./roohctl -enable
```

3. On Oracle Real Application Clusters (Oracle RAC) installations, run the preceding `roohctl` script on every node of the cluster. Alternatively, run the `roohctl` script with the `nodeList` option and provide the list of cluster nodes:

```
$ ./roohctl -enable -nodelist comma_separated_list_of_nodes
```

Run Oracle Database Configuration Assistant

1. Ensure that you are still in the `bin` directory and run Oracle DBCA.

```
$ ./dbca
```

2. In the Select Database Operation screen, select **Create a Database**.
3. The configuration screens vary depending on the options you select. Respond to the prompts as needed.

Note

Click **Help** if you have any questions about the information you are asked to submit during database creation.

Related Topics

- *Oracle Database 2 Day DBA*

Copying demo Directories to Oracle Base Home

In a read-only mode ORACLE_HOME, you must copy the demo directories listed in this topic from ORACLE_HOME to ORACLE_BASE_HOME.

Oracle Database contains various demo directories that include a variety of examples and product demonstrations that you can use to learn about the products, options, and features of Oracle Database. In a read-only mode ORACLE_HOME, you cannot use these demo directories in ORACLE_HOME because writes are performed to these demo directories when they are used.

Many of the demo directories are not available by default. You must install Oracle Database Examples to view and use the examples and product demonstrations.

Copy the respective demo directory to the corresponding location in ORACLE_BASE_HOME. Now, you can use this copy of the demo directory.

You must copy the following demo directories from ORACLE_HOME to ORACLE_BASE_HOME:

- jdbc/demo
- odbc/demo
- ord/http/demo
- precomp/demo
- rdbms/demo
- sqlj/demo
- sqlplus/demo
- xdk/demo

You must also create symbolic links for the odbc/demo, precomp/demo, rdbms/demo, and xdk/demo demo directories. See the "Creating Symbolic Links" section in this topic.

Copying demo Directories

For example, to copy the rdbms/demo directory from ORACLE_HOME to ORACLE_BASE_HOME, perform the following:

1. Login as the Oracle software owner user (oracle).
2. Check if the rdbms/demo directory is copied to ORACLE_BASE_HOME.

```
$ ls -l -d $(orabasehome)/rdbms/demo
```

3. If the rdbms/demo directory has not been copied, then create it and copy it.

```
$ mkdir -p $(orabasehome)/rdbms
$ cp -r $ORACLE_HOME/rdbms/demo $(orabasehome)/rdbms/demo
```

Similarly, copy all the demo directories listed earlier from ORACLE_HOME to ORACLE_BASE_HOME.

Creating Symbolic Links

You must create symbolic links for the odbc/demo, precomp/demo, rdbms/demo, and xdk/demo demo directories.

For rdbms/demo, replace \$ORACLE_HOME/rdbms/demo with a symbolic link to the copy.

1. Ensure that the symbolic link does not already exist.

```
$ ls -l -d $ORACLE_HOME/rdbms/demo
```

2. If \$ORACLE_HOME/rdbms/demo is still the original demo directory, rename it and replace it with the symbolic link.

```
$ cd $ORACLE_HOME/rdbms
$ mv demo demo.installed
$ ln -s $(orabasehome)/rdbms/demo $ORACLE_HOME/rdbms/demo
```

For odbc/demo, replace \$ORACLE_HOME/odbc/demo with a symbolic link to the copy.

1. Ensure that the symbolic link does not already exist.

```
$ ls -l -d $ORACLE_HOME/odbc/demo
```

2. If \$ORACLE_HOME/odbc/demo is still the original demo directory, rename it and replace it with the symbolic link.

```
$ cd $ORACLE_HOME/odbc
$ mv demo demo.installed
$ ln -s $(orabasehome)/odbc/demo $ORACLE_HOME/odbc/demo
```

For precomp/demo, replace \$ORACLE_HOME/precomp/demo with a symbolic link to the copy.

1. Ensure that the symbolic link does not already exist.

```
$ ls -l -d $ORACLE_HOME/precomp/demo
```

2. If \$ORACLE_HOME/precomp/demo is still the original demo directory, rename it and replace it with the symbolic link.

```
$ cd $ORACLE_HOME/precomp
$ mv demo demo.installed
$ ln -s $(orabasehome)/precomp/demo $ORACLE_HOME/precomp/demo
```

The xdk/demo directory requires a symbolic link at \$ORACLE_HOME/xdk/include pointing to \$(orabasehome)/xdk/include after you copy the xdk/demo directory.

1. Ensure that the symbolic link does not already exist:

```
$ ls -l -d $ORACLE_HOME/xdk/include
```

2. If the symbolic link does not exist, then, run the following command:

```
$ ln -s $ORACLE_HOME/xdk/include $(orabasehome)/xdk/include
```

Note

In the `plsql/demo` directory, `ncmpdemo.sql` is unusable in read-only mode.

Copying the `init.ora` File

Copy the `init.ora` file from `ORACLE_HOME` to `ORACLE_BASE_HOME`.

1. Login as the Oracle software owner user (`oracle`).
2. Check if the `init.ora` file exists in `ORACLE_BASE_HOME`.

```
$ ls $(orabasehome)/init.ora
```

If an `init.ora` file exists in `ORACLE_BASE_HOME`, then update this `init.ora` file to be in-sync with the `$ORACLE_HOME/init.ora` file.

3. If the `init.ora` file does not exist in `ORACLE_BASE_HOME`, then copy it from `ORACLE_HOME`.

```
$ cp $ORACLE_HOME/init.ora $(orabasehome)/init.ora
```

Related Topics

- *Oracle Database Examples Installation Guide*

Determining if an Oracle Home is Read-Only

Run the `orabasehome` command to determine if your Oracle home is a read/write or read-only Oracle home.

If the output of the `orabasehome` command is the same as `$ORACLE_HOME`, then your Oracle home is in read/write mode. If the output displays the path `ORACLE_BASE/homes/HOME_NAME`, then your Oracle home is in read-only mode.

1. Set the `ORACLE_HOME` environment variable:

Bourne, Bash or Korn shell:

```
$ ORACLE_HOME=/u01/app/oracle/product/19.0.0/dbhome_1  
$ export ORACLE_HOME
```

C shell:

```
% setenv ORACLE_HOME /u01/app/oracle/product/19.0.0/dbhome_1
```

- Go to the bin directory and run the orabasehome command:

```
$ cd $ORACLE_HOME/bin
$ ./orabasehome
/u01/app/oracle/homes/OraDB19Home1
```

In this example, the Oracle home is in read-only mode.

File Path and Directory Changes in Read-Only Oracle Homes

Examples of hierarchical file mappings in a read-only Oracle home as compared to a read/write Oracle home.

This example shows an Optimal Flexible Architecture-compliant Oracle Database installation, for the user oracle, with the ORACLE_HOME, ORACLE_BASE, ORACLE_BASE_HOME, and ORACLE_BASE_CONFIG logical locations. The database files are mounted on /u01 and the HOME_NAME is OraDB19Home1.

This example also shows the changes in the Oracle Database software defined paths of configuration files, log files, and other directories in a read-only Oracle home when compared to a read/write Oracle home.

Table C-1 read/write and Read-Only Oracle Home File Path Examples

Directory	Read/Write Oracle Home File Path	Read-Only Oracle Home File Path
ORACLE_HOME	/u01/app/oracle/product/19.0.0/dbhome_1	/u01/app/oracle/product/19.0.0/dbhome_1
ORACLE_BASE	/u01/app/oracle/	/u01/app/oracle/
ORACLE_BASE_HOME	ORACLE_HOME (or) /u01/app/oracle/product/19.0.0/dbhome_1	ORACLE_BASE/homes/HOME_NAME (or) /u01/app/oracle/homes/OraDB19Home1
ORACLE_BASE_CONFIG	ORACLE_HOME (or) /u01/app/oracle/product/19.0.0/dbhome_1	ORACLE_BASE (or) /u01/app/oracle/
network	ORACLE_HOME/network (or) /u01/app/oracle/product/19.0.0/dbhome_1/network	ORACLE_BASE_HOME/network (or) /u01/app/oracle/homes/OraDB19Home1/network
dbs	ORACLE_HOME/dbs (or) /u01/app/oracle/product/19.0.0/dbhome_1/dbs	ORACLE_BASE/dbs (or) /u01/app/oracle/dbs

D

Managing Oracle Database Port Numbers

Review default port numbers. If needed, use these steps to change assigned ports after installation.

- [About Managing Ports](#)
During installation, Oracle Universal Installer assigns port numbers to components from a set of default port numbers.
- [Oracle Database Component Port Numbers and Protocols](#)
This table lists the port numbers and protocols configured for Oracle Database components during a single-instance installation. By default, the first port in the range is assigned to the component, if it is available.

About Managing Ports

During installation, Oracle Universal Installer assigns port numbers to components from a set of default port numbers.

Many Oracle Database components and services use ports. As an administrator, it is important to know the port numbers used by these services, and to ensure that the same port number is not used by two services on your host. Enter the following command to identify the ports currently used on your computer:

```
$/bin/netstat -a
```

Most port numbers are assigned during installation. Every component and service has an allotted port range, which is the set of port numbers Oracle Database attempts to use when assigning a port. Oracle Database starts with the lowest number in the range and performs the following checks:

- Is the port used by another Oracle Database installation on the host?
The installation may be up or down at the time. Oracle Database can still detect if the port is used.
- Is the port used by a process that is currently running?
This can be any process on the host, even a non-Oracle Database process.
- Is the port listed in the `/etc/services` file?

If the answer to any of the preceding questions is yes, then Oracle Database moves to the next highest port in the allotted port range, and continues checking until it finds a free port.

Oracle Database Component Port Numbers and Protocols

This table lists the port numbers and protocols configured for Oracle Database components during a single-instance installation. By default, the first port in the range is assigned to the component, if it is available.

Table D-1 Protocols and Default Port Numbers for Oracle Database Components

Component	Description	Default Port Number	Port Range	Protocol
Oracle Net Services Listener	Enables Oracle client connections to the database over the Oracle Net Services protocol. You can configure it during installation. To reconfigure this port, use Net Configuration Assistant.	1521	Port number changes to the next available port. Modifiable manually to any available port.	TCP
Oracle Connection Manager	Listening port for Oracle client connections to Oracle Connection Manager. It is not configured during installation, but can be configured manually by editing the <code>cman.ora</code> parameter file. This file is located under the <code>/network/admin</code> directory.	1630	1630	TCP
Oracle XML DB	The Oracle XML DB HTTP port is used if web-based applications must access an Oracle database from an HTTP listener. You must configure this port manually.	0	Configured Manually	HTTP
Oracle XML DB	The Oracle XML DB FTP is used when applications must access an Oracle database from an FTP listener. You must configure this port manually.	0	Configured Manually	FTP

Related Topics

- Using HTTP(S) on a Standard Port
- Using FTP on the Standard Port
- *Oracle Real Application Clusters Installation Guide for Linux and UNIX*

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